

BINARY FOUNDATIONS AND AUXILIARY PROOFS

QIAN QI

ABSTRACT. This supplement gives the complete binary and auxiliary proofs used in Finite-Use Geometry and Learning of Ordered Quantum Measurements. All inherited proof sections and their hypotheses are retained. The article and this source-reconstructible supplement form one proof package; the independent structural paper is not a dependency.

S1. INTRODUCTION

A measurement used repeatedly is not determined, at a fixed accuracy, by its one-use distance alone. Its spectral noise controls how errors accumulate, and normalization couples the effects of different outcomes. We study these two features for ordered, memoryless measurements which consume their input and return a classical label. The tester may retain an arbitrary reference and quantum memory, use common adaptive control, and stop publicly after at most N calls. The resulting operational distance D_N is the supremum of the unhalved final-state trace norm. Nonadaptive experiments may use entangled inputs and retained references; they do not condition subsequent inputs on earlier outcomes.

S1.1. Finite-outcome interior geometry and learning. For $d, N \geq 1$ and $k \geq 2$, consider the balanced interior

$$\mathfrak{P}_{d,k} = \{(E_j)_{j=1}^k : \sum_j E_j = I_d, I_d/(2k) \preceq E_j \preceq 3I_d/(2k)\}.$$

No common eigenbasis is assumed. With $r(E, F) = \max_j \|E_j - F_j\|_{\text{op}}$ and $p = (k-1)d^2$, Theorems S22.1 and S22.2 give

$$\begin{aligned} \frac{1}{128} \min\{1, \sqrt{N}r(E, F)\} &\leq D_N^{\text{na}}(E, F) \leq D_N(E, F) \leq \min\{2, 2k\sqrt{N}r(E, F)\}, \\ \log_2 \mathcal{C}_{N,k}(\delta) &= \frac{p}{2} \log_2 N + p \log_2(1/\delta) + O(p \log(k+1)). \end{aligned}$$

The geometric comparison holds without an accuracy restriction. The covering statement assumes $0 < \delta < 1/128$, permits arbitrary legal centres, and has an absolute implied constant in its logarithmic remainder. A normalization-preserving rational grid yields a finite exact public code at this fixed-length order.

The common learner in Theorem S23.1 satisfies

$$M_{d,k}^*(N, \delta, \eta) = \Theta_k\left(N\delta^{-2}[d^2 + d \log(1/\eta)]\right), \quad 0 < \delta \leq 2^{-24}, \quad 0 < \eta \leq 1/8.$$

This is a joint law in d, N, δ, η for each *fixed* outcome count k . Its explicit upper bound is $Ck^3N\delta^{-2}[d^2 + d \log(k/\eta)]$; the lower bound does not contain a growing- k factor. The upper uses independent one-call probe-reference acquisitions and collective processing of completed outputs. The lower permits coherent adaptive training and bounded public stopping. Every training record satisfies the call budget. The eigenvectors used in pairwise lower witnesses are not inputs to this common learner.

Date: 6 October 2026.

2020 Mathematics Subject Classification. Primary 81P15, 81P47; Secondary 94A17, 81P70.

S1.2. A coupled covariance on the complete body. Let $\mathcal{P}_{d,k}$ instead impose only $E_j \succeq 0$ and $\sum_j E_j = I_d$. On zero-sum Hermitian tuples define

$$S_E(K) = \sum_j E_j K_j, \quad (\mathcal{C}_E K)_j = \frac{1}{2}(E_j K_j + K_j E_j - E_j S_E(K) - S_E(K)^* E_j).$$

The operator is positive and concave in the tuple E . Its quadratic form is the variance

$$\langle K, \mathcal{C}_E K \rangle = \sum_j \text{tr}(E_j K_j^2) - \left\| \sum_j E_j K_j \right\|_{\text{HS}}^2.$$

Put $H = F - E$, $M = (E + F)/2$, and

$$\mathcal{Q}_N(E, F)^2 = N \langle H, (\mathcal{C}_M + N^{-1} \text{Id})^{-1} H \rangle.$$

Theorem 2.1 proves the closed-body bound

$$D_N(E, F) \leq \min\{2, 2\sqrt{k+1}\} \mathcal{Q}_N(E, F),$$

including zero effects, changes of rank, and noncommuting supports. A regularized horizontal energy identity separates the tangent into a part accumulating at rate $\sqrt{N}$ and a residual accumulating at rate N . Concavity then converts a path estimate to the displayed midpoint bound. For two outcomes the operator is exactly the binary Sylvester variance, with $\mathcal{Q}_N^2 = 2Q_N^2$.

This covariance also gives a data-processing inequality for its unregularized dual energy, and it vanishes identically precisely at projection-valued measurements. Under admixture of uniform outcome noise of size ϵ , Theorem 2.5 obtains a finite-use crossover proportional to $N/\sqrt{1+N\epsilon}$, sharp up to fixed- k constants on a rotating projective family. Proposition 2.6 evaluates the squared modulus by one exact rational linear system of dimension p . This is an upper certificate, not a formula for the exact optimal tester or a matching classification of the whole multi-outcome boundary.

S1.3. Legalization, descriptions, and proof organization. Theorem S24.1 replaces tuple-grid legalization by a rational affine map with polynomial bit complexity. The map preserves normalization, satisfies both balanced spectral bounds, and increases component error by at most a factor four. Its guarded version is legal on every record. Corollary S24.2 preserves the joint query and description orders. This removes enumeration from *legalization*, not from dictionary construction or general collective-readout synthesis.

Sections S22–S24 contain the finite-outcome statements and the coupled closed-body extension. Section S3 gives quantitative comparisons with current tomography, together with the distinct resource conventions. The appendices supply the complete binary boundary geometry, entropy, learning, confidence, finite-control, and certificate proofs used by these results. In particular, no proof of a cited binary subprocedure is left outside this article. The stochastic-realization companion has its own hypotheses and proofs and is a separate mathematical object.

S2. BINARY FOUNDATIONS AND THEIR FINITE-OUTCOME APPLICATION

We study the finite-use geometry of input-consuming quantum measurements with ordered classical outputs and no residual quantum system. There are two complementary parameter families: the complete body of binary effects, including singular boundaries, and balanced interiors of arbitrary finite outcome alphabets. In the second family no simultaneous diagonalization is assumed. The results join operational comparison, description entropy, common learning, and finite exact realization under one declared interface.

Repeated uses of an ordered binary quantum measurement resolve different perturbations at different scales. An interior change of an eigenvalue is detected at scale $N^{-1/2}$, whereas a rotation of a projective measurement can be detected at scale N^{-1} . These scales meet at singular

effects and repeated eigenvalues. We determine the resulting geometry on the compact body of matrix effects, calculate its description entropy, and construct common learners whose loss is distinguishability by all future experiments with at most N uses.

For an input dimension d , let

$$\mathfrak{E}_d = \{E = E^* \in M_d(\mathbb{C}) : 0 \preceq E \preceq I_d\}.$$

An effect E specifies the memoryless ordered binary measurement

$$\mathcal{M}_E(\rho) = \text{tr}(E\rho)|1\rangle\langle 1| + \text{tr}((I_d - E)\rho)|0\rangle\langle 0|.$$

The input is consumed; the device returns a classical outcome and no residual quantum system. The distance d_N is the supremum of the unhalved trace norm between the final states of a common experiment with at most N calls. Such an experiment may entangle inputs, retain an arbitrary reference and quantum memory, use feedback, and stop at a public bounded time. Thus $0 \leq d_N \leq 2$; classical total variation is one half of the corresponding trace norm. The nonadaptive restriction d_N^{na} allows entangled input blocks and retained references, without feedback between calls.

S2.1. Geometry and entropy of the closed effect body. For $E, F \in \mathfrak{E}_d$, put

$$M = \tfrac{1}{2}(E + F), \quad H = F - E, \quad V = M(I_d - M),$$

and define the comparison modulus

$$Q_N(E, F)^2 = N \langle H, (L_V + R_V + N^{-1} \text{Id})^{-1} H \rangle_{\text{HS}}.$$

The inverse acts on the real Hilbert space of Hermitian matrices. For every $d, N \geq 1$, Theorem S4.1 gives

$$(S2.1) \quad \frac{\min\{1, Q_N(E, F)\}}{8192d} \leq d_N^{\text{na}}(\mathcal{M}_E, \mathcal{M}_F) \leq d_N(\mathcal{M}_E, \mathcal{M}_F) \leq \min\{2, 8Q_N(E, F)\}.$$

The comparison holds at every spectral rank and multiplicity, with no margin from the support endpoints. In particular, $d_N \leq 65536d d_N^{\text{na}}$ uniformly in N . The modulus provides an explicit comparison of the operational distances; its use requires no triangle inequality for Q_N .

A horizontal measurement dilation gives the adaptive tangent bound. A regularized Sylvester equation separates the tangent into a horizontal part and a remainder controlled by the hybrid inequality. Operator concavity of $G(I_d - G)$ then integrates the bound along a finite segment, including boundary endpoints. Bernoulli inputs and exact two-dimensional compressions give the converse. These pair-dependent experiments establish separation; the common learning construction below is a separate argument.

Let $\mathcal{C}_N(\mathfrak{E}_d, \delta)$ be the least number of legal memoryless binary centres in a d_N -cover. For every fixed d , Theorem S5.1 proves

$$(S2.2) \quad \mathcal{C}_N(\mathfrak{E}_d, \delta) \asymp_d N^{d^2/2} [\log(N+2)]^{\lfloor d/2 \rfloor} \delta^{-d^2}, \quad N \geq 1, \quad 0 < \delta \leq \delta_d.$$

The positive cap δ_d and comparison constants may depend on d . The proof bounds the weighted volume of actual operational balls uniformly over all boundary strata. The total volume is a spectral integral: its ordered endpoint sectors have cumulative exponents $S_j^2/2$, where S_j is a signed prefix sum. Balanced prefixes, together with matching nested endpoint boxes, give the logarithmic multiplicity $\lfloor d/2 \rfloor$.

S2.2. Common learning and independent acquisition blocks. The learner receives calls to an unknown effect, without its matrix entries. Theorem S7.1 constructs one procedure common to all $E \in \mathfrak{E}_d$ such that

$$(S2.3) \quad \sup_E \mathbb{P}_E \{d_N(\mathcal{M}_E, \mathcal{M}_{\hat{E}}) > \delta\} \leq \eta, \quad M \leq Cd^4 N \delta^{-2} \log(d/\eta),$$

where C is absolute, $N \geq 1$, $0 < \delta \leq \delta_0$ for an absolute $\delta_0 > 0$, and $0 < \eta \leq 1/8$. The training-call bound holds on every record, and each entangled block uses at most N calls. A scalar Bernoulli subfamily gives a lower bound of order $N\delta^{-2} \log(1/\eta)$ even for arbitrary reference-assisted adaptive training. Thus the horizon, accuracy, and confidence orders are optimal for fixed d ; the displayed d^4 factor is an upper bound.

The qubit construction learns unknown bias, contrast, and direction. Independent fair signs remove the bias from GHZ parity observations. A dyadic amplitude search maintains a small-cone invariant and terminates at the first rejected scale. Fresh observations after each data-dependent choice complete the direction and endpoint estimates. Conditional confidence allocation controls exceptional records as well as the maximum training cost.

The matrix extension first calibrates both support boundaries in relative Loewner order at scale N^{-1} and the absolute matrix error at scale $N^{-1/2}$. In the calibrated eigenbasis, these estimates control the additional variance produced by every two-dimensional compression. Qubit learning on the compressions then supplies compatible coordinates in the regularized quadratic form. Projection onto the legal effect body, followed by rational selection, completes the estimate. Conditional summation of the calibration and compression budgets introduces no $\log N$ factor.

The maximal independent block length is a further acquisition resource. Within a fresh block, reference-assisted quantum controls are allowed. At its boundary, completed outputs may be processed jointly, and the classical record may determine the next preparation. The next block, including its reference and unused inputs, is isolated from earlier quantum registers until its final call. Thus collective final estimation and classical feedback are allowed, while older quantum memory cannot feed unfinished queries.

For a declared cap $1 \leq b \leq N$ and every fixed $d \geq 2$, Theorem S9.2 proves

$$(S2.4) \quad M_b^*(d, N, \delta, \eta) = \Theta_d \left(\frac{N^2}{b\delta^2} \log \frac{1}{\eta} \right)$$

for an absolute small-error range and $0 < \eta \leq 1/8$. The explicit upper bound is

$$Cd^4(N^2/b)\delta^{-2} \log(d/\eta).$$

Horizon blocking gives the common learner; a history-weighted root-fidelity bound gives the converse with arbitrary storage and collective processing of completed blocks. In dimension one the device is a coin, and the cost is $\Theta(N\delta^{-2} \log(1/\eta))$ for every b .

S2.3. Joint learning laws in the interior. On the full-dimensional interior

$$\mathcal{I}_d = \{E : I_d/4 \preceq E \preceq 3I_d/4\},$$

the operational distance has a dimension-free comparison with $\min\{1, \sqrt{N}\|E - F\|_{\text{op}}\}$. The binary Choi estimator of Mele–Bittel [5], analysed under this loss, gives

$$(S2.5) \quad M_{\text{int}}^*(d, N, \delta, 1/8) = \Theta(d^2 N \delta^{-2})$$

with absolute constants, using independent one-call probes. For the converse, a family within operator radius r of $I/2$ supplies at most $4r^2$ information about its finite unknown label per call. Matrix packing and Fano's inequality give Theorem S10.2, which permits arbitrary coherent adaptive training.

The confidence dependence is also determined jointly. For all $d, N \geq 1$, $0 < \delta \leq 2^{-13}$ and $0 < \eta \leq 1/8$, Theorem S12.1 proves

$$(S2.6) \quad M_{\text{int}}^*(d, N, \delta, \eta) = \Theta\left(\frac{N}{\delta^2} \left(d^2 + d \log \frac{1}{\eta}\right)\right).$$

The constants are absolute. The upper bound uses independent one-call Choi acquisitions, and the lower bound allows coherent adaptive training; hence the order is the same under every fresh-block cap. At $N = 1$ it gives the joint dimension, accuracy, and confidence law for binary operator-norm learning, including the fixed-confidence order $\Theta(d^2 \epsilon^{-2})$.

The confidence converse compares $I_d/2$ with d diagonal single-coordinate perturbations. Revealing the sampled basis index gives a relative-entropy chain rule through retained quantum memory. Under the central hypothesis, the expected occupation counts sum to the entire budget, so binary testing forces the $d \log(1/\eta)$ term. Taking the maximum with the packing bound gives the sum in (S2.6) up to an absolute factor. For the upper bound, the estimator of [5, Corollary III.9] is run at a rational failure probability exponentially small in d . Independent repetition and metric-majority selection give the required confidence at the same additive cost.

The resulting operator estimator also applies to the closed body. At operator accuracy $\delta/(2N)$, the hybrid inequality gives future loss at most δ . Selection between this one-call procedure and the block learner, using only public parameters, yields

$$M_b^* \leq C \frac{N^2}{\delta^2} \min \left\{ d^2 + d \log(1/\eta), \frac{d^4}{b} \log(d/\eta) \right\}$$

in their common range; see (S12.9). The corresponding joint lower obstructions are given in (S12.8). In particular, the projective boundary in (S2.4) accounts for the different horizon cost of one-call acquisition on the closed body.

S2.4. Exact descriptions and learned words. Theorem S8.1 constructs a deterministic finite rational codebook at the order in (S2.2). Exact comparisons of Q_N^2 define a greedy selection on a finite legal matrix grid. Separation in the actual operational metric controls its cardinality, and inward rounding gives coverage at every rank. Encoder and decoder reconstruct the same dictionary from public parameters; every fixed-length word decodes to a legal effect. Its payload is

$$\frac{d^2}{2} \log_2 N + \lfloor d/2 \rfloor \log_2 \log(N+2) + d^2 \log_2(1/\delta) + O_d(1).$$

The direct qubit code also gives detailed resource bounds. Corollary S8.2 combines the common learner and exact encoder at their respective optimal fixed-dimensional orders, without further device calls. Corollary S9.6 gives the same conclusion under a prescribed independent-block cap.

The interior has a dimension-uniform description law. For all $d, N \geq 1$ and $0 < \delta \leq 2^{-13}$, Theorem S11.1 gives

$$(S2.7) \quad \left(\frac{\sqrt{N}}{2048\delta} \right)^{d^2} \leq \mathcal{C}_N(\mathfrak{I}_d, \delta) \leq \left(\frac{25\sqrt{N}}{\delta} \right)^{d^2}.$$

The lower bound allows arbitrary legal binary centres. Theorem S11.2 realizes the upper bound by finite exact rational comparison and public reconstruction. With

$$A_d(N, \delta) = \frac{1}{2} d^2 \log_2 N + d^2 \log_2(1/\delta),$$

its length is $A_d(N, \delta) + O(d^2)$, uniformly in all parameters; there is no $d^2 \log d$ entrywise-transmission penalty. The absence of the endpoint logarithm reflects the interior geometry.

With shared randomness and worst-target failure probability $0 \leq \eta \leq 1/8$, Theorem S11.4 gives optimal expected one-way prefix length $(1 - \eta)A_d(N, \delta) + O(d^2)$. Its converse permits a

randomized decoder and counts all nonpublic information. Its upper bound uses a public target-independent abort event, whose occurrence requires no transmitted flag. For learned descriptions, Theorem S11.5 attains the call order in (S2.5) and this deterministic index length at failure $1/8$; Corollary S12.5 attains (S2.6) at every stated confidence.

S2.5. Finite risk certificates and trusted controls. For rational public errors, these joint query and description orders admit a complete finite classical specification before the first device call. The algebraic construction in Theorem S14.2 fixes the public dictionary and a collective readout on independent Choi records. The subnormalized record is polynomial in the unknown effect, so positivity, completeness, and uniform success form a first-order formula over the reals with rational coefficients. Effective quantifier elimination decides feasibility, and algebraic-point selection returns exact readout data. The statistical existence bound controls the first feasible call count.

Section S15 gives a rational construction with finite certificates of uniform risk. Theorem S15.1 transfers verified inequalities on a finite rational grid to every real effect in $\mathfrak{J}_d$. If the grid is an operator-norm r -net, and its tests bound failure by α at operator radius a , then an m -call readout satisfies

$$\sup_{E \in \mathfrak{J}_d} \mathbb{P}_E\{\|E - C_J\|_{\text{op}} > a + r\} \leq \alpha + mr.$$

An analytic continuity bound for the complete Choi record proves the transfer with one fixed set of good labels at each grid point. The prescribed parameters certify operational loss $5\delta/8$ with failure at most $5\eta/16$. Proposition S15.2 supplies rational POVMs with exact positivity and completeness while preserving the risk margin. Bounded-denominator enumeration and a doubling schedule then give the computable learner of Theorem S15.3, with the call order in (S2.6) and payload $A_d(N, \delta) + O(d^2)$. This construction uses finite rational tests in place of real quantifier elimination.

The finite-control theorem approximates trusted preparations and readouts with a pathwise norm budget. It preserves the operational loss, confidence, and every-record call orders, and it charges classical specifications and trusted realization errors explicitly. The interior constructions retain fresh one-call acquisitions and collective processing of completed outputs. Exhaustive dictionary reconstruction, readout synthesis, quantum storage, and known-control work are separate resources from device calls and transmitted bits; the finite constructions assert computability without an efficiency bound for those resources.

S2.6. Antecedents and organization. Horizontal Kraus gauges, channel-extension metrology, and integration of local distinguishability underlie the path argument [22, 7, 23, 24, 25, 26]. Inverse-Sylvester geometry and spectral matrix integration have classical antecedents [27, 28]. Here these methods enter the explicit closed-body midpoint comparison and the nested endpoint entropy. Mele–Bittel [5] and Zambrano–Ramos-Calderer–Kueng [29] provide common estimators under one-use measurement losses. Their precise access, dimension, rank, and confidence statements, and the future-loss analysis of those acquisitions, are compared in Section S18. The phase-estimation components of the full-body learner use ideas from [31, 32], with the conditional construction proved here.

The matrix geometry and entropy precede the qubit learning construction, its matrix extension, and the exact dictionary. The independent-block theorem and the interior comparison then lead to the joint learning, confidence, and description laws. Finite-control synthesis and rational risk certification complete the construction. The subsequent qubit geometry and the final prerequisite appendix provide the remaining proofs needed by the article.

S2.7. A finite outcome alphabet. For $k \geq 2$ let $\mathfrak{P}_{d,k}$ consist of the ordered tuples $\mathbf{E} = (E_1, \dots, E_k)$ satisfying $\sum_j E_j = I_d$ and $I_d/(2k) \preceq E_j \preceq 3I_d/(2k)$. Put $p = (k-1)d^2$ and $r(\mathbf{E}, \mathbf{F}) = \max_j \|E_j - F_j\|_{\text{op}}$. Theorems S22.1–S22.4 give, with constants independent of the input dimension,

$$\frac{1}{128} \min\{1, \sqrt{Nr}\} \leq D_N^{\text{na}} \leq D_N \leq \min\{2, 2k\sqrt{Nr}\},$$

$$\log_2 \mathcal{C}_{N,k}(\delta) = \frac{p}{2} \log_2 N + p \log_2(1/\delta) + O(p \log(k+1)).$$

The entropy statement holds for $0 < \delta < 1/128$ and allows arbitrary legal decoding centres. A rational inward grid preserves the sum of all effects and both spectral inequalities. An exact public code attains the displayed length, with finite but unoptimized reconstruction. The horizontal differential equation $\dot{X}_j = X_j A_j^{-1} \dot{A}_j/2$ provides the upper geometry; its cross-slot cancellation uses $\sum_j \dot{A}_j = 0$ rather than commutation. A single-label Bernoulli test supplies the lower geometry.

For every fixed k , Theorem S23.1 proves

$$M_{d,k}^*(N, \delta, \eta) = \Theta_k\left(N\delta^{-2}[d^2 + d \log(1/\eta)]\right)$$

jointly for $d, N \geq 1$, $0 < \delta \leq 2^{-24}$ and $0 < \eta \leq 1/8$. The explicit upper factor is Ck^3 with confidence logarithm $\log(k/\eta)$; the lower constant is absolute. Thus no optimal claim in a growing outcome count is implicit. Independent one-call Choi preparations attain the upper bound, with a simultaneous optimal-order fixed public index. Public random relabeling reduces each component to an interior binary learner. Exact legalization restores the common normalization. Conversely, a label-splitting embedding of binary effects transfers the coherent adaptive lower bound, including its dimension-dependent confidence contribution.

Theorem S23.2 gives finite risk certificates for collective k -outcome Choi records. A complete tuple r_0 -net at radius a and failure α transfers to every real target at radius $a + r_0$ and failure $\alpha + Mkr_0/2$. The statement uses a fixed set of successful output labels; no continuity of the decision boundary is assumed.

These finite-outcome results concern the balanced interior, not the full spectral boundary of arbitrary POVMs. The complete binary-body results and their endpoint logarithms remain unchanged. Neither family includes a device-returned postmeasurement quantum state. The underlying one-use tomography primitive is credited to Mele–Bittel; the transfer of its guarantees to the present future-use loss, the matching coherent converse, and the explicit description and certificate constructions are distinguished in the comparison section. Human specialist priority assessment, exact adaptive optima, and algorithmically efficient synthesis are not consequences of source reconstruction or finite arithmetic checks.

S3. TOMOGRAPHY COMPARISONS AND RESOURCE DISTINCTIONS

The learner above uses established tomography as a statistical primitive. Its contribution is the future-use loss, the coherent converse under that loss, and the simultaneous description guarantee, not the introduction of finite-outcome measurement tomography.

S3.1. One-use guarantees and balanced legalization. Write ϵ for one-use estimation accuracy and η for failure. Mele–Bittel [5, Theorem III.3] learn a channel with input size d , output size k , and Kraus rank at most r using $O(dkr\epsilon^{-2})$ calls at constant confidence, in halved diamond distance. Their stated high-confidence theorem includes an additional $O(d\epsilon^{-2} \log(1/\eta))$ term and a lower admissible failure threshold; arbitrary smaller failure can instead be obtained by independent repetition and metric-majority selection. For a measurement channel one may take $r \leq dk$. Dephasing the estimated output gives a legal POVM B . If the halved diamond error is at

most ϵ , then $r(E, B) \leq \epsilon$: on every input the outcome differences sum to zero, so the magnitude of any one difference is at most their total variation.

The estimator need not belong to the balanced interior. Applying Theorem S24.1, rather than silently using the interior metric outside its domain, gives

$$r(E, L_\epsilon(B)) \leq 4\epsilon, \quad D_N(E, L_\epsilon(B)) \leq 8k\sqrt{N}\epsilon.$$

Consequently $\epsilon = \delta/(8k\sqrt{N})$ supplies a sufficient constant-confidence future-loss budget

$$(S3.1) \quad O(d^2 k^4 N \delta^{-2}).$$

This is a parameter substitution, not a lower bound or a claim that this analysis is optimal for that estimator. The component reduction used in Theorem S23.1 instead invokes Mele–Bittel’s binary operator-norm primitive [5, Corollary III.9], together with the confidence argument in Lemma S12.3. It gives the stated k^3 upper factor and its joint confidence term.

For comparison, let d_{op} be the worst-input outcome total variation without a retained reference. Zambrano–Ramos-Calderer–Kueng [29, Theorem 2] give a legal projected least-squares estimate with $d_{\text{op}}(E, B) \leq \epsilon$ and failure at most η provided

$$(S3.2) \quad m \geq \frac{8[d^2 + \epsilon(d^2 + 1)/6]}{\epsilon^2} \log \frac{2^{k+1} 9^{2d}}{\eta}.$$

Their protocol uses nonadaptive single-copy probes from a global state 2-design and classical reconstruction. The subset-event formula for d_{op} implies $r(E, B) \leq d_{\text{op}}(E, B)$. The same affine repair and the same choice of ϵ therefore supply the sufficient bound

$$(S3.3) \quad O(k^2 N \delta^{-2} d^2 [d + k + \log(1/\eta)])$$

for the balanced-interior future-use problem. These probes differ from independent Choi acquisitions followed by collective processing. Neither the source’s lower-bound class nor its one-use loss is identified with the coherent adaptive future-use converse proved here.

S3.2. Geometry and the scope of the new principle. Horizontal purification and cancellation of cross-slot terms have antecedents in channel geometry and channel-extension metrology [22, 7, 24, 25, 26]. Orthogonal projection, quadratic energy minimization, and inverse-order comparison are also standard methods. What is established in Section 2 is the particular normalized effect-tuple covariance, its exact binary restriction, and its global regularized adaptive bound with an explicit rank-stable midpoint formula. The proof writes out these identities rather than attributing them to a generic Fisher-information argument.

In particular, the covariance includes inter-outcome terms through $S_E(K)$. Its zero operator characterizes projection-valued measurements, while uniform outcome noise supplies a positive floor. The finite-angle crossover in Theorem 2.5 exhibits the change from linear to square-root accumulation. The classical-output, input-consuming interface is essential: a device returning a residual quantum system is not covered by these theorems.

S3.3. Four distinct computational resources. Unknown-device calls and the future-use horizon are different quantities. The common learner bounds the former on every record; the latter appears only in the loss. A public code charges the fixed-length transmitted index, not a target-dependent dictionary. All dictionary parameters and reconstruction rules are public and fixed independently of the unknown device.

The covariance certificate and guarded affine repair have polynomial bit complexity on Gaussian-rational input. This does not make the full public dictionary or an arbitrary collective learning readout polynomial-time. The finite-risk theorem concerns a fixed ideal readout and a complete parameter net; the actual-control learning theorem separately budgets errors of the implemented trusted controls. An additional certified implementation-law error must be added when transferring

an ideal certificate. A finite scalar replay illustrates that theorem and checks the software contract; it is not execution of a general matrix learner.

S3.4. Further questions. The closed-body upper bound suggests a coupled boundary classification. A matching nonadaptive lower certificate and the corresponding full multi-outcome volume and entropy laws remain to be determined. The fixed- k learning law also leaves a genuine growing-outcome problem: its binary lower reduction does not match the explicit k^3 upper factor. For the complete binary body, the displayed d^4 learning upper is not a growing-dimensional min-max classification. Finally, polynomial-time public dictionary construction and general readout synthesis are distinct from the polynomial evaluator and legalizer proved here. These questions do not alter the ranges of the results already established.

S4. MATRIX GEOMETRY IN ARBITRARY INPUT DIMENSION

The binary effect body has a finite-use description that does not require eigenvector coordinates. Its regularization is the same at every spectral rank, and two-dimensional compressions supply lower witnesses for the full matrix problem. Throughout this section $d, N \geq 1$ are integers and

$$(S4.1) \quad \mathfrak{E}_d = \{E \in \text{End}(\mathbb{C}^d) : E = E^*, 0 \preceq E \preceq I_d\}.$$

The measurement $\mathcal{M}_E$ has the ordered effects $E, I_d - E$ and no residual quantum output. The distance d_N has the same reference-assisted adaptive meaning as in Theorem S16.1. Write d_N^{na} for its restriction to nonadaptive experiments, allowing entanglement across the calls and a retained reference. These experiments may depend on the known pair of hypotheses.

Put $\tau = N^{-1}$ and $V_E = E(I_d - E)$. On the real Hilbert space of Hermitian matrices with Hilbert–Schmidt inner product, define

$$(S4.2) \quad \mathcal{A}_E(H) = V_E H + H V_E + \tau H,$$

$$(S4.3) \quad g_{N,E}(H)^2 = N \langle H, \mathcal{A}_E^{-1}(H) \rangle_{\text{HS}}.$$

The operator $\mathcal{A}_E$ is strictly positive, including at a singular effect. For a pair E, F set

$$(S4.4) \quad M = \frac{1}{2}(E + F), \quad H = F - E, \quad Q_N(E, F) = g_{N,M}(H).$$

In any orthonormal eigenbasis of M , with eigenvalues m_i and $v_i = m_i(1 - m_i)$, this is

$$(S4.5) \quad Q_N(E, F)^2 = N \sum_{i,j=1}^d \frac{|H_{ij}|^2}{v_i + v_j + N^{-1}}.$$

Definition (S4.4) makes the expression independent of all choices at repeated eigenvalues. We use it as a comparison modulus; a triangle inequality for Q_N is not required.

The inverse Sylvester operator is familiar from the Bures metric g^{B} on the positive definite cone [27]. Its normalization gives the algebraic identity

$$g_{N,E}(H)^2 = 2N g_{V_E + \tau I_d/2}^{\text{B}}(H, H).$$

Here H remains the effect tangent; the differential of the variance map $E \mapsto V_E$ is $H - EH - HE$.

Theorem S4.1 (Finite-use geometry of the matrix effect body). *For every $E, F \in \mathfrak{E}_d$,*

$$(S4.6) \quad \frac{1}{8192d} \min\{1, Q_N(E, F)\} \leq d_N^{\text{na}}(\mathcal{M}_E, \mathcal{M}_F) \leq d_N(\mathcal{M}_E, \mathcal{M}_F) \leq \min\{2, 8Q_N(E, F)\}.$$

The constants are uniform in the horizon and in both effects, without a spectral margin or a condition on eigenvalue multiplicities. The adaptive upper bound permits arbitrary retained references, common feedback, and bounded public stopping. Each lower witness uses inputs in one fixed subspace of dimension at most two. Moreover,

$$(S4.7) \quad d_1(\mathcal{M}_E, \mathcal{M}_F) = 2\|E - F\|_{\text{op}}.$$

The theorem includes every rank of projection and all of the intermediate singular strata. The variance $E(I_d - E)$, rather than the smallest eigenvalue of either outcome, determines the matrix weights. The factor N^{-1} makes the formula finite on the entire closed body. The proof first establishes a path inequality valid for every common adaptive tester, and then obtains the converse from one- and two-dimensional experiments.

S4.1. A horizontal measurement dilation. Kraus-gauge bounds for channel estimation were developed in [22, 7, 23]. Their integration along channel paths appears in [25, Section III, equation (26)] and, with general adaptive bounds, in [26, Section IV.A]. We use an explicit horizontal lift for binary effects and control the finite-horizon regularization by a trace-norm remainder.

Lemma S4.2 (Horizontal derivative and regularized path bound). *Every continuously differentiable path $G : [0, 1] \rightarrow \mathfrak{E}_d$ satisfies*

$$(S4.8) \quad d_N(\mathcal{M}_{G(0)}, \mathcal{M}_{G(1)}) \leq 4 \int_0^1 g_{N,G(t)}(G'(t)) dt.$$

Proof. We begin with an interior effect G , so that G and $I_d - G$ are invertible. Set $S = \sqrt{G(I_d - G)}$. For a Hermitian matrix K put $H_0 = SK + KS$. At the two factors $A_1 = \sqrt{G}$, $A_0 = \sqrt{I_d - G}$ prescribe derivatives

$$(S4.9) \quad B_1 = \sqrt{I_d - G} K, \quad B_0 = -\sqrt{G} K.$$

They obey

$$(S4.10) \quad \begin{aligned} A_1^* B_1 + B_1^* A_1 &= H_0, & A_0^* B_0 + B_0^* A_0 &= -H_0, \\ \sum_{y=0}^1 A_y^* B_y &= 0, & \sum_{y=0}^1 B_y^* B_y &= K^2. \end{aligned}$$

These tangent equations can be realized by genuine dilations. Let $G_1(s) = G + sH_0$, $G_0(s) = I_d - G - sH_0$ for s in a sufficiently small open interval, and put $R_y(s) = \sqrt{G_y(s)}$. The first line of (S4.10) implies that

$$T_y = (B_y - R'_y(0))R_y(0)^{-1}$$

is anti-Hermitian. Indeed, multiplying $T_y + T_y^*$ on the left and right by $R_y(0)$ gives, with $R = R_y(0)$,

$$R(T_y + T_y^*)R = RB_y + B_y^*R - (RR'_y(0) + R'_y(0)R) = 0,$$

by differentiating $R_y(s)^2 = G_y(s)$. Thus $\tilde{A}_y(s) = \exp(sT_y)R_y(s)$ has derivative B_y at zero and satisfies $\tilde{A}_y(s)^* \tilde{A}_y(s) = G_y(s)$ exactly. A measurement isometry is

$$W(s)\psi = \sum_{y=0}^1 |y\rangle_C |y\rangle_L \tilde{A}_y(s)\psi.$$

Here C is the observed outcome, and L and the factor space of $\tilde{A}_y$ are environments. Tracing the environments implements $\mathcal{M}_{G(s)}$, also on inputs entangled with a reference. At zero, (S4.10) gives

$$(S4.11) \quad W^* \dot{W} = 0, \quad \dot{W}^* \dot{W} = K^2.$$

The horizontal Kraus condition has the antecedents [22, Theorems 4–5]. Its adaptive channel-extension form is $F_N \leq 4N\|\alpha\|$ when $\beta = 0$; see [23, equation (9)] and [24, equations (5)–(10)]. In the present lift $\alpha = K^2$ and $\beta = 0$. We give the cancellation explicitly for this binary measurement.

Fix a common tester with N slots. Purify its initial state and all intervening operations, and retain the measurement environments only for this comparison; neither the outcome copy nor the factor environment is accessible to the tester. The derivative of its final purification in the

direction H_0 is the sum of N terms, each having one differentiated occurrence of W . The terms are mutually orthogonal. For two different differentiated slots, cancel the common isometries after the later slot; the remaining overlap contains $W^* \dot{W} \otimes I = 0$. Each term has norm at most $\|\dot{W}\|_{\text{op}} = \|K\|_{\text{op}}$, independently of the reference and memory dimensions. Hence the derivative of the purification has norm at most $\sqrt{N}\|K\|_{\text{op}}$. The pure-state trace-norm derivative bound, followed by contraction under the partial trace, bounds the observed output derivative by

$$(S4.12) \quad 2\sqrt{N}\|K\|_{\text{op}}.$$

A publicly stopped tester is padded to N slots by calls on fixed dummy inputs, whose outcomes and environments are both discarded. The same purification argument then applies to its original output. Classical feedback is included by purifying the common controlled operations.

Now let H be an arbitrary Hermitian tangent at G , and set $\kappa = N^{-1/2}$. There is a unique Hermitian solution of

$$(S4.13) \quad SK + KS + \kappa K = H.$$

Write $H = H_0 + R$, with $H_0 = SK + KS$ and $R = \kappa K$. Both are legitimate tangents at the interior effect G . The derivative of a fixed tester's output is linear in the inserted channel tangent. The part arising from H_0 is bounded by (S4.12). The part arising from R is bounded by $2N\|R\|_{\text{op}}$. To verify the latter bound, the one-call tangent has two opposite reference blocks

$$B_\rho = \text{tr}_A[(R \otimes I)\rho], \quad -B_\rho.$$

Duality against Hermitian contractions gives $\|B_\rho\|_1 \leq \|R\|_{\text{op}}$ for every input state. Insert this tangent into each of the N slots in turn; all subsequent common channels contract the Hermitian trace norm. This proves the bound for its sum. Thus the actual tangent H has output derivative bounded by

$$(S4.14) \quad 2\sqrt{N}\|K\|_{\text{op}} + 2N\|R\|_{\text{op}} = 4\sqrt{N}\|K\|_{\text{op}}.$$

The splitting is used only for the differential at an interior effect; it does not postulate a physical mixture realizing the entire path.

In an eigenbasis of G , with $v_i = g_i(1 - g_i)$, equation (S4.13) reads

$$K_{ij} = \frac{H_{ij}}{\sqrt{v_i} + \sqrt{v_j} + \kappa}.$$

Since $(\sqrt{v_i} + \sqrt{v_j} + \kappa)^2 \geq v_i + v_j + \tau$,

$$(S4.15) \quad \sqrt{N}\|K\|_{\text{op}} \leq \sqrt{N}\|K\|_{\text{HS}} \leq g_{N,G}(H).$$

The integration of local distinguishability has direct antecedents: [25, Section III, equation (26), PDF version 3] treats the parallel path bound, and [26, Section IV.A, equations (14)–(22)] applies adaptive metrological estimates. Here the regularized binary Sylvester split and its residual produce the explicit integrand. For an interior path, integrate the resulting output derivative bound for each fixed tester, and then take the supremum. This proves (S4.8) without exchanging a supremum with a derivative.

For a path in the closed body, apply the result to $G_\epsilon(t) = (1 - 2\epsilon)G(t) + \epsilon I_d$, where $0 < \epsilon < 1/2$. At fixed N , the form $g_{N,G}(H)$ is continuous on the whole body and bounded above by $N\|H\|_{\text{HS}}$. The integrals therefore converge as $\epsilon \downarrow 0$. The hybrid inequality $d_N(\mathcal{M}_E, \mathcal{M}_F) \leq 2N\|E - F\|_{\text{op}}$ gives continuity of the distances at both endpoints. Passing to the limit proves the asserted closed-body statement. $\square$

S4.2. The midpoint comparison.

Proof of Theorem S4.1. For the upper bound use $G(t) = (1-t)E + tF$. The map $G \mapsto V_G = G - G^2$ is operator concave. For $0 < t \leq 1/2$, write $G(t) = (1-2t)E + 2tM$ and discard the positive multiple of V_E in the concavity inequality. Applying the same argument from the other endpoint gives

$$(S4.16) \quad V_{G(t)} \succeq a(t)V_M, \quad a(t) = 2 \min\{t, 1-t\}.$$

For a positive matrix D and a Hermitian matrix X ,

$$\langle X, (L_D + R_D)X \rangle_{\text{HS}} = 2 \operatorname{tr}(DX^2) \geq 0.$$

Thus left-plus-right multiplication preserves the stated order on the real Hilbert space of Hermitian matrices. Since $a(t) \leq 1$, $\mathcal{A}_{G(t)} \succeq a(t)\mathcal{A}_M$. Inverting positive operators reverses their order, whence

$$g_{N,G(t)}(H) \leq a(t)^{-1/2}Q_N(E, F).$$

The scalar factor has integral 2 on $[0, 1]$. Lemma S4.2 yields $d_N \leq 8Q_N$; the trace-norm bound $d_N \leq 2$ gives the stated upper estimate.

For the lower bound choose any orthonormal eigenbasis e_i of M and put

$$(S4.17) \quad q_{ij} = |H_{ij}| \sqrt{\frac{N}{v_i + v_j + \tau}}, \quad Q_N^2 = \sum_{i,j} q_{ij}^2.$$

Repeated input e_i gives Bernoulli probabilities $s = m_i - H_{ii}/2$, $t = m_i + H_{ii}/2$. Their variances satisfy

$$\max\{s(1-s), t(1-t)\} \leq s(1-s) + t(1-t) \leq 2m_i(1-m_i).$$

Lemma S16.2 therefore gives

$$(S4.18) \quad d_N^{\text{na}}(\mathcal{M}_E, \mathcal{M}_F) \geq \frac{1}{64} \min\{1, q_{ii}\}.$$

For $i \neq j$, restrict each input to $\mathcal{S} = \operatorname{span}\{e_i, e_j\}$. The resulting channels have the compressed effects $E_{\mathcal{S}}, F_{\mathcal{S}}$, including on inputs entangled with a retained reference. Their midpoint is $M_{\mathcal{S}} = \operatorname{diag}(m_i, m_j)$. Write their spectral data as (p, q, u) and (p', q', v) in the notation of Section S16, with gaps r, r' and angle h when both gaps are positive. Put

$$V_E^{\mathcal{S}} = p(1-p) + q(1-q), \quad V_F^{\mathcal{S}} = p'(1-p') + q'(1-q').$$

For the compressed matrices the exact identity

$$V_{M_{\mathcal{S}}} = \frac{1}{2}V_{E_{\mathcal{S}}} + \frac{1}{2}V_{F_{\mathcal{S}}} + \frac{1}{4}(F_{\mathcal{S}} - E_{\mathcal{S}})^2$$

shows, on taking traces, that

$$(S4.19) \quad v_i + v_j + \tau \geq \frac{1}{2}(V_E^{\mathcal{S}} + \tau), \quad v_i + v_j + \tau \geq \frac{1}{2}(V_F^{\mathcal{S}} + \tau).$$

The scalar part of $F_{\mathcal{S}} - E_{\mathcal{S}}$ has zero off-diagonal entry. Its traceless part has operator norm $|ru - r'v|/2$. Decomposing this vector along the smaller radius gives

$$(S4.20) \quad \begin{aligned} |H_{ij}| &\leq \frac{1}{2}|r - r'| + \min\{r, r'\} \sin(h/2) \\ &\leq \frac{1}{2}(|p - p'| + |q - q'| + \min\{r, r'\}h). \end{aligned}$$

When one radius is zero, the angular term is zero and the same inequality holds without choosing its direction. Each endpoint variance is bounded by the appropriate $V_E^{\mathcal{S}}$ or $V_F^{\mathcal{S}}$. Equations (S4.19)–(S4.20) therefore imply

$$(S4.21) \quad q_{ij} \leq \frac{1}{\sqrt{2}} (T_N(p, p') + T_N(q, q') + \min\{K_N(p, q), K_N(p', q')\}h).$$

The nonadaptive assertion of Theorem S16.1 applied to this compressed pair yields

$$(S4.22) \quad d_N^{\text{na}}(\mathcal{M}_E, \mathcal{M}_F) \geq \frac{1}{8192} \min\{1, q_{ij}\}.$$

Every such witness is a legitimate experiment for the original effects by a fixed isometric embedding of $\mathcal{S}$. Since $\max_{i,j} q_{ij} \geq Q_N/d$, taking the strongest of (S4.18) and (S4.22) proves the lower bound. For $d = 1$ only the diagonal argument is needed.

Finally, a one-use reference-assisted difference has output blocks B_ρ and $-B_\rho$, with $B_\rho = \text{tr}_A[(E - F) \otimes I]\rho$. Trace-norm duality against Hermitian contractions gives $\|B_\rho\|_1 \leq \|E - F\|_{\text{op}}$. An eigenstate of $E - F$ at an extremal eigenvalue attains equality. This proves (S4.7). $\square$

Corollary S4.3 (Uniform comparison with nonadaptive experiments). *For all $E, F \in \mathfrak{E}_d$ and all $N \geq 1$,*

$$(S4.23) \quad d_N(\mathcal{M}_E, \mathcal{M}_F) \leq 65536d d_N^{\text{na}}(\mathcal{M}_E, \mathcal{M}_F).$$

Here nonadaptive experiments may entangle all inputs and retain a reference; they use no feedback between calls. The dimensional constant comes from the maximum of d^2 weighted entries and is not optimized.

Proof. If $Q_N \leq 1$, divide the upper estimate $8Q_N$ by the lower estimate $Q_N/(8192d)$. If $Q_N \geq 1$, the upper estimate is 2 and the lower estimate is $1/(8192d)$, which gives a smaller constant. The identical pair needs no division. $\square$

S4.3. All spectral ranks and local forms.

Lemma S4.4 (Spectral noncancellation in every rank). *Let $\lambda_1(E) \geq \dots \geq \lambda_d(E)$ and $\lambda_1(F) \geq \dots \geq \lambda_d(F)$ be the ordered spectra. For every $1 \leq k \leq d$,*

$$(S4.24) \quad B_N(\lambda_k(E), \lambda_k(F)) \leq d_N^{\text{na}}(\mathcal{M}_E, \mathcal{M}_F).$$

For effects diagonal in the same ordered orthonormal basis,

$$(S4.25) \quad d_N(\mathcal{M}_E, \mathcal{M}_F) \leq \sum_{k=1}^d B_N(\lambda_k(E), \lambda_k(F)).$$

Proof. Suppose $\lambda_k(E) \geq \lambda_k(F)$. The span of the first k eigenvectors of E intersects the orthogonal complement of the first $k - 1$ eigenvectors of F in a nonzero subspace. A unit vector in that intersection has success probabilities $s \geq \lambda_k(E) \geq \lambda_k(F) \geq t$ under the two hypotheses. The monotonicity in Lemma S16.2 implies $B_N(s, t) \geq B_N(\lambda_k(E), \lambda_k(F))$. Repeated use of this vector proves the first assertion; reverse the hypotheses for the opposite ordering. Multiplicities cause no change in the dimension argument.

For the upper bound, first measure the common eigenbasis and then produce a bit whose success probability is its corresponding eigenvalue. Supply N independent bits in each of the d rows before the experiment. Every common adaptive tester is a common channel of these row arrays, including its reference blocks and all publicly stopped histories. The tensor-product triangle inequality gives (S4.25); unused row bits are discarded. $\square$

The following local comparison will permit volume estimates at every boundary stratum. It concerns matrices themselves and does not require a coordinate chart for eigenvectors.

Lemma S4.5 (Uniform local comparison of the variance forms). *Set $W_G = G(I_d - G) + \tau I_d/2$. For a Hermitian H , put $q = g_{N,G}(H)$. Then*

$$(S4.26) \quad \|W_G^{-1/2} H W_G^{-1/2}\|_{\text{HS}} \leq 2q.$$

For either sign for which $G \pm H/2 \in \mathfrak{E}_d$, one has

$$(S4.27) \quad \|W_G^{-1/2} (W_{G \pm H/2} - W_G) W_G^{-1/2}\|_{\text{op}} \leq q + \frac{3}{4}q^2.$$

In particular, for $q \leq 1/4$,

$$(S4.28) \quad \frac{1}{2}W_G \preceq W_{G \pm H/2} \preceq 2W_G.$$

Thus, if either $Q_N(E, F) \leq 1/4$ or $g_{N,E}(F - E) \leq 1/4$, the midpoint and the fixed-endpoint quadratic forms are comparable within factors 2.

Proof. In an eigenbasis of W_G , with eigenvalues $w_i \geq \tau/2$,

$$\frac{w_i + w_j}{Nw_iw_j} \leq 4.$$

Multiplying this inequality by $N|H_{ij}|^2/(w_i + w_j)$ and summing proves (S4.26). Put $Z = W_G^{-1/2}HW_G^{-1/2}$. The exact expansion

$$W_{G \pm H/2} - W_G = \pm \frac{1}{2}(H - GH - HG) - \frac{1}{4}H^2$$

has normalized linear part $\pm((I_d - 2G)Z + Z(I_d - 2G))/4$, since G commutes with W_G . Its operator norm is at most $\|Z\|_{\text{op}}/2 \leq q$. The normalized quadratic part is $-ZW_GZ/4$. Since $\tau \leq 1$, one has $\|W_G\|_{\text{op}} \leq 3/4$, and its norm is at most $3q^2/4$. This proves (S4.27) and hence (S4.28). The same order holds for left plus right multiplication, and inversion gives the quadratic-form comparison. Apply it first at $G = M$ or at $G = E$, respectively, to obtain the final assertion. $\square$

S5. DESCRIPTION ENTROPY IN ARBITRARY MATRIX DIMENSION

The intrinsic comparison of Theorem S4.1 permits the full effect body to be treated without choosing eigenvector coordinates near repeated spectra. Throughout this section the input dimension $d \geq 1$ is fixed. Constants may depend on d , but not on the integer horizon $N \geq 1$ or on the accuracy parameter.

Write $\mathcal{C}_N(\mathfrak{E}_d, \delta)$ for the covering number under d_N , with centres in the same legal memoryless ordered binary interface, and write $\mathcal{C}_N^{\text{rat}}$ when the real and imaginary parts of every matrix entry of each centre are rational.

Theorem S5.1 (Entropy of the full effect body). *For every integer $d \geq 1$ there are constants $c_d, C_d, \delta_d > 0$ such that, for all integers $N \geq 1$ and all $0 < \delta \leq \delta_d$,*

$$(S5.1) \quad \begin{aligned} c_d N^{d^2/2} [\log(N+2)]^{\lfloor d/2 \rfloor} \delta^{-d^2} &\leq \mathcal{C}_N(\mathfrak{E}_d, \delta) \\ &\leq \mathcal{C}_N^{\text{rat}}(\mathfrak{E}_d, \delta) \\ &\leq C_d N^{d^2/2} [\log(N+2)]^{\lfloor d/2 \rfloor} \delta^{-d^2}. \end{aligned}$$

Consequently the optimum fixed-length reusable description has length

$$(S5.2) \quad \frac{d^2}{2} \log_2 N + \lfloor d/2 \rfloor \log_2 \log(N+2) + d^2 \log_2(1/\delta) + O_d(1).$$

The estimates include all support and spectral multiplicities. The rational-centre assertion here follows from a covering argument. Theorem S8.1 gives a deterministic finite exact encoder and decoder attaining the same order in every dimension. Theorem S17.2 retains its separate direct construction in dimension two. Computational cost is accounted for separately from the number of transmitted bits.

For $d = 1$, the result is the Bernoulli covering law $N^{1/2}\delta^{-1}$. For $d = 2$, it recovers Theorem S17.1. The next two cases have orders $N^{9/2} \log(N+2)\delta^{-9}$ and $N^8 [\log(N+2)]^2 \delta^{-16}$. The logarithmic power counts independent nested approaches of paired eigenvalues to the two support endpoints. We first establish a uniform measure estimate for actual operational balls, and then evaluate the total measure.

S5.1. Operational balls and a regularized volume form. Set $n = d^2$, $\tau = N^{-1}$ and

$$W_E = E(I - E) + \frac{\tau}{2}I.$$

As in Section S4, let

$$(S5.3) \quad g_{N,E}(H)^2 = N \langle H, (L_{W_E} + R_{W_E})^{-1} H \rangle_{\text{HS}}$$

on the real Hilbert space of Hermitian matrices. Let $G_{N,E}$ denote this positive quadratic form in Hilbert–Schmidt orthonormal coordinates, and define

$$(S5.4) \quad \rho_N(E) = \sqrt{\det G_{N,E}}, \quad d\mu_N(E) = \rho_N(E) dE,$$

The real orthonormal basis consists of e_{ii} and, for $i < j$, $(e_{ij} + e_{ji})/\sqrt{2}$ and $i(e_{ij} - e_{ji})/\sqrt{2}$; dE is Lebesgue measure in these coordinates. The cutoff makes ρ_N positive and continuous on the whole closed body for each N .

Lemma S5.2 (Uniform volume of operational balls). *For fixed d there are $a_d, A_d, r_d > 0$ such that*

$$(S5.5) \quad a_d r^{d^2} \leq \mu_N(B_{d_N}(E, r)) \leq A_d r^{d^2}$$

for every $N \geq 1$, every $E \in \mathfrak{E}_d$, and every $0 < r \leq r_d$. Balls are taken inside $\mathfrak{E}_d$.

Proof. We make explicit the two local consequences of Lemma S4.5 that will be used. If H is Hermitian and $q = g_{N,E}(H)$, diagonalizing W_E gives

$$(S5.6) \quad \|W_E^{-1/2} H W_E^{-1/2}\|_{\text{HS}}^2 \leq 4q^2,$$

since its eigenvalues w_i are at least $\tau/2$ and

$$\frac{w_i + w_j}{N w_i w_j} = \tau(w_i^{-1} + w_j^{-1}) \leq 4.$$

For any legal $E + tH$, direct expansion yields

$$W_{E+tH} - W_E = \frac{t}{2}((I - 2E)H + H(I - 2E)) - t^2 H^2.$$

The matrices E and W_E commute, $\|I - 2E\|_{\text{op}} \leq 1$, and $\|W_E\|_{\text{op}} \leq 3/4$. Thus the norm of the normalized difference is at most $2|t|q + 3t^2q^2$. For $|t| \leq 1$ and q below an absolute constant, it follows that

$$(S5.7) \quad \frac{1}{2}W_E \preceq W_{E+tH} \preceq 2W_E.$$

Loewner comparison passes to $L_W + R_W$ and, by inversion, to the quadratic forms. It also gives

$$(S5.8) \quad 2^{-n/2} \rho_N(E) \leq \rho_N(E + tH) \leq 2^{n/2} \rho_N(E).$$

Apply this comparison from an endpoint to the midpoint and from the midpoint to either endpoint. Theorem S4.1 then provides constants $A, B, r_0 > 0$, depending only on d , such that

$$(S5.9) \quad d_N(E, F) \leq r \implies g_{N,E}(F - E) \leq Ar,$$

$$(S5.10) \quad g_{N,E}(F - E) \leq Br \implies d_N(E, F) \leq r,$$

for legal E, F and $0 < r \leq r_0$. Decrease r_0 so that Ar_0 lies in the small-form range of (S5.7).

Let ω_n be the Euclidean volume of the unit ball in $\mathbb{R}^n$. The ellipsoid $g_{N,E}(H) \leq u$ has Lebesgue volume $\omega_n u^n / \rho_N(E)$. Equations (S5.9) and (S5.8) therefore imply

$$\mu_N(B_{d_N}(E, r)) \leq 2^{n/2} \omega_n (Ar)^n.$$

For the lower bound we construct a full legal ellipsoid, including when E lies on a support boundary. Choose a small constant $a > 0$ depending only on d , set $s = ar\tau$, and move towards the scalar interior point:

$$(S5.11) \quad E_s = (1 - 2s)E + sI.$$

Take r_0 sufficiently small that $s \leq 1/4$. Then $sI \preceq E_s \preceq (1 - s)I$. Since the displacement commutes with E , its form is bounded by

$$g_{N,E}(E_s - E)^2 = N \sum_i \frac{s^2(1 - 2\lambda_i)^2}{2\lambda_i(1 - \lambda_i) + \tau} \leq d \frac{s^2}{\tau^2} = da^2 r^2.$$

Choose $a\sqrt{d} \leq B/4$. By (S5.10), $d_N(E, E_s) \leq r/4$.

Let $X = X^*$ satisfy $g_{N,E_s}(X) \leq br$, with $b > 0$ to be chosen. Write ℓ_i for the eigenvalues of E_s and $w_i = \ell_i(1 - \ell_i) + \tau/2$. Since $\ell_i \geq s$ and $w_i \leq \ell_i + \tau/2$, we have

$$\begin{aligned} \|E_s^{-1/2} X E_s^{-1/2}\|_{\text{HS}}^2 &\leq g_{N,E_s}(X)^2 \max_{i,j} \frac{w_i + w_j}{N\ell_i\ell_j} \\ &\leq b^2 r^2 \left(\frac{2\tau}{s} + \frac{\tau^2}{s^2} \right) = b^2 \left(\frac{2r}{a} + \frac{1}{a^2} \right). \end{aligned}$$

The same bound holds with $I - E_s$ in place of E_s , because $1 - \ell_i \geq s$ and $w_i \leq 1 - \ell_i + \tau/2$. Taking $b \leq a/4$ and $ar_0 \leq 1$ makes both operator norms at most $1/2$. Consequently every point of the entire ellipsoid

$$\mathcal{A}_s = \{E_s + X : g_{N,E_s}(X) \leq br\}$$

is a legal effect. Choose also $b \leq B/4$. Equation (S5.10) and the triangle inequality give $\mathcal{A}_s \subset B_{d_N}(E, r)$. The density comparison about E_s and the exact ellipsoid volume now imply

$$\mu_N(B_{d_N}(E, r)) \geq 2^{-n/2} \omega_n (br)^n.$$

All choices depend only on d . This proves the assertion on every boundary stratum as well as in the interior. $\square$

S5.2. The total regularized volume.

Proposition S5.3 (Nested endpoint accumulation). *For fixed $d \geq 1$, uniformly over $N \geq 1$,*

$$(S5.12) \quad \mu_N(\mathfrak{E}_d) \asymp_d N^{d^2/2} [\log(N+2)]^{\lfloor d/2 \rfloor}.$$

Proof. In a spectral basis, the quadratic form has weight $N/(2w_i)$ on the i th diagonal direction and weight $N/(w_i + w_j)$ on each of the two Hilbert–Schmidt orthonormal real directions belonging to $i < j$. Hence

$$(S5.13) \quad \rho_N(E) = 2^{-d/2} N^{d^2/2} \prod_i w_i^{-1/2} \prod_{i < j} (w_i + w_j)^{-1}.$$

We use the classical spectral change of variables [28]. For a simple spectrum, differentiation of $E = U \text{diag}(\lambda) U^*$ gives

$$U^*(dE)U = d \text{diag}(\lambda) + [U^* dU, \text{diag}(\lambda)].$$

Each complex off-diagonal pair therefore contributes $(\lambda_i - \lambda_j)^2$ to the real Jacobian. Integration over the compact unitary orbit and over permutations supplies a positive constant depending only on d . The repeated-spectrum set is Lebesgue null. This spectral change of variables, together with (S5.13), yields

$$(S5.14) \quad \mu_N(\mathfrak{E}_d) \asymp_d N^{d^2/2} I_d(\tau),$$

where

$$(S5.15) \quad I_d(\tau) = \int_{[0,1]^d} \prod_i (v_i + \tau)^{-1/2} \prod_{i < j} \frac{(\lambda_i - \lambda_j)^2}{v_i + v_j + \tau} d\lambda, \quad v_i = \lambda_i(1 - \lambda_i).$$

Replacing $v_i + \tau/2$ by $v_i + \tau$ in the single-coordinate factors changes the integral by a factor between fixed positive constants depending only on d .

We prove

$$(S5.16) \quad I_d(\tau) \asymp_d [\log(2 + \tau^{-1})]^{[d/2]}, \quad 0 < \tau \leq 1.$$

For the upper bound, partition each coordinate into $[0, 1/4]$, $[1/4, 3/4]$, and $[3/4, 1]$. Factors involving an eigenvalue in the middle interval are bounded above by constants, so these coordinates can be integrated out. For each of the remaining $l \leq d$ coordinates put $x_i = \lambda_i$ near zero and $x_i = 1 - \lambda_i$ near one. On a sector $0 \leq x_1 \leq \dots \leq x_l \leq 1/4$, let $\epsilon_i \in \{1, -1\}$ record the endpoint and put

$$S_j = \sum_{i=1}^j \epsilon_i, \quad S_0 = 0, \quad y_j = x_j + \tau.$$

The individual factor is at most $Cy_j^{-1/2}$. For $i < j$, a same-endpoint pair factor is at most Cy_j , whereas an opposite-endpoint factor is at most C/y_j . The sector integrand is consequently bounded by

$$(S5.17) \quad C_d \prod_{j=1}^l y_j^{a_j-1}, \quad a_j = \frac{1}{2} + \epsilon_j S_{j-1} = \frac{S_j^2 - S_{j-1}^2}{2}.$$

Its cumulative exponents satisfy

$$(S5.18) \quad A_j = \sum_{i=1}^j a_i = \frac{S_j^2}{2} \geq 0.$$

Set $T = 1/4 + \tau$ and $L = \log(T/\tau)$. Substitute $y_j = T \exp(-z_j)$, where $z_1 \geq \dots \geq z_l \geq 0$, and then $u_j = z_j - z_{j+1}$ for $j < l$, $u_l = z_l$. The sector integral is bounded by

$$(S5.19) \quad C_d \int_{u_j \geq 0, \sum_j u_j \leq L} \exp\left(-\sum_j A_j u_j\right) du.$$

The omitted factor T^{A_l} is bounded above uniformly in $0 < \tau \leq 1$. Each $A_j > 0$ is at least $1/2$, so its variable has an integrable exponential tail. Each vanishing A_j contributes at most one factor $1 + L$. A vanishing prefix sum requires an even j , and there are at most $[l/2]$ such indices. Summing the finitely many sectors proves the upper estimate in (S5.16). The case $l = 0$ contributes only a bounded constant.

For the lower bound let $m = [d/2]$. Choose scales

$$(S5.20) \quad \tau \leq t_1, \quad 8t_j \leq t_{j+1} \ (j < m), \quad t_m \leq 1/32,$$

and restrict $2m$ eigenvalues in pairs by

$$(S5.21) \quad \lambda_j^- = a_j \in [t_j, 2t_j], \quad \lambda_j^+ = 1 - b_j, \quad b_j \in [t_j, 2t_j], \quad 1 \leq j \leq m.$$

If d is odd, restrict the remaining eigenvalue to $[3/8, 5/8]$; all its factors are bounded above and below by positive constants. For one pair the two individual spectral factors and its opposite-endpoint factor have product comparable to t_j^{-2} . For two pairs at scales $t_i \leq t_j/8$, the two same-endpoint factors are comparable to t_j and the two opposite-endpoint factors to t_j^{-1} . Their product is therefore bounded above and below by fixed constants. Thus the integrand on the box is comparable to $\prod_j t_j^{-2}$ and its volume to $\prod_j t_j^2$. Each box contributes at least $c_d > 0$.

Take the scales from $\tau, 8\tau, 8^2\tau, \dots$ up to $1/32$. The boxes for distinct increasing m -tuples are disjoint. Their number is a binomial coefficient bounded below by $c_d[\log(\tau^{-1})]^m$ once $\log(\tau^{-1})$ is sufficiently large depending on d . In the remaining bounded range, a fixed box of distinct eigenvalues strictly inside $(0, 1)$ has a positive uniform integral and supplies the lower estimate. For $m = 0$ this interior argument suffices on the full range. This proves (S5.16) and, by (S5.14), the proposition. $\square$

The balanced-prefix identity (S5.18) accounts for all nested endpoint scales. A simultaneous approach of all eigenvalues to the endpoints detects only one possible critical scaling; the separated pairs in (S5.21) show why the logarithmic multiplicity can be larger.

S5.3. Covering, rational centres, and description length.

Proof of Theorem S5.1. Choose δ_d smaller than the radius in Lemma S5.2. A cover by L balls of radius δ satisfies

$$\mu_N(\mathfrak{E}_d) \leq LA_d \delta^{d^2},$$

which gives the required lower bound. Conversely, take a maximal δ -separated subset of $\mathfrak{E}_d$. Its balls of radius $\delta/3$ are pairwise disjoint, and the lower bound in (S5.5) bounds its cardinality by

$$C_d \mu_N(\mathfrak{E}_d) \delta^{-d^2}.$$

It is a radius- δ cover by maximality. Such finite maximal sets exist: the effect body is compact, d_N is continuous by the hybrid bound, and $d_1(E, F) = 2\|E - F\|_{\text{op}}$ separates distinct effects. Proposition S5.3 now gives both estimates for legal centres.

Rational legal effects are dense. Indeed, first move towards $I/2$ by (S5.11), and then approximate all independent real matrix coordinates by rational numbers while remaining strictly between 0 and I . For fixed N , continuity in the operational distance follows again from the hybrid bound. Replace every centre of a radius- $\delta/2$ cover by a rational legal effect at distance less than $\delta/2$. The resulting radius- δ cover has the same cardinality. This proves the rational-centre upper bound.

A fixed-length payload indexes the centres of a cover, and conversely each legal reusable program with a fixed payload supplies a centre. Taking base-two logarithms of (S5.1) and rounding the index length gives (S5.2). $\square$

S6. LEARNING THE COMPLETE BINARY QUBIT EFFECT BODY

We now pass from supplied descriptions to an unknown device. Throughout this section the input dimension is two. Write

$$E = \frac{1}{2}((1+b)I + ru \cdot \sigma), \quad 0 \leq r \leq 1, \quad |b| + r \leq 1,$$

with the direction absent when $r = 0$. The bias, contrast, and direction are all unknown. The outputs are ordered and classical, and each call consumes its input without a residual quantum output. A learning procedure may prepare entangled input blocks, retain the unused input qubits, and use classical feedback between blocks. It has no access to a measurement environment. Its choices depend only on the public parameters and the observed record.

The integer N below is the horizon of a future comparison, measured by d_N . The number M of training calls is a separate resource. The output is a classical description of a legal effect. For a unit vector w set $P_w = (I + w \cdot \sigma)/2$ and write $\mathcal{M}_{p,q,w}$ for the measurement associated with $pP_w + q(I - P_w)$, where $p = (1 + b + r)/2$ and $q = (1 + b - r)/2$. We use the spectral quantities

$$V = p(1 - p) + q(1 - q), \quad K_N(p, q) = r\sqrt{N/(V + N^{-1})}.$$

Theorem S6.1 (Common learning at the finite-use scale). *There are absolute constants $c, C, \delta_0 > 0$ such that, for every $N \geq 1$, $0 < \delta \leq \delta_0$, and $0 < \eta \leq 1/8$, one common learning procedure returns an effect $\hat{E} \in \mathfrak{E}_2$ with*

$$(S6.1) \quad \sup_{E \in \mathfrak{E}_2} \mathbb{P}_E\{d_N(\mathcal{M}_E, \mathcal{M}_{\hat{E}}) > \delta\} \leq \eta.$$

On every record it uses at most

$$(S6.2) \quad M \leq CN\delta^{-2} \log(1/\eta)$$

calls, and every entangled input block has length at most N . Conversely, every common procedure satisfying (S6.1), with at most M calls on every record, obeys

$$(S6.3) \quad M \geq cN\delta^{-2} \log(1/\eta).$$

The converse permits arbitrary reference-assisted adaptive training and bounded public stopping. The constants are uniform over the closed effect body, including scalar, deterministic, and projective effects.

The budget in (S6.2) is a deterministic upper bound for each completed record, including unsuccessful records. It counts calls during training; N specifies only the subsequent loss. The construction uses independent input blocks, retaining at most $N - 1$ unused input qubits after the next qubit has been supplied, and no quantum system between blocks. Its preparation model permits ideal trusted finite-state preparation and fresh fair random signs. Preparation cost, classical running time, and classical workspace are separate resources. The reusable payload is charged separately in Corollary S6.6. The proof below specifies a written statistical algorithm; the accompanying finite identity checks do not constitute a software implementation of its risk guarantee.

The proof constructs one decision tree for the entire parameter body. Its experiments are not selected from a known pair of hypotheses. The angular part uses multiscale phase estimation, a method with antecedents in [31, 32]. We give the phase and confidence bounds explicitly. A data-dependent amplitude threshold selects the block length without prior calibration of the spectrum.

S6.1. Bernoulli errors and estimated axes. For $s, t \in [0, 1]$ put

$$(S6.4) \quad \mathfrak{h}(s, t)^2 = (\sqrt{s} - \sqrt{t})^2 + (\sqrt{1-s} - \sqrt{1-t})^2.$$

Write $B_N(s, t)$ for the unhalved distance between the two N -fold Bernoulli product laws. We use the following elementary estimates at both support endpoints.

Lemma S6.2 (Endpoint estimation). *For every $s, t \in [0, 1]$,*

$$(S6.5) \quad B_N(s, t) \leq 2\sqrt{N} \mathfrak{h}(s, t).$$

For each fixed $A > 0$ there is C_A such that, if $0 < \lambda \leq 1$ and

$$(S6.6) \quad |s - t| \leq A(\sqrt{t(1-t)\lambda} + \lambda),$$

then $\mathfrak{h}(s, t) \leq C_A\sqrt{\lambda}$. If two estimates of an ordered pair (p, q) each have $\mathfrak{h}$ error at most ε , sorting the estimates preserves this bound up to an absolute constant.

Proof. The root fidelity is $1 - \mathfrak{h}^2/2$. Tensorization and the trace-fidelity inequality give

$$B_N(s, t) \leq 2\sqrt{1 - (1 - \mathfrak{h}(s, t)^2/2)^{2N}} \leq 2\sqrt{N} \mathfrak{h}(s, t).$$

For the first square root in (S6.4), if $t \geq \lambda$ divide the difference by $\sqrt{t}$ in (S6.6). If $t < \lambda$, both t and s are at most $C_A \lambda$. Apply the same argument to their complements. For sorting, set $\phi(t) = \arcsin \sqrt{t}$. Then

$$\mathfrak{h}(s, t) = 2 \sin(|\phi(s) - \phi(t)|/2).$$

The two distances are uniformly comparable on $[0, \pi/2]$. Sorting is nonexpansive for the sup norm against an already ordered pair of real numbers. Apply this fact in the ϕ coordinate. $\square$

If $\hat{t}$ is the empirical frequency in n Bernoulli trials, Bernstein's inequality gives, with failure at most $2e^{-L}$,

$$(S6.7) \quad |\hat{t} - t| \leq \sqrt{2t(1-t)L/n} + \frac{2L}{3n}.$$

Thus Lemma S6.2 applies with $\lambda = L/n$, including at $t = 0, 1$.

For a unit estimate w of u , write

$$(S6.8) \quad h = \angle(u, w), \quad \gamma = \frac{r}{2}(1 - u \cdot w).$$

The input states on the axes $+w$ and $-w$ have respective success probabilities

$$(S6.9) \quad s_+ = p - \gamma, \quad s_- = q + \gamma, \quad p = \frac{1}{2}(1 + b + r), \quad q = \frac{1}{2}(1 + b - r).$$

If $\hat{x}$ estimates $x = ru$ with Euclidean error at most e , normalizing $\hat{x}$ gives

$$(S6.10) \quad rh \leq \pi \min(r, e), \quad \gamma \leq \min(r, e^2/r) \quad (r > 0).$$

Indeed $\|u - w\| \leq 2e/r$, $h \leq (\pi/2)\|u - w\|$, and $\gamma = r\|u - w\|^2/4$. If $\hat{x} = 0$, choose a fixed direction; then $e \geq r$ gives the same bounds. At $r = 0$ both quantities vanish.

Lemma S6.3 (Product learning away from unit contrast). *Suppose $r \leq 7/8$. There is a product-input procedure, not requiring b, r, u as input, which uses eight groups of n calls and returns a legal effect F such that, for $1 \leq L \leq n$,*

$$(S6.11) \quad d_N(\mathcal{M}_E, \mathcal{M}_F) \leq C(\sqrt{NL/n} + NL/n)$$

outside an event of probability at most Ce^{-L} .

Proof. First use the positive and negative eigenstates of each of the three Pauli axes, with n calls per state. Encode an observed outcome as $Z \in \{-1, 1\}$. The two means on axis j are $b + x_j$ and $b - x_j$. Estimate x_j by half their empirical difference. Put

$$w_0 = 1 - b^2, \quad \lambda = L/n.$$

The two variances on axis j sum to $2(w_0 - x_j^2)$. Bernstein's inequality and a union bound therefore give

$$(S6.12) \quad e = \|\hat{x} - x\| \leq 3(\sqrt{w_0 \lambda} + \lambda).$$

More explicitly, the coordinate error is at most $\sqrt{2w_0 \lambda} + 4\lambda/3$; the sign observation has range length two. Taking the Euclidean norm gives (S6.12), since $\sqrt{6} < 3$ and $4\sqrt{3}/3 < 3$.

Normalize to a direction w . Feasibility and $r \leq 7/8$ imply

$$(S6.13) \quad w_0 \geq r(2 - r), \quad V = \frac{1}{2}(w_0 - r^2) \geq w_0/9.$$

Use (S6.10) and Theorem S16.4. The square-root term in (S6.12) contributes at most $C\sqrt{N\lambda}$ to $K_N(p, q)h$, and its additive term contributes at most $CN\lambda$. Consequently,

$$(S6.14) \quad d_N(\mathcal{M}_{p,q,u}, \mathcal{M}_{p,q,w}) \leq C(\sqrt{N\lambda} + N\lambda).$$

This calculation does not require a lower bound on r or w_0 .

It remains to control both endpoint probabilities in (S6.9). For either $t = p$ or $t = q$ put $v_t = t(1 - t)$. Since

$$|p(1 - p) - q(1 - q)| = r|b| \leq r, \quad w_0 = 2V + r^2 \leq 4v_t + 3r,$$

we claim

$$(S6.15) \quad \gamma \leq 9\sqrt{v_t\lambda} + 72\lambda.$$

Equation (S6.12) gives $e^2 \leq 18(w_0\lambda + \lambda^2)$. If $r \leq \lambda$, use $\gamma \leq r$. If $r > \lambda$, use

$$\gamma \leq 72v_t\lambda/r + 72\lambda, \quad \gamma \leq r,$$

and $\min(r, 72v_t\lambda/r) \leq \sqrt{72v_t\lambda}$. This proves (S6.15).

Make two fresh groups of n calls, along $+w$ and $-w$. Conditional on the previous record, these are Bernoulli trials with the parameters in (S6.9). Their variances differ from v_t by at most γ . Equations (S6.7) and (S6.15) imply

$$|\hat{t}_{\text{raw}} - t| \leq C(\sqrt{v_t\lambda} + \lambda).$$

For the additional term use $\sqrt{\gamma\lambda} \leq C(\sqrt{v_t\lambda} + \lambda)$, which follows from $\sqrt{AB} \leq (A + B)/2$ with $A = \sqrt{v_t\lambda}$ and $B = \lambda$. Lemma S6.2 gives root-distance error $C\sqrt{\lambda}$ for both endpoints. Sort the two empirical parameters to obtain $\hat{p} \geq \hat{q}$ and return $F = \hat{p}P_w + \hat{q}(I - P_w)$. The aligned programme bound (S16.9) and (S6.5) give a spectral contribution at most $C\sqrt{N\lambda}$. Combine with (S6.14). All concentration bounds for the fresh groups hold conditionally, so their failure probabilities may be added to those of the first six groups. $\square$

S6.2. A common entangled experiment with unknown bias.

Lemma S6.4 (Two-plane phase observation). *Fix an orthonormal frame (e_1, e_2, w) known to the experimenter. For each $i = 1, 2$ and integer $m \geq 1$, there is an m -call experiment with an observation $Y \in \{-1, 1\}$ whose mean, for a chosen phase φ , is*

$$(S6.16) \quad \mathbb{E}_E Y = \Re(e^{i\varphi} Z_i), \quad Z_i = [x \cdot w + ix \cdot e_i]^m, \quad x = ru.$$

Two phases estimate the complex number Z_i . For $0 < \zeta \leq 1$ and $0 < \epsilon \leq 1/2$, four settings, using the two planes, estimate both complex numbers with error at most ζ and failure at most ϵ , with at most $C\zeta^{-2} \log(C/\epsilon)$ independent blocks per setting.

Fix $a = 1/64$. If $r > 0$ and $h = \angle(u, w) \leq 2a/m$, then

$$(S6.17) \quad .99r^m \leq |Z_i| \leq r^m, \quad \text{Arg } Z_i = m \arctan \frac{u \cdot e_i}{u \cdot w},$$

where the principal argument is used. If the amplitudes are bounded below by a fixed positive constant and ζ is sufficiently small, these observations produce a direction estimate w' with

$$(S6.18) \quad \angle(u, w') \leq C\zeta/m.$$

Proof. Choose a computational basis along the axis perpendicular to the plane spanned by w, e_i , with phase convention such that the off-diagonal matrix element of $O = 2E - I = bI + ru \cdot \sigma$ is $r(u \cdot w + iu \cdot e_i)$. Prepare

$$|G_{\varphi, s}\rangle = \frac{|0\rangle^{\otimes m} + se^{i\varphi}|1\rangle^{\otimes m}}{\sqrt{2}},$$

where s is a fresh fair sign for this block. Conditional on the entire preceding record, the two signs have probability $1/2$ each; the random draw is independent of all earlier preparation randomness. Pass its qubits through the measurement and record $Y = s \prod_{j=1}^m Z_j$, where each output is encoded as $Z_j \in \{-1, 1\}$. Memorylessness makes the joint outcome law the product POVM on the entangled input; the qubits may be supplied sequentially while the unused ones are retained.

The expansion of $\langle G_{\varphi,s} | O^{\otimes m} | G_{\varphi,s} \rangle$ has diagonal term

$$\frac{1}{2}((b + ru \cdot f_i)^m + (b - ru \cdot f_i)^m),$$

where f_i is the chosen perpendicular axis. It is independent of s and disappears after multiplication by s and averaging. The off-diagonal terms give (S6.16). At $x = 0$ the mean is zero and no direction is needed. In particular the experiment removes the unknown bias exactly. The phases $0, \pi/2$ give the real part and the negative imaginary part. More precisely, at each of the four settings use

$$(S6.19) \quad B(\zeta, \epsilon) = \left\lceil 4\zeta^{-2} \log(8/\epsilon) \right\rceil$$

fresh independent blocks. Hoeffding's inequality bounds the error of each real sample mean by $\zeta/\sqrt{2}$ outside an event of probability $2\exp(-B\zeta^2/4)$. The union bound over the four means is at most ϵ , and each complex error is at most ζ . This bound holds conditional on every preceding record: the frame and the setting are then fixed, and the memoryless device acts on fresh blocks. Each estimate batch uses its prescribed number of blocks without stopping within the batch.

For the phase conclusions assume $r > 0$ and put $\theta_i = \arctan((u \cdot e_i)/(u \cdot w))$. Then $|m\theta_i| \leq 2a$ and

$$|Z_i| = r^m (1 - (u \cdot f_i)^2)^{m/2} \geq r^m \cos(h)^m \geq r^m (1 - 2a^2/m).$$

This proves (S6.17), with no phase ambiguity. For the estimates set

$$\hat{\theta}_i = \text{Arg}(\hat{Z}_i)/m, \quad w' = \frac{w + \tan(\hat{\theta}_1)e_1 + \tan(\hat{\theta}_2)e_2}{\|w + \tan(\hat{\theta}_1)e_1 + \tan(\hat{\theta}_2)e_2\|}.$$

The exact formula with θ_i recovers u . For $|\hat{Z}_i - Z_i| \leq |Z_i|/2$,

$$|\text{Arg}(\hat{Z}_i) - \text{Arg}(Z_i)| \leq 2|\hat{Z}_i - Z_i|/|Z_i|.$$

On $|\theta_i| \leq 1/4$ the displayed map from two angles to a unit vector is Lipschitz; a bound of 16 for angular distance against the largest coordinate error follows from differentiation of $\tan$ and normalization. This proves (S6.18). Our fixed cone lies strictly inside the principal-argument interval, and the argument uses no unwrapping threshold from a separate phase-estimation theorem. $\square$

We make the reconstruction rule total. Fix a coordinate basis of $\mathbb{R}^3$ and a tie-breaking order. Given a unit vector w , choose the first coordinate vector minimizing its absolute inner product with w , normalize its projection onto $w^\perp$ to obtain e_1 , and put $e_2 = w \times e_1$. The projection has norm at least $\sqrt{2/3}$, so the frame is defined at every w . For complex estimates z_1, z_2 use the phase formula in the preceding proof only when both pass the guard

$$(S6.20) \quad \Re z_i > 0, \quad 8|\Im z_i| \leq \Re z_i \quad (i = 1, 2).$$

Write $\mathcal{R}_m(w; z_1, z_2) = (\text{ok}, w')$ for the resulting update, and set $\mathcal{R}_m = (\text{fail}, w)$ otherwise. A passed guard gives $|\text{Arg } z_i| \leq \arctan(1/8) < 1/4$; the vector to be normalized has inner product one with w and cannot vanish. Zero complex estimates, the branch cut, and an out-of-cone phase therefore have prescribed outcomes before an argument or tangent is evaluated.

Lemma S6.5 (Selecting a block length from the observations). *Suppose $r \geq 5/8$ and an initial known direction w satisfies $\angle(u, w) \leq a = 1/64$. A common adaptive procedure selects a length m_* and a direction w_* , for $N \geq 1$ and $0 < \eta \leq 1/8$, using at most $CN \log(C/\eta)$ calls on every record. With failure at most $\eta/16$,*

$$(S6.21) \quad \frac{1}{8} \min\{N, (1-r)^{-1}\} \leq m_* \leq N,$$

$$(S6.22) \quad r^{m_*} \geq 1/5, \quad \angle(u, w_*) \leq a/m_*.$$

The reciprocal at $r = 1$ is interpreted as infinity. New observations at m_*, w_* have both complex amplitudes at least $1/6$.

Proof. Fix $\kappa = a/1024$, and let H be the largest power of two not exceeding N . At a tested length m , estimate both complex numbers in Lemma S6.4 to error at most κ with $B(\kappa, \epsilon_m)$ blocks per setting, where

$$(S6.23) \quad \epsilon_m = \frac{\eta m}{32N}.$$

The following procedure stops at the first rejection; it never tests a larger length after that rejection. Here w_0 is the input direction and $\perp$ denotes an empty register.

SELECTBLOCK(N, η, w_0)

1. Set $m \leftarrow 1$, $w \leftarrow w_0$, and $A \leftarrow \perp$.
2. Obtain z_1, z_2 from the four fresh settings at length m and frame w , with budgets (S6.23).
3. If $\min_i |z_i| < 1/4$, return A if $A \neq \perp$, and return $(1, w_0)$ otherwise.
4. Set $(s, v) \leftarrow \mathcal{R}_m(w; z_1, z_2)$. If $s = \text{fail}$, return A if $A \neq \perp$, and return $(1, w_0)$ otherwise.
5. Set $A \leftarrow (m, v)$. If $m = H$, return A .
6. Set $w \leftarrow v$, $m \leftarrow 2m$, and return to step 2.

Equality in the amplitude threshold is accepted, and equality in the second phase guard is allowed. All registers, decisions, and fallbacks are functions of the observed record.

We prove the assertions on the event that the successive complex estimates obey their bounds. Before a length m is tried the angular error is at most $2a/m$: this holds initially, and is preserved by each accepted update. If the amplitude test passes, each true amplitude $\rho = |Z_i|$ is at least $1/4 - \kappa > 1/5$. Since $|\text{Arg } Z_i| \leq 2a$,

$$\Re z_i \geq \rho(1 - 2a^2) - \kappa > 8(2a\rho + \kappa) \geq 8|\Im z_i|.$$

The strict inequality follows directly from $a = 1/64$, $\kappa = a/1024$, and $\rho \geq 1/4 - \kappa$. Thus the guard cannot reject such a stage on this event. The argument bound and the Lipschitz constant in Lemma S6.4 give an updated angular error at most

$$\frac{32\kappa}{(1/4 - \kappa)m} \leq \frac{a}{4m}.$$

Furthermore $r^m \geq 1/4 - \kappa$, whence $m(1 - r) \leq 2$.

On rejection, the amplitude test must have failed, and (S6.17) gives $r^m < (1/4 + \kappa)/.99 < 1/3$. Since $r \geq 5/8$,

$$-\log r \leq (1 - r)/r \leq 2(1 - r), \quad m(1 - r) \geq (\log 3)/2.$$

The procedure returns the preceding accepted length $m/2$, which is therefore at least $(\log 3)/(4(1 - r))$. Length one is always accepted on the good event, since its true amplitudes exceed $.99(5/8)$. If the cap is reached, the selected length is $H \geq N/2$. These facts prove (S6.21) and (S6.22). At the updated direction the amplitude is at least $r^{m*} \cos(a/m_*)^{m*} > 1/6$.

For completeness, let $\mathcal{F}_m$ contain the record immediately before a possible stage m , let $T_m \in \mathcal{F}_m$ be the event that this stage is reached, and let G_m be its complex-error event. On every reached record the concentration bound gives $\mathbb{P}_E(G_m^c \mid \mathcal{F}_m) \leq \epsilon_m$. Consequently

$$(S6.24) \quad \mathbb{P}_E\left(\bigcup_m (T_m \cap G_m^c)\right) \leq \sum_m \mathbb{E}_E[\mathbf{1}_{T_m} \mathbb{P}_E(G_m^c \mid \mathcal{F}_m)] \leq \sum_{m=1,2,4,\dots,H} \epsilon_m \leq \eta/16.$$

The sample-mean bound holds for every preceding record, even one whose frame has left the cone; the cone is needed only to turn those sample means into directional information. Independence

between stage-failure events is not assumed. By Lemma S6.4, the number of calls is bounded by an absolute constant times

$$(S6.25) \quad \sum_{m=1,2,4,\dots,H} m \log \frac{CN}{\eta m} \leq C' N \log(C'/\eta).$$

To see this, write $m = H/2^k$ and use the convergent sums of 2^{-k} and $k2^{-k}$, together with $1 \leq N/H < 2$. The public cap and these budgets bound the cost on every record, including records on which an estimate fails. Rejection can only shorten the sum. $\square$

S6.3. The learning bound and its converse. All statistical batches use fresh inputs and fresh randomization. Table 1 allocates their error probabilities. Each conditional bound is uniform over the preceding record. A row belonging to a branch not taken contributes no failure event.

TABLE 1. Confidence allocation for one execution of the learner.

Batch	Failure allowance	Conditioning and scope
Coarse Pauli vector	$\eta/8$	Unconditional; all six input groups jointly.
Product-branch Pauli vector	$\eta/8$	Given the coarse record; six fresh input groups jointly.
Dyadic stage m	$\eta m/(32N)$	Given the record before that stage; all reached stages together have allowance at most $\eta/16$.
Fine two-plane estimate	$\eta/8$	Given the complete search record, including the selected m_*, w_* .
Fresh endpoint groups	$\eta/16$ each	Given the final estimated axis; the two allowances sum to $\eta/8$, in either branch.

The conservative sum of all rows is $9\eta/16 < \eta$. The conditional union calculation (S6.24) applies to every row, so this sum requires no independence between the events attached to different batches.

Proof of Theorem S6.1. We choose constants in the following order. Fix the local cone $a = 1/64$ and $\kappa = a/1024$ as above. Choose a sufficiently small absolute constant $c_* > 0$ in the final phase tolerance below. Then choose the absolute multipliers in the Bernoulli sample counts sufficiently large. Finally reduce δ_0 , if needed, so that $\delta_0 \leq 1/4096$ and $c_*\delta_0 \leq a/100$. No optimized value of these constants is used. Set

$$(S6.26) \quad L = \log(C_2/\eta), \quad n_0 = \lceil C_0 L \rceil, \quad n = \lceil C_1 N \delta^{-2} L \rceil,$$

where C_0, C_1, C_2 are absolute constants large enough for the concentration bounds and loss estimates below. The common procedure is the following.

- (1) Make the six coarse Pauli experiments, with n_0 calls per input, and form their half-difference vector $\hat{x}_0$.
- (2) If $\|\hat{x}_0\| < 3/4$, make six fresh Pauli groups of n calls per input and form $\hat{x}$. Set $w_f = \hat{x}/\|\hat{x}\|$, using the fixed third coordinate direction if $\hat{x} = 0$, and proceed to step 4.
- (3) Otherwise set $w_0 = \hat{x}_0/\|\hat{x}_0\|$ and run SELECTBLOCK(N, η, w_0). At its returned (m_*, w_*) , obtain two fresh complex estimates with tolerance $\zeta_* = c_*\delta\sqrt{m_*/N}$ and failure allowance $\eta/8$. Apply $\mathcal{R}_{m_*}$ to them; take its updated direction as w_f on success and retain $w_f = w_*$ on failure.
- (4) Make n fresh calls along each of $+w_f$ and $-w_f$. If the empirical success frequencies are s_+, s_- , return

$$\hat{E} = \max(s_+, s_-)P_{w_f} + \min(s_+, s_-)(I - P_{w_f}).$$

The empirical frequencies already lie in $[0, 1]$. Ties in the coarse comparison use step 3; ties between endpoint frequencies give a scalar effect. Together with the frame, phase, and first-rejection conventions above, these rules define an output for every record, with no retry or additional calibration step.

We verify the risk. With failure at most $\eta/8$, the coarse half-difference estimate satisfies

$$\|\hat{x}_0 - x\| \leq \varepsilon_0, \quad \varepsilon_0 \leq a/100.$$

On this event the two decisions have the explicit overlap

$$\begin{aligned} \|\hat{x}_0\| < 3/4 &\implies r < 3/4 + 1/6400 < 7/8, \\ \|\hat{x}_0\| \geq 3/4 &\implies r \geq 3/4 - 1/6400 > 5/8. \end{aligned}$$

For the product branch, Lemma S6.3 gives error at most $\delta/2$ with the sample sizes in (S6.26), after increasing C_1 . Its first six groups have joint allowance $\eta/8$ conditional on the coarse record. After their axis is fixed, each endpoint group has allowance $\eta/16$. These are the product-branch and endpoint rows of Table 1, and all samples here are fresh relative to the coarse test.

For the entangled branch, the normalized initial estimate has angular error at most $\pi\varepsilon_0/r \leq 8\pi a/500 < a$ by (S6.10). Apply Lemma S6.5. At the selected length and updated direction, make fresh two-plane observations with complex error tolerance

$$(S6.27) \quad \zeta_* = c_*\delta\sqrt{m_*/N},$$

and failure allowance $\eta/8$, conditional on the full selection record. On the preceding good event, the two true amplitudes are at least $1/6$ and their phases have absolute value at most a . On the current complex-error event, $\zeta_* \leq a/100$ and the inequalities $|\sin t| \leq |t|$ and $\cos t \geq 1 - t^2/2$ show that both phase guards pass. The resulting direction w_f satisfies

$$(S6.28) \quad h_f = \angle(u, w_f) \leq Cc_*\delta/\sqrt{Nm_*}.$$

The calls used in this step are at most

$$(S6.29) \quad Cm_*\zeta_*^{-2} \log(C/\eta) \leq C'N\delta^{-2} \log(C/\eta).$$

Put $\nu = 1 - r$. Since $|b| \leq \nu$ and $r \geq 5/8$,

$$r\nu \leq V = \frac{1}{2}(1 - r^2 - b^2) \leq \nu.$$

Equation (S6.21) implies

$$(S6.30) \quad m_*(V + N^{-1}) \geq c_0 > 0$$

for an absolute c_0 . Thus (S6.28) and Theorem S16.4 give

$$(S6.31) \quad d_N(\mathcal{M}_{p,q,u}, \mathcal{M}_{p,q,w_f}) \leq Cc_*\delta.$$

The endpoint misalignment is also bounded:

$$(S6.32) \quad \gamma = \frac{r}{2}(1 - u \cdot w_f) \leq rh_f^2/4 \leq Cc_*^2\delta^2/N.$$

The final fresh groups have the public size n in (S6.26). Conditional on the record fixing w_f , their success probabilities are exactly $p - \gamma$ and $q + \gamma$, and each group obeys Bernstein's bound with allowance $\eta/16$. Conditional Bernstein bounds, Lemma S6.2, and $\mathfrak{h}(s, t)^2 \leq 2|s - t|$ give

$$(S6.33) \quad \begin{aligned} \mathfrak{h}(\hat{p}_{\text{raw}}, p), \mathfrak{h}(\hat{q}_{\text{raw}}, q) &\leq C\sqrt{\log(C/\eta)/n} + C\sqrt{\gamma} \\ &\leq c'\delta/\sqrt{N}. \end{aligned}$$

The constant c' can be made as small as required by the preceding choices of c_* and the sample multiplier. Sort the estimates and return $\hat{E} = \hat{p}P_{w_f} + \hat{q}(I - P_{w_f})$. Equations (S16.9) and (S6.5)

make the spectral loss at most $\delta/4$; choose c_* so that (S6.31) contributes at most $\delta/4$. Table 1 now gives the stated risk.

Finally we check the budget for every record. The product branch uses exactly $6n_0 + 8n$ calls. The entangled branch uses at most

$$(S6.34) \quad 6n_0 + 4 \sum_{m=1,2,4,\dots,H} mB(\kappa, \epsilon_m) + 4m_*B(\zeta_*, \eta/8) + 2n.$$

This is a pathwise upper bound: the sum includes even the stages not reached after the first rejection. Its dyadic term is $O(N \log(C/\eta))$ by (S6.25), including the ceilings in (S6.19). For every returned $1 \leq m_* \leq N$, even on a failed record,

$$4m_*B(\zeta_*, \eta/8) \leq 4N + 16c_*^{-2}N\delta^{-2} \log(64/\eta).$$

Here $m_*\zeta_*^{-2} = c_*^{-2}N\delta^{-2}$ exactly. Since $N \geq 1$, $\delta \leq \delta_0$, and $\eta \leq 1/8$, both branches satisfy (S6.2). The prescribed fallbacks always leave a unit direction and a legal effect, and every block has at most $H \leq N$ inputs. This proves the upper bound with the same worst-record convention as the converse.

For the converse consider the scalar effects

$$(S6.35) \quad E_0 = \tfrac{1}{2}I, \quad E_1 = (\tfrac{1}{2} + \Delta)I, \quad \Delta = 256\delta/\sqrt{N}.$$

Since $\delta \leq 1/4096$, both parameters lie in $[1/2, 9/16]$. The finite Bernoulli lower bound in Lemma S16.2 yields

$$(S6.36) \quad \begin{aligned} d_N(\mathcal{M}_{E_0}, \mathcal{M}_{E_1}) &= B_N(1/2, 1/2 + \Delta) \\ &\geq \tfrac{1}{64} \min\{1, \Delta\sqrt{N/(5/4)}\} > 2\delta. \end{aligned}$$

For these devices every call gives a fresh Bernoulli bit independent of the input. On an entangled input, each conditional reference block is the same reduced state multiplied by the scalar outcome probability. Thus any adaptive learning record, including its final classical estimate, is a common stochastic processing of M independent Bernoulli bits and parameter-independent resources. Publicly stopped records may be padded. The relative entropy before processing is at most

$$(S6.37) \quad MD(\text{Bern}(1/2) \parallel \text{Bern}(1/2 + \Delta)) = -\tfrac{M}{2} \log(1 - 4\Delta^2) \leq 3M\Delta^2.$$

The two radius- δ success balls are disjoint by (S6.36). Testing whether the estimated effect belongs to the first ball therefore gives two errors at most η . For classical laws P, Q , Jensen's inequality gives root fidelity at least $e^{-D(P\parallel Q)/2}$. The trace-fidelity inequality then implies that the sum of testing errors is at least $\frac{1}{2}e^{-D(P\parallel Q)}$. Consequently

$$2\eta \geq \tfrac{1}{2}e^{-3M\Delta^2}, \quad M \geq \frac{N}{3 \cdot 256^2 \delta^2} \log \frac{1}{4\eta}.$$

For $\eta \leq 1/8$ this proves (S6.3). □

Corollary S6.6 (A learned joint description). *For rational δ in the range of Theorem S6.1, the learner may return one word of the rational joint code, with the same risk and call-count bounds. Its payload length is*

$$(S6.38) \quad 2\log_2 N + \log_2 \log(N+2) + 4\log_2(1/\delta) + O(1), \quad 0 < \delta \leq \delta_0.$$

The decoding centre is a legal rational effect. This is a statement about device calls and payload length; classical arithmetic, workspace, and state-preparation costs are separate resources.

Proof. Use the learner with loss budget $\delta/2$. Its output is a finite computable expression determined by a finite record. In particular, the empirical frequencies are rational, and the guarded phase

roots and normalized frames specify algebraic directions. Choose a positive rational $\varepsilon \leq \delta/(100N)$ and rational approximations (b_0, x_0) to its bias and Bloch vector with

$$|b_0 - \widehat{b}| + \|x_0 - \widehat{x}\| \leq \varepsilon.$$

Set $(b_1, x_1) = (b_0, x_0)/(1 + 2\varepsilon)$. Then

$$|b_1| + \|x_1\| \leq \frac{1 + \varepsilon}{1 + 2\varepsilon} \leq 1,$$

and the exact one-use formula (S16.6) bounds the distance from the unrounded hypothesis by 3ε . The N -use hybrid inequality adds at most $3N\varepsilon$. Apply Theorem S17.2 to this rational estimated effect with public accuracy $\delta/2$. Its certified radius is $\delta/4$, so the total loss is less than δ . The capacity estimate in that theorem, at a constant rescaling of accuracy, gives (S6.38). This postprocessing uses no additional device calls and receives no exact description of the unknown target. $\square$

The loss in Theorem S6.1 is a future adaptive distance. One-use measurement tomography is studied in [5, 29]; combining the stated one-use accuracy bounds only with the hybrid inequality gives a quadratic dependence on the future horizon. The argument here uses finite-use geometry and data-dependent entangled blocks to obtain the linear dependence in (S6.2), with a matching scalar converse. Learning a stored programme from which a measurement is later retrieved, as in [30], has a different output resource. The present theorem gives a classical estimate over the entire ordered binary qubit body. Section S7 uses this common qubit procedure after a separate calibration to obtain the corresponding learning law on the full matrix effect body.

S7. COMMON LEARNING IN ARBITRARY INPUT DIMENSION

The two-dimensional learner extends to the full effect body through a calibration step. The calibration must resolve the two support endpoints simultaneously. A bound in operator norm alone does not provide this information: after compression, a poorly aligned subspace can introduce noise at a scale larger than N^{-1} . The procedure below learns a basis in which this additional noise is bounded by the finite-horizon cutoff.

Throughout this section, $\mathcal{M}_E$ is the ordered binary measurement associated with $E \in \mathfrak{E}_d$. Training may use entangled input blocks and classical feedback, but it has no access to the measurement environment. Its output is a classical description of a legal effect. The future horizon N and the number of training calls remain separate resources.

Theorem S7.1 (Common learning on the matrix effect body). *There are absolute constants $c, C, \delta_0 > 0$ with the following property. For every $d, N \geq 1$, $0 < \delta \leq \delta_0$, and $0 < \eta \leq 1/8$, one common procedure returns a legal effect $\widehat{E} \in \mathfrak{E}_d$ such that*

$$(S7.1) \quad \sup_{E \in \mathfrak{E}_d} \mathbb{P}_E\{d_N(\mathcal{M}_E, \mathcal{M}_{\widehat{E}}) > \delta\} \leq \eta.$$

On every record its number of calls satisfies

$$(S7.2) \quad M \leq Cd^4 N \delta^{-2} \log(d/\eta),$$

and every entangled input block has length at most N . Conversely, any common procedure satisfying (S7.1) with at most M calls on every record satisfies

$$(S7.3) \quad M \geq cN \delta^{-2} \log(1/\eta).$$

The converse permits arbitrary reference-assisted adaptive training and bounded public stopping. In particular, for each fixed d , the minimax number of calls is $\Theta_d(N \delta^{-2} \log(1/\eta))$ on the complete closed effect body. The displayed dependence on d in the upper bound is not asserted to be optimal.

We first give the calibration, including its finite budgets. All matrix orders below are Loewner orders. The use of a basis selected from earlier observations is always followed by fresh samples.

Lemma S7.2 (Simultaneous endpoint calibration). *Let $\tau = N^{-1}$ and $0 < \epsilon \leq 1/8$. There is a common product-input procedure returning $A \in \mathfrak{E}_d$ with at most $Cd^4N \log(d/\epsilon)$ calls on every record. With probability at least $1 - \epsilon$, it satisfies*

$$(S7.4) \quad \frac{1}{2}(A + \tau I) \preceq E + \tau I \preceq 2(A + \tau I),$$

$$(S7.5) \quad \frac{1}{2}(I - A + \tau I) \preceq I - E + \tau I \preceq 2(I - A + \tau I),$$

$$(S7.6) \quad \|E - A\|_{\text{op}} \leq \sqrt{\tau}.$$

The input states depend only on the public parameters and the preceding record; no spectral data about E are supplied.

Proof. Set $J = \lceil \log_2 N \rceil$, $H = 2^J$, and $t_j = 2^{-j}$ for $0 \leq j \leq J$. Thus $N \leq H < 2N$ unless $N = 1$, and $\tau/2 < t_J \leq \tau$. Start from $A_0 = I/2$. If $J = 0$, return A_0 : all three conclusions hold without samples.

We use

$$h_d = d + 4 \binom{d}{2} = 2d^2 - d, \quad K_d = (320d)^2, \quad \alpha = 1/16.$$

At stage $j \geq 1$, assign failure allowance and sample size

$$(S7.7) \quad \epsilon_j = \frac{\epsilon 2^j}{4H}, \quad L_j = \log \frac{2h_d}{\epsilon_j}, \quad n_j = \left\lceil K_d t_j^{-1} L_j \right\rceil.$$

Choose an orthonormal eigenbasis $e_1, \dots, e_d$ of the current legal estimate A_{j-1} . The choice is deterministic and measurable, including at multiple eigenvalues. More explicitly, fix a public input basis. All calibration matrices have algebraic entries in that basis: the sample frequencies are rational, and spectral projections, clipping at rational endpoints, and normalization preserve algebraic entries. Order the distinct eigenvalues, project the public basis onto each eigenspace, and apply Gram–Schmidt in that fixed order, discarding exactly zero vectors. Exact algebraic arithmetic therefore supplies a finite choice at every stage and requires no oracle for arbitrary real spectral data. Make n_j independent calls on each of the h_d states

$$(S7.8) \quad e_i, \quad \frac{e_i + e_j}{\sqrt{2}}, \quad \frac{e_i - e_j}{\sqrt{2}}, \quad \frac{e_i + ie_j}{\sqrt{2}}, \quad \frac{e_i - ie_j}{\sqrt{2}} \quad (i < j).$$

Here and below a unit vector denotes its pure input state. Estimate each diagonal entry of E by its empirical outcome frequency. Estimate the real part of each off-diagonal entry by half the frequency difference between the positive and negative real superpositions. Its imaginary part is half the frequency on $(e_i - ie_j)/\sqrt{2}$ minus half the frequency on $(e_i + ie_j)/\sqrt{2}$. These formulas define a Hermitian matrix B_j .

To define the procedure on every record, clip the eigenvalues of B_j to $[-\alpha t_j, 1 + \alpha t_j]$, obtaining $\tilde{B}_j$, and set

$$(S7.9) \quad A_j = \frac{\tilde{B}_j + \alpha t_j I}{1 + 2\alpha t_j}.$$

The returned matrix is always a legal effect. In particular, a failed concentration event cannot affect the subsequent call budget or require an additional experiment.

We prove the invariant

$$(S7.10) \quad \frac{1}{2}(A_j + t_j I) \preceq E + t_j I \preceq 2(A_j + t_j I), \quad \frac{1}{2}(I - A_j + t_j I) \preceq I - E + t_j I \preceq 2(I - A_j + t_j I)$$

on the successive good events. It holds for $j = 0$ because $0 \preceq E \preceq I$.

Fix a preceding good record and abbreviate $t = t_j$, $A = A_{j-1}$, $B = B_j$, $D = B - E$, and $q = L_j/n_j$. All expectations and probabilities in the next calculation are conditional on that record. Bernstein's inequality and a union bound over the h_d groups show, with failure at most $2h_d e^{-L_j} = \epsilon_j$, that every empirical frequency has error at most

$$\sqrt{2p(1-p)q} + 2q/3.$$

For a pair of opposite superposition states, the sum of the outcome probabilities is $E_{ii} + E_{jj}$. Cauchy–Schwarz in each half-difference therefore gives the following two simultaneous bounds, also valid for $i = j$ after enlarging the constants:

$$(S7.11) \quad |D_{ij}| \leq \sqrt{2(E_{ii} + E_{jj})q} + 2q, \quad |D_{ij}| \leq \sqrt{2(2 - E_{ii} - E_{jj})q} + 2q.$$

For an off-diagonal complex entry, apply the real estimate to the two quadratures and take their Euclidean norm. The factor $\sqrt{2}$ in (S7.11) includes this operation.

Write a_i for the eigenvalues of A . The preceding invariant, whose regularization is $2t$, gives

$$E_{ii} \leq 2a_i + 2t, \quad 1 - E_{ii} \leq 2(1 - a_i) + 2t, \quad q \leq t/K_d.$$

Since

$$t(a_i + a_j + 2t) \leq 2(a_i + t)(a_j + t), \quad t \leq \sqrt{(a_i + t)(a_j + t)},$$

the first bound in (S7.11) yields

$$(S7.12) \quad \frac{|D_{ij}|}{\sqrt{(a_i + t)(a_j + t)}} \leq \frac{2\sqrt{2}}{\sqrt{K_d}} + \frac{2}{K_d} \leq \frac{5}{\sqrt{K_d}}.$$

Replacing a_i by $1 - a_i$ gives the same estimate for the complement. The Hilbert–Schmidt norm bounds the operator norm, so our choice of K_d implies

$$(S7.13) \quad \begin{aligned} -\frac{1}{64}(A + tI) &\preceq D \preceq \frac{1}{64}(A + tI), \\ -\frac{1}{64}(I - A + tI) &\preceq D \preceq \frac{1}{64}(I - A + tI). \end{aligned}$$

The preceding invariant and $t_{j-1} = 2t$ also imply

$$A + tI \preceq 4(E + tI), \quad I - A + tI \preceq 4(I - E + tI).$$

Consequently

$$(S7.14) \quad -\alpha(E + tI) \preceq D \preceq \alpha(E + tI), \quad -\alpha(I - E + tI) \preceq D \preceq \alpha(I - E + tI).$$

These inequalities place the spectrum of B inside $[-\alpha t, 1 + \alpha t]$. Thus the clipping in (S7.9) does nothing on this event. On it,

$$A_j - E = \frac{D + \alpha t(I - 2E)}{1 + 2\alpha t}.$$

Since $-I \preceq I - 2E \preceq I$, equations (S7.14) imply

$$\begin{aligned} -2\alpha(E + tI) &\preceq A_j - E \preceq 2\alpha(E + tI), \\ -2\alpha(I - E + tI) &\preceq A_j - E \preceq 2\alpha(I - E + tI). \end{aligned}$$

Adding the corresponding regularized endpoint gives factors $1 - 2\alpha = 7/8$ and $1 + 2\alpha = 9/8$. This proves the weaker factor-two invariant (S7.10) for stage j .

The unweighted entry estimate also gives the operator-norm bound needed later. Because $E_{ii} + E_{jj} \leq 2$, we have

$$\|D\|_{\text{op}} \leq 4d\sqrt{t/K_d} = \sqrt{t}/80.$$

It follows that

$$(S7.15) \quad \|A_j - E\|_{\text{op}} \leq \sqrt{t}/80 + \alpha t \leq \sqrt{t}.$$

At the final stage, adding $(\tau - t_J)I$ preserves the factor-two orders, and (S7.15) gives (S7.6).

The conditional failure allowances sum to less than $\epsilon/2$, since $\sum_{j=1}^J 2^j < 2H$. This proves the asserted probability bound. Finally, the deterministic call count is $h_d \sum_j n_j$. Writing $2^j = H/2^k$ yields

$$\sum_{j=1}^J 2^j L_j \leq H \sum_{k=0}^{J-1} 2^{-k} \left(\log \frac{8h_d}{\epsilon} + k \log 2 \right) \leq CH \log(d/\epsilon).$$

The ceiling terms contribute at most $h_d J \leq h_d H$. As $h_d K_d \leq Cd^4$ and $H < 2N$ when $N > 1$, the stated bound holds on every record. In particular, the number of dyadic stages does not introduce a factor $\log N$. $\square$

Lemma S7.3 (Calibrated variance and compression). *Suppose $A, E \in \mathfrak{C}_d$ satisfy (S7.4)–(S7.6). Set $W_G = G(I - G) + \tau I/2$. Then*

$$(S7.16) \quad \frac{1}{12} W_A \preceq W_E \preceq 12 W_A.$$

If P is any orthogonal projection commuting with A and $C = PEP|_{P\mathbb{C}^d}$, then, on $P\mathbb{C}^d$,

$$(S7.17) \quad \frac{1}{12} P W_A P \preceq W_C \preceq 14 P W_A P.$$

All constants are independent of the spectrum, its multiplicities, and the rank of P .

Proof. For a legal effect G , define the harmonic mean without a factor of two by

$$R_G = ((G + \tau I)^{-1} + (I - G + \tau I)^{-1})^{-1}.$$

The two factors commute because both are functions of G , so

$$(S7.18) \quad R_G = \frac{G(I - G) + (\tau + \tau^2)I}{1 + 2\tau}, \quad \frac{1}{3}(G(I - G) + \tau I) \preceq R_G \preceq G(I - G) + \tau I.$$

The inequalities use only $0 < \tau \leq 1$. Inversion reverses Loewner order. Apply it to both endpoint comparisons, add, and invert again to get $R_E \preceq R_A$ with factors two. Equation (S7.18) therefore compares $E(I - E) + \tau I$ and $A(I - A) + \tau I$ with factors six. Replacing the regularization τI by $\tau I/2$ proves (S7.16).

The compression identity is exact:

$$(S7.19) \quad C(I - C) = PE(I - E)P + PE(I - P)EP.$$

Because P commutes with A , $PE(I - P) = P(E - A)(I - P)$. Thus

$$0 \preceq PE(I - P)EP \preceq \|E - A\|_{\text{op}}^2 P \preceq \tau P.$$

The first term in (S7.19), after adding $\tau P/2$, is $PW_E P$. Apply (S7.16) and use $PW_A P \succeq \tau P/2$ to absorb the final τP with a factor two. This gives (S7.17). $\square$

Proof of Theorem S7.1. For $d = 1$, Bernoulli sampling and Lemma S6.2 give the upper bound directly. Assume $d \geq 2$ and put $s = \binom{d}{2}$. Apply Lemma S7.2 with failure allowance $\eta/2$. Conditional on its good event, fix an orthonormal eigenbasis of the returned A , with eigenvalues a_i , and write

$$w_i = a_i(1 - a_i) + \tau/2.$$

For each $i < j$, let P_{ij} project onto $\text{span}(e_i, e_j)$ and let C_{ij} be the corresponding compression of E . An input qubit, including its entanglement with other input qubits or a retained reference, can be embedded isometrically into this two-dimensional subspace. The resulting observed channel is exactly $\mathcal{M}_{C_{ij}}$.

Run the common qubit learner of Theorem S6.1 on each compression, using fresh records, accuracy

$$(S7.20) \quad \varepsilon = c_* \delta / d,$$

and failure allowance $\eta/(2s)$. The constant $c_* > 0$ will be chosen below. This returns legal 2×2 effects $\hat{C}_{ij}$. Every guarantee is conditional on the calibration record and is uniform over the resulting compression. A union bound therefore shows that, with total failure at most η , calibration succeeds and

$$(S7.21) \quad d_N(\mathcal{M}_{C_{ij}}, \mathcal{M}_{\hat{C}_{ij}}) \leq \varepsilon \quad \text{for every } i < j.$$

We record explicitly how an operational error controls an endpoint form. Theorem S4.1 in dimension two and Lemma S4.5 imply that there are absolute $a_0, A_0 > 0$ such that, for legal $C, D \in \mathfrak{E}_2$,

$$(S7.22) \quad d_N(\mathcal{M}_C, \mathcal{M}_D) \leq u \leq a_0 \implies g_{N,C}(D - C) \leq A_0 u.$$

Indeed, for $u < 1/16384$ the lower metric bound forces $Q_N(C, D) \leq 16384u$; decreasing a_0 makes this at most $1/4$, where the local comparison applies. Choose c_* small enough that $\varepsilon \leq a_0$ and lies within the qubit learner's accuracy range.

For a Hermitian matrix X on $\text{span}(e_i, e_j)$ put

$$\gamma_{ij}(X)^2 = N \left(\frac{|X_{ii}|^2}{2w_i} + \frac{|X_{jj}|^2}{2w_j} + \frac{2|X_{ij}|^2}{w_i + w_j} \right).$$

Equation (S7.17), applied to left plus right multiplication and then inverted, gives

$$\gamma_{ij}(X) \leq \sqrt{14} g_{N,C_{ij}}(X).$$

Hence (S7.21) and (S7.22) imply

$$(S7.23) \quad \gamma_{ij}(\hat{C}_{ij} - C_{ij}) \leq \sqrt{14} A_0 \varepsilon.$$

Assemble a Hermitian matrix B by taking each off-diagonal entry from its corresponding estimate $\hat{C}_{ij}$. For each diagonal entry choose once and for all one incident pair and use that pair's estimate. No consistency among different estimates of the same diagonal entry is assumed. In the diagonal coordinates of A , the form $g_{N,A}$ is the sum of the nonnegative coordinate terms displayed above. Each coordinate used in $B - E$ occurs in at least one of the pair forms, while unused duplicate diagonal terms can be discarded. Therefore

$$(S7.24) \quad g_{N,A}(B - E)^2 \leq \sum_{i < j} \gamma_{ij}(\hat{C}_{ij} - C_{ij})^2 \leq 14sA_0^2\varepsilon^2.$$

The assembled matrix need not be legal. Define the estimate by minimizing its distance to the legal effect body in the known positive quadratic norm $g_{N,A}$. The minimum is attained and is unique. An exact minimizer F satisfies

$$g_{N,A}(F - E) \leq 2g_{N,A}(B - E),$$

because E is a feasible competitor. To obtain a finite procedure with a rational legal output, it suffices instead to choose $F \in \mathfrak{E}_d$ with rational entries such that

$$(S7.25) \quad g_{N,A}(F - B) \leq \inf_{G \in \mathfrak{E}_d} g_{N,A}(G - B) + \varepsilon.$$

This changes the bound only to

$$(S7.26) \quad g_{N,A}(F - E) \leq 2\sqrt{14s}A_0\varepsilon + \varepsilon \leq C_0d\varepsilon$$

for an absolute C_0 .

For completeness, (S7.25) is an effective finite selection, not a query to the unknown effect. Since $W_A \succeq \tau I/2$, $g_{N,A}(X) \leq N\|X\|_{\text{HS}}$. A finite rational grid covering $\mathfrak{E}_d$ in Hilbert–Schmidt norm to $\varepsilon/(4N)$ therefore covers it in $g_{N,A}$ to $\varepsilon/4$. Such a legal grid is obtained by rounding after the inward move $G \mapsto (1 - 2r)G + rI$: entrywise rounding with Hilbert–Schmidt error at most $r/2$ preserves positivity of both outcomes, and the total rounding displacement is at most $C_d r$.

Concretely, choose positive rational $r \leq \varepsilon/[4N(\sqrt{d}+1)]$ and a rational entry mesh at most $r/(2d)$. The inward displacement is at most $r\sqrt{d}$, and the rounding displacement is at most $r/2$, so their sum is at most $\varepsilon/(4N)$. The grid and its rational entries are defined in the fixed public input basis, independently of the random eigenbasis used for calibration. Enumerate this bounded grid, retain legal matrices using exact positivity tests, and evaluate its finitely many objective values to tolerance $\varepsilon/4$. Selecting the smallest computed value proves (S7.25) with slack. The calibration matrices are algebraic, and the qubit subroutine's outputs are finite computable expressions in its record, so these evaluations require no oracle for the unknown effect. Only the observed matrices, the public parameters, and finite classical calculations are used. Arithmetic and state preparation are not included in the device-call bound.

Finally, (S7.16) gives

$$g_{N,E}(F-E) \leq \sqrt{12} g_{N,A}(F-E) \leq \sqrt{12} C_0 d \varepsilon.$$

Choose the absolute c_* in (S7.20) so that this is at most $\delta/(8\sqrt{2})$, and reduce δ_0 if necessary so that it is also at most $1/4$. The endpoint-to-midpoint comparison in Lemma S4.5 and the upper bound in Theorem S4.1 then give

$$d_N(\mathcal{M}_E, \mathcal{M}_F) \leq 8Q_N(E, F) \leq 8\sqrt{2} g_{N,E}(F-E) \leq \delta.$$

This proves the risk bound.

The calibration costs at most $Cd^4N \log(d/\eta)$ calls. There are $s \leq d^2/2$ qubit learners, each with accuracy $c_*\delta/d$ and confidence $\eta/(2s)$. Their combined call count is at most $Cd^4N\delta^{-2} \log(d/\eta)$ by Theorem S6.1. All budgets hold on every record, all compressed entangled blocks have length at most N , and postprocessing uses no additional calls.

The confidence and resource allocation can be recorded as follows; the constants in the call bounds are absolute.

Stage	Failure allowance	Calls on every record
Endpoint calibration	$\eta/2$	$Cd^4N \log(d/\eta)$
Each of $s = \binom{d}{2}$ pair learners	$\eta/(2s)$	$CN(d/\delta)^2 \log(d/\eta)$
Legal rational selection	0	0

The conditional guarantees of the pair learners allow their allowances to be added after calibration, despite the dependence of the selected basis on its record.

For the lower bound take

$$E_0 = \frac{1}{2}I_d, \quad E_1 = (\frac{1}{2} + \Delta)I_d, \quad \Delta = 256\delta/\sqrt{N}.$$

Choose $\delta_0 \leq 1/4096$. The Bernoulli separation in (S6.36) is unchanged by the input dimension, and the two effects are separated by more than 2δ under d_N . Every use of either scalar device produces a fresh Bernoulli bit independent of the input. Its conditional reference state is the same reduced state multiplied by that scalar outcome probability. Thus an arbitrary adaptive training record with at most M calls, padded if necessary, is a common processing of M independent Bernoulli bits and parameter-independent resources. Its relative entropy is at most $3M\Delta^2$ by (S6.37). The two disjoint success balls turn the estimated effect into a test with both errors at most η . The same fidelity inequality used in the proof of Theorem S6.1 gives

$$2\eta \geq \frac{1}{2}e^{-3M\Delta^2}, \quad M \geq \frac{N}{3 \cdot 256^2\delta^2} \log \frac{1}{4\eta}.$$

For $\eta \leq 1/8$ this is (S7.3). □

The common learner now has the same dimension and boundary scope as the metric and entropy theorems. Its calibration is a sequence of product experiments, while the compression learners provide the enhanced angular resolution when an estimated two-dimensional subspace is close

to opposite support endpoints. No pair of candidate hypotheses is supplied to either part of the procedure.

S8. A FINITE RATIONAL CODE IN EVERY DIMENSION

The volume argument can be made into a single deterministic code whose decoder reconstructs its dictionary from the public parameters. The construction is finite and exact. Its purpose is to settle the constructive description problem; the enumeration below can be very large.

Theorem S8.1 (Deterministic rational matrix code). *Let $d, N \geq 1$ be integers and let $0 < \delta \leq 1$ be rational. There is a deterministic finite procedure, using rational arithmetic, that constructs an ordered list*

$$\mathcal{C}_{d,N,\delta} = (C_0, \dots, C_{M-1}) \subset \mathfrak{E}_d$$

of Gaussian-rational effects with the following properties. Every $E \in \mathfrak{E}_d$ has an assigned index $j(E)$ satisfying

$$(S8.1) \quad d_N(\mathcal{M}_E, \mathcal{M}_{C_{j(E)}}) \leq \frac{11}{16}\delta.$$

For fixed d ,

$$(S8.2) \quad M \leq C_d N^{d^2/2} [\log(N+2)]^{\lfloor d/2 \rfloor} \delta^{-d^2},$$

uniformly in N and δ . The encoder on a supplied Gaussian-rational effect and the decoder on its index both terminate using exact arithmetic. They require no matrix-dependent advice or preassigned list of centres. With (d, N, δ) public, the payload length is $\lceil \log_2 M \rceil$, and for $0 < \delta \leq \delta_d$ it is

$$(S8.3) \quad \frac{d^2}{2} \log_2 N + \lfloor d/2 \rfloor \log_2 \log(N+2) + d^2 \log_2(1/\delta) + O_d(1).$$

Here δ_d may be the small-error constant in Theorem S5.1. The procedure makes no assertion of polynomial running time in the payload length.

Proof. Set

$$(S8.4) \quad K = \left\lceil \frac{32dN}{\delta} \right\rceil, \quad s = \frac{2d}{K}, \quad \eta = \frac{\delta}{16}.$$

Fix the order of independent real coordinates to be the d diagonal entries, followed by the real and imaginary parts of G_{ij} for $i < j$ in lexicographic order. Let $\mathcal{G}_{d,K}$ consist of the legal Hermitian matrices whose diagonal coordinates belong to $\{0, 1/K, \dots, 1\}$ and whose remaining coordinates belong to $\{-1, -1 + 1/K, \dots, 1\}$. List these matrices in the induced lexicographic order. This list is finite, and it contains zero.

Traverse the list once. Retain its first element. Retain a subsequent element G exactly when

$$(S8.5) \quad Q_N(G, C)^2 > \eta^2 \quad \text{for every previously retained element } C.$$

The retained elements, in their order of appearance, form $\mathcal{C}_{d,N,\delta}$. Every grid element either is retained or has a retained predecessor C with $Q_N(G, C) \leq \eta$. No triangle inequality for Q_N is used.

We first verify that the whole procedure is exact. Legality of a Gaussian-rational Hermitian matrix is decidable by exact Schur elimination, applied to G and $I - G$. A negative pivot excludes positive semidefiniteness. A zero pivot in a positive semidefinite matrix forces the corresponding row and column of the remaining matrix to vanish; otherwise the test rejects. A positive pivot reduces the question to the exact Schur complement. These rules terminate after at most d pivots,

including on singular strata. For a legal rational pair, its midpoint A and variance $A(I - A)$ are Gaussian rational. The equation

$$A(I - A)X + XA(I - A) + N^{-1}X = G - C$$

has an invertible Gaussian-rational coefficient matrix. Solving it gives $Q_N(G, C)^2 = N \operatorname{tr}((G - C)X) \in \mathbb{Q}_{\geq 0}$. Consequently (S8.5) is an exact rational comparison. It requires neither spectral coordinates nor a numerical eigenvalue threshold.

To encode E , first form the inward displacement

$$E_s = (1 - 2s)E + sI.$$

Since $s \leq 1/16$, its spectrum lies in $[s, 1 - s]$. Round each independent real coordinate of E_s to its nearest multiple of $1/K$, choosing the larger multiple at a tie, and call the resulting Hermitian matrix $G(E)$. A diagonal entry changes by at most $1/(2K)$ and the modulus of an off-diagonal entry changes by at most $\sqrt{2}/(2K)$. Thus

$$\|G(E) - E_s\|_{\text{op}} \leq \frac{d}{K}.$$

In particular,

$$(S8.6) \quad \frac{d}{K}I \preceq G(E) \preceq \left(1 - \frac{d}{K}\right)I.$$

Every coordinate lies in the prescribed grid range, and hence $G(E) \in \mathcal{G}_{d,K}$. Moreover,

$$(S8.7) \quad \|E - G(E)\|_{\text{op}} \leq s + \frac{d}{K} = \frac{3d}{K} \leq \frac{3\delta}{32N}.$$

Choose the first retained centre with $Q_N(G(E), C_j)^2 \leq \eta^2$. Such a centre exists by the completed greedy traversal. The hybrid inequality and Theorem S4.1 yield

$$\begin{aligned} d_N(\mathcal{M}_E, \mathcal{M}_{C_j}) &\leq 2N\|E - G(E)\|_{\text{op}} + 8Q_N(G(E), C_j) \\ &\leq \frac{3\delta}{16} + \frac{\delta}{2} = \frac{11\delta}{16}. \end{aligned}$$

This proves (S8.1). If E is Gaussian rational, the prescribed rounding and every subsequent operation are exact finite rational calculations.

For two distinct retained centres, Theorem S4.1 gives

$$(S8.8) \quad d_N(\mathcal{M}_C, \mathcal{M}_{C'}) \geq \frac{1}{8192d} \min\{1, Q_N(C, C')\} > \frac{\delta}{131072d},$$

where $\delta \leq 1$ was used. Let a_d, r_d be the constants of Lemma S5.2, and put

$$b_d = \min\left\{r_d, \frac{1}{393216d}\right\}, \quad r = b_d\delta.$$

The operational balls of radius r about distinct retained centres are disjoint, by (S8.8) and the triangle inequality for d_N . Their measures are each at least $a_d(b_d\delta)^{d^2}$. Hence

$$Ma_d(b_d\delta)^{d^2} \leq \mu_N(\mathfrak{E}_d).$$

Proposition S5.3 proves (S8.2).

Both parties reconstruct this same finite list from (d, N, δ) . The encoder transmits the binary representation of j in $\lceil \log_2 M \rceil$ bits. The decoder reconstructs the list and returns C_j for $j < M$; every unused word of the same length is assigned the legal effect $I/2$. The codeword contains no additional centre coordinates. Equation (S8.2) gives the upper bound in (S8.3); the covering lower bound of Theorem S5.1 gives the matching lower bound, since this list is a radius- δ cover by (S8.1). The description of the fixed reconstruction algorithm contributes only a constant. As

usual, if the parameters are not public, their explicitly chosen header representation must be charged separately. $\square$

Real input and finite precision. The covering statement concerns the entire real effect body. To interpret its encoder computationally on a real input, a representation of that input must be supplied. A certified rational approximation of each independent real coordinate to error at most $1/(8K)$ is sufficient. Apply the inward displacement to this approximation and then use the same exact nearest-grid rounding. Relative to the true E_s , each independent coordinate then has error at most $5/(8K)$; each complex off-diagonal error is at most $5\sqrt{2}/(8K) < 1/K$. Therefore (S8.6)–(S8.7) and the full error bound remain valid. The exact supplied-matrix implementation takes Gaussian-rational input; no physical-device learning operation is implicit in that interface. Computational scope. The ambient grid contains at most

$$(S8.9) \quad (K+1)^d(2K+1)^{d(d-1)}$$

candidate coordinate tuples. The literal algorithm performs exact legality tests and, for each legal tuple, comparisons with the retained list. Retaining the list uses $O(Md^2)$ rational entries, and the decoder repeats the same construction. These are finite bounds, but their dependence on the requested horizon and precision prevents an efficiency assertion in the payload length. The exact kernel `matrix_codec.py` implements this specified enumeration, greedy test, rounding, and replay. A resource cutoff is reported as an incomplete construction and never as a codebook. Regression checks of small complete cases and of matrix kernels certify those computations; they do not assert that a full large-dimensional dictionary has been enumerated.

Corollary S8.2 (Learning into an optimal-order description). *For every fixed $d \geq 1$, every integer $N \geq 1$, every rational $0 < \delta \leq \min\{1, \delta_0, \delta_d\}$, and every $0 < \eta \leq 1/8$, there is one learner, common to all $E \in \mathfrak{E}_d$, which returns a fixed-length word whose deterministic public decoder produces a legal effect C with*

$$\sup_{E \in \mathfrak{E}_d} \mathbb{P}_E\{d_N(\mathcal{M}_E, \mathcal{M}_C) > \delta\} \leq \eta.$$

The training budget on every record is at most $Cd^4N\delta^{-2}\log(d/\eta)$, and each entangled block has at most N calls. The word has length

$$(S8.10) \quad \frac{d^2}{2} \log_2 N + \lfloor d/2 \rfloor \log_2 \log(N+2) + d^2 \log_2(1/\delta) + O_d(1).$$

For fixed d , both the training order $N\delta^{-2}\log(1/\eta)$ and the payload order in (S8.10) are optimal. Payload optimality is among fixed-length words with a fixed public decoder into legal memoryless ordered binary effects. Encoding after learning uses no additional calls to the unknown device.

Proof. Apply Theorem S7.1 with accuracy $\delta/2$. It returns a legal effect F with rational entries in the public input basis. Encode F using Theorem S8.1 with accuracy $\delta/2$. On the learning success event, the triangle inequality gives

$$d_N(\mathcal{M}_E, \mathcal{M}_C) \leq \frac{\delta}{2} + \frac{11\delta}{32} = \frac{27\delta}{32} < \delta.$$

The second operation is a deterministic computation on the recorded estimate. It uses no device calls and does not increase the failure probability. Replacing δ by $\delta/2$ changes the training constant and adds only $O_d(1)$ to the payload length.

The scalar subfamily converse in Theorem S7.1 applies to this learning task as well. For the payload converse, a B -bit word and fixed decoder have at most 2^B possible decoded effects. Every E has a strictly positive success probability, so at least one of these effects lies within distance δ of E . The decoded family is therefore a radius- δ cover of $\mathfrak{E}_d$. The lower bound in Theorem S5.1 gives (S8.10) with the opposite inequality up to $O_d(1)$. $\square$

S9. THE COST OF LIMITING INDEPENDENT QUANTUM BLOCKS

The training law changes when the number of calls in one independent quantum experiment is prescribed. The relevant restriction concerns access to the next queried input. Quantum systems retained from completed experiments may still be stored and measured together. This distinction allows a direct comparison with tomography based on independently prepared probe–reference pairs.

Definition S9.1 (Independent blocks with retained outputs). A training protocol has independent blocks of length at most b if it has the following form. At a block boundary it may apply an arbitrary common quantum instrument to all its retained systems. The resulting classical history h , including its classical random choices and the number of calls already charged, determines whether to stop, or else a length $1 \leq t(h) \leq b$ and a fresh $t(h)$ -use tester. Conditional on h , this tester starts in a parameter-independent state tensor-separated from all retained systems. It may use arbitrary references, entangled inputs, and adaptive control within its $t(h)$ calls.

From its preparation until its last call, the entire fresh tester, including its reference and unused inputs, is isolated from the retained systems. Those systems are available again at the next block boundary, when the new outputs may be joined to them and processed collectively. Thus classical feedback may select the next tester, while no retained quantum register can be coupled into a block that still has calls to make. A block length is declared before that block and charged in full; an internal early exit is padded within its declared length. The sum of declared lengths is at most M on every record. Stopping at block boundaries may depend on the full classical history. The final estimate may use an arbitrary joint measurement of all retained systems. Classical record spaces are standard Borel and quantum registers are separable.

The retained quantum systems belong to the tester; the unknown measurement still consumes its input and returns only its ordered binary classical label. The definition permits an unbounded amount of retained quantum storage and classical computation. It also permits all probes to be prepared in advance when their states factor into the prescribed independent blocks: their preparation can be deferred without changing the experiment. In particular, independent one-call Choi preparations followed by a collective measurement belong to the case $b = 1$. The distinction between classical input control and quantum memory on the output side has an antecedent in the channel-discrimination model of [9, Section II.2]. The familiar quadratic information gain within a block and additive gain across independent blocks also occur in the metrological bounds of [33, 34]. We give a finite-budget fidelity argument because a local information bound alone does not supply the confidence dependence needed here.

Write $M_b^*(d, N, \delta, \eta)$ for the least every-record call budget of a protocol in Definition S9.1 whose legal estimate satisfies

$$(S9.1) \quad \sup_{E \in \mathfrak{E}_d} \mathbb{P}_E \{d_N(\mathcal{M}_E, \mathcal{M}_{\hat{E}}) > \delta\} \leq \eta.$$

The future distance d_N retains its unrestricted adaptive definition and its unhalved trace-norm normalization. The restriction on b applies only to training.

Theorem S9.2 (Sharp tradeoff for independent training blocks). *There are absolute constants $c, C, \delta_* > 0$ such that, for every $d \geq 2$, $N \geq 1$, $1 \leq b \leq N$, $0 < \delta \leq \delta_*$, and $0 < \eta \leq 1/8$,*

$$(S9.2) \quad c \frac{N^2}{b\delta^2} \log \frac{1}{\eta} \leq M_b^*(d, N, \delta, \eta) \leq C d^4 \frac{N^2}{b\delta^2} \log \frac{d}{\eta}.$$

The upper bound is attained by one procedure common to the entire closed effect body. The lower bound allows classical feedback, arbitrary quantum storage and collective processing at block

boundaries, and every-record bounded stopping. Consequently, for each fixed $d \geq 2$,

$$(S9.3) \quad M_b^*(d, N, \delta, \eta) = \Theta_d \left(\frac{N^2}{b\delta^2} \log \frac{1}{\eta} \right).$$

In dimension $d = 1$ the corresponding order is $\Theta(N\delta^{-2} \log(1/\eta))$, independently of b .

The endpoints $b = 1$ and $b = N$ differ by a factor of order N . The claim is uniform in the block cap, horizon, accuracy, and confidence; the upper factor d^4 is not asserted to be sharp in growing dimension.

Lemma S9.3 (Changing the future horizon). *For all effects E, F and integers $N, b \geq 1$,*

$$(S9.4) \quad d_N(\mathcal{M}_E, \mathcal{M}_F) \leq \left\lceil \frac{N}{b} \right\rceil d_b(\mathcal{M}_E, \mathcal{M}_F).$$

Proof. Fix an N -use tester and partition its calls into consecutive groups of at most b . Replace the device in one group at a time. For each replacement the common prefix prepares a reference-assisted input to a tester with at most b calls, including all memory carried from the prefix. The common suffix is a channel on that tester's output and cannot increase trace distance. Each replacement therefore costs at most d_b . Sum these costs and take the supremum over the original tester. Public stopping is included by padding to N calls with discarded dummy experiments. $\square$

Upper bound in Theorem S9.2. Put $q = \lceil N/b \rceil$. Apply Theorem S7.1 with future horizon b , accuracy δ/q , and failure allowance η . Its construction uses fresh independent blocks of length at most b , with classical records between them, so it satisfies Definition S9.1. Lemma S9.3 converts its success event into (S9.1). On every record its call count is at most

$$Cd^4 b q^2 \delta^{-2} \log(d/\eta) \leq 4Cd^4 \frac{N^2}{b\delta^2} \log(d/\eta),$$

because $1 \leq b \leq N$. Choose δ_* no larger than the accuracy constant in Theorem S7.1. $\square$

For positive, possibly subnormalized operators define root fidelity by

$$\mathfrak{f}(A, B) = \|\sqrt{A}\sqrt{B}\|_1.$$

We use its tensor-product identity, its additivity on direct sums, and its monotonicity under a common channel. In particular, for a common instrument $(\mathcal{J}_z)_z$,

$$(S9.5) \quad \sum_z \mathfrak{f}(\mathcal{J}_z(A), \mathcal{J}_z(B)) \geq \mathfrak{f}(A, B).$$

These facts hold for subnormalized operators by homogeneity. For states, the Bures angle $\arccos \mathfrak{f}$ is a contractive metric; see [1, 25].

Lemma S9.4 (Fidelity with adaptive choices of fresh blocks). *Fix a two-dimensional subspace and, for $\phi \in \mathbb{R}$, set*

$$(S9.6) \quad E_\phi = \frac{1}{2}(I_2 + \cos \phi \sigma_1 + \sin \phi \sigma_2) \oplus \frac{1}{2}I_{d-2},$$

with the second summand absent when $d = 2$. If $b|\theta| \leq 1$, the final states Γ_0, Γ_θ of any protocol in Definition S9.1, applied to E_0, E_θ with every-record budget M , satisfy

$$(S9.7) \quad \mathfrak{f}(\Gamma_0, \Gamma_\theta) \geq \exp(-Mb\theta^2/4).$$

Furthermore, if $N|\theta| \leq \pi$, then

$$(S9.8) \quad d_N^{\text{na}}(\mathcal{M}_{E_0}, \mathcal{M}_{E_\theta}) = d_N(\mathcal{M}_{E_0}, \mathcal{M}_{E_\theta}) = 2 \sin(N|\theta|/2).$$

Proof. For qubits, (S9.8) follows from [16, Corollary 1 and Theorem 2]. We give the calculation to specify the normalization and the fidelity estimate needed below.

Let $U_\theta = e^{-i\theta\sigma_3/2} \oplus I_{d-2}$. Then $E_\theta = U_\theta E_0 U_\theta^*$ and $\mathcal{M}_{E_\theta} = \mathcal{M}_{E_0} \circ \text{Ad}_{U_\theta^*}$. If $|\theta| \leq \pi$, every purified input ψ , including an arbitrary reference, satisfies

$$|\langle \psi, (U_\theta^* \otimes I) \psi \rangle| \geq \text{Re} \langle \psi, (U_\theta^* \otimes I) \psi \rangle \geq \cos(|\theta|/2).$$

The last inequality follows from the spectrum of $(U_\theta + U_\theta^*)/2$. Applying the common measurement and discarding a purification can only increase fidelity. Thus one call changes the output Bures angle by at most $|\theta|/2$, uniformly over its input and reference.

For a common adaptive tester, apply the triangle inequality before each query: the angle between the two current states is contracted by the same channel, and changing the channel on the second state adds at most $|\theta|/2$. The interposed controls are common channels and also contract the angle. Induction from the common initial state proves that every t -use tester has output states ρ_0, ρ_θ satisfying

$$(S9.9) \quad \mathfrak{f}(\rho_0, \rho_\theta) \geq \cos(t|\theta|/2) \quad \text{when } t|\theta| \leq \pi.$$

This argument permits arbitrary adaptive quantum memory inside the tester. Its output includes all internal outcomes before any postselection. For $t \leq b$ and $b|\theta| \leq 1$, $\cos x \geq e^{-x^2}$ for $0 \leq x \leq 1/2$ gives

$$(S9.10) \quad \mathfrak{f}(\rho_0, \rho_\theta) \geq e^{-t^2\theta^2/4} \geq e^{-at}, \quad a = b\theta^2/4.$$

We next retain the whole classical history, including all branch lengths and stopping decisions. At a boundary write the two classical-quantum states as direct sums of subnormalized retained states A_h, B_h . Their traces include the respective probabilities of h . Let $m(h)$ be the charged calls in that history, and consider

$$(S9.11) \quad \Psi = \sum_h e^{-a(M-m(h))} \mathfrak{f}(A_h, B_h).$$

Initially $\Psi = e^{-aM}$, since the initial resources are the same. A common boundary instrument preserves $m(h)$ along each of its children, so (S9.5) shows that it cannot decrease Ψ . The history is kept until the end; in particular, records with different charged call counts are never merged in this calculation.

At a continuing history h , the prescribed fresh tester has the same specification under both hypotheses. Conditional on h , its output states ρ_0^h, ρ_θ^h are normalized and are tensor-separated from the old retained state. Immediately after this block the contribution of this history is therefore

$$\begin{aligned} & e^{-a(M-m(h)-t(h))} \mathfrak{f}(A_h \otimes \rho_0^h, B_h \otimes \rho_\theta^h) \\ & \geq e^{-a(M-m(h)-t(h))} e^{-at(h)} \mathfrak{f}(A_h, B_h) = e^{-a(M-m(h))} \mathfrak{f}(A_h, B_h). \end{aligned}$$

The subsequent joint instrument is covered by (S9.5). A stopped history is carried unchanged. Since each continuing block charges at least one call, all histories are terminal after at most M such generations. At that time all weights in (S9.11) are at most one. Additivity of root fidelity on the terminal histories therefore gives

$$\mathfrak{f}\left(\bigoplus_h A_h, \bigoplus_h B_h\right) = \sum_h \mathfrak{f}(A_h, B_h) \geq \Psi \geq e^{-aM}.$$

Any final collective processing or discarding of history again increases fidelity. This proves (S9.7) without asserting independence of the unconditional records or imposing a common support condition on their laws.

For general measurable classical outcomes, choose a measure dominating the two history laws and interpret A_h, B_h as their operator-valued densities. Direct-sum additivity becomes the corresponding integral identity, and the same proof uses integrals in (S9.5) and (S9.11). These identities follow equivalently by finite coarse-graining of the classical record and refinement. The charged call count has only the values $0, \dots, M$ and is retained in every such coarse-graining.

Finally, use the first two coordinates for a parallel N -input state

$$\frac{|0\rangle^{\otimes N} + e^{i\alpha}|1\rangle^{\otimes N}}{\sqrt{2}}, \quad \alpha = N\theta/2 + \pi/2.$$

The parity of the ordered output signs has mean $\cos(N\phi - \alpha)$ under E_ϕ , hence means $-\sin(N\theta/2)$ and $\sin(N\theta/2)$ under the two hypotheses. Its unhalved distance is $2|\sin(N\theta/2)|$. For $N|\theta| \leq \pi$, apply (S9.9) with $t = N$ to any future adaptive tester and use $\|\rho - \sigma\|_1 \leq 2\sqrt{1 - \mathfrak{f}(\rho, \sigma)^2}$. The resulting upper bound is $2\sin(N|\theta|/2)$, equal to the parallel witness. This proves (S9.8). $\square$

Lower bound in Theorem S9.2. Reduce δ_* further so that $\delta_* \leq 1/4$, and set $\theta = 4\delta/N$ in (S9.6). Then $b|\theta| \leq 1$, while (S9.8) gives

$$d_N(\mathcal{M}_{E_0}, \mathcal{M}_{E_\theta}) = 2\sin(2\delta) > 2\delta.$$

The two radius- δ success balls are disjoint. Thus any estimate satisfying (S9.1) determines a test with both error probabilities α, β at most η : choose the first hypothesis precisely when the estimate lies in its radius- δ ball. The classical root fidelity of that test is

$$\sqrt{(1-\alpha)\beta} + \sqrt{\alpha(1-\beta)} \leq 2\sqrt{\eta}.$$

Fidelity is nondecreasing under the map from the training output to this test. Lemma S9.4 consequently implies

$$e^{-Mb\theta^2/4} \leq 2\sqrt{\eta}, \quad M \geq \frac{N^2}{8b\delta^2} \log \frac{1}{4\eta} \geq \frac{N^2}{24b\delta^2} \log \frac{1}{\eta},$$

where the last inequality uses $\eta \leq 1/8$. This proves the asserted lower bound in every $d \geq 2$.

For $d = 1$ every effect is scalar. Bernoulli endpoint estimation in Lemma S6.2 gives the upper order with independent one-call samples. The scalar converse in Theorem S7.1 gives the matching lower order even with unrestricted adaptive quantum training. Thus increasing b has no effect in dimension one. $\square$

Corollary S9.5 (The limitation of independent one-call acquisition). *Fix $d \geq 2$. Every learner that acquires its data using fresh one-call probes and fresh references, with classical feedback and arbitrary subsequent collective processing as in Definition S9.1, requires $\Omega_d(N^2\delta^{-2} \log(1/\eta))$ calls for (S9.1). The complete closed effect body admits the same risk with $O_d(N\delta^{-2} \log(1/\eta))$ calls when independent blocks of length up to N are permitted.*

Proof. Use $b = 1$ and $b = N$ in Theorem S9.2. $\square$

This conclusion applies to the acquisition interfaces of [5, 29]. The channel learner in [5, Section III.2 and Algorithm I.2] prepares independent Choi copies before their collective processing. The measurement tomography procedure in [29, Section 3] queries separately prepared known inputs. Both are contained in $b = 1$ above. Any estimator retaining these interfaces and required to satisfy the present worst-case future- N loss therefore incurs the quadratic horizon order, including when its final estimator is collective. The block lower bound settles this interface comparison directly; it does not rely on substituting δ/N into a sufficient one-use accuracy guarantee.

Corollary S9.6 (Learning with bounded blocks into an optimal-order word). *For fixed $d \geq 2$, $1 \leq b \leq N$, rational $0 < \delta \leq \min\{1, \delta_*, \delta_d\}$, and $0 < \eta \leq 1/8$, there is a common learner in*

Definition S9.1 that returns a fixed-length word whose deterministic public decoder produces a legal effect C with

$$\sup_{E \in \mathfrak{E}_d} \mathbb{P}_E \{d_N(\mathcal{M}_E, \mathcal{M}_C) > \delta\} \leq \eta.$$

Its every-record training budget is at most $Cd^4N^2(b\delta^2)^{-1} \log(d/\eta)$ and its payload has length

$$(S9.12) \quad \frac{d^2}{2} \log_2 N + \lfloor d/2 \rfloor \log_2 \log(N+2) + d^2 \log_2(1/\delta) + O_d(1).$$

For each fixed d , both the training order in this block model and the payload order among fixed public decoders into legal effects are optimal. Encoding uses no additional device calls.

Proof. Run the learner of Theorem S9.2 at accuracy $\delta/2$. The upper construction inherits the legal rational output of Theorem S7.1. Apply the exact codec of Theorem S8.1 at accuracy $\delta/2$. On the learning success event its decoded effect obeys

$$d_N(\mathcal{M}_E, \mathcal{M}_C) \leq \delta/2 + 11\delta/32 = 27\delta/32 < \delta.$$

The codec is a deterministic computation on the recorded rational estimate. Theorem S8.1 gives (S9.12). The training converse is Theorem S9.2. For the payload converse, positive success probability for every E forces the decoded family to cover $\mathfrak{E}_d$ by radius- δ balls; apply Theorem S5.1. As before, reconstruction time, workspace, and state-preparation complexity are separate from the two resource counts. $\square$

S10. DIMENSION AND ACCURACY IN THE UNIFORMLY NOISY INTERIOR

The block converse concerns projective boundary effects. A different obstruction is present in the full-dimensional interior: an effect close to a fair coin supplies little information per call, even when the learner retains coherent quantum memory. We quantify this obstruction before analysing a one-use estimator under the future loss. Entropies and relative entropies in this section use natural logarithms. The quantum information identities used below are in [1]; the particular conditional bound is proved here.

Lemma S10.1 (Information supplied by a weak binary measurement). *Let X have a finite alphabet. Conditional on $X = x$, an input register A and a retained register R have an arbitrary joint state ρ_{AR}^x . Apply the consuming binary measurement with effect $E_x = I/2 + H_x$, where $\|H_x\|_{\text{op}} \leq r < 1/2$, to A , and call its classical outcome Y . Then*

$$(S10.1) \quad I(X : YR)_{\text{out}} \leq I(X : AR)_{\text{in}} + \log(1 + 4r^2) \leq I(X : AR)_{\text{in}} + 4r^2.$$

The input states need not be independent of x or of one another.

Proof. Fix x , abbreviate $\rho = \rho_R^x$, and let Δ be the partial trace of $(H_x \otimes I)\rho_{AR}^x$ over A . Positivity of $rI \pm H_x$ gives

$$-r\rho \preceq \Delta \preceq r\rho, \quad \omega_{YR}^x = |1\rangle\langle 1| \otimes (\rho/2 + \Delta) + |0\rangle\langle 0| \otimes (\rho/2 - \Delta).$$

On the support of ρ , write $\Delta = \rho^{1/2}K\rho^{1/2}$ with $\|K\|_{\text{op}} \leq r$. Set $\sigma = (I_Y/2) \otimes \rho$. The support inclusion $\text{supp } \omega^x \subseteq \text{supp } \sigma$ holds, and direct multiplication gives

$$\begin{aligned} \text{tr}[(\omega^x)^2 \sigma^{-1}] &= 1 + 4 \text{tr}(\Delta^2 \rho^{-1}) \\ &= 1 + 4 \text{tr}(\rho K^2) \leq 1 + 4r^2. \end{aligned}$$

All inverses in this calculation are support inverses. For density matrices a, b with the same support inclusion, $D(a||b) \leq \log \text{tr}(a^2 b^{-1})$ follows directly from scalar Jensen: in eigenbases of a and b , use the probability weights $a_i |\langle u_i, v_j \rangle|^2$ and apply the concavity of $\log$ to a_i/b_j . Consequently

$$D(\omega_{YR}^x || (I_Y/2) \otimes \rho_R^x) \leq \log(1 + 4r^2).$$

Average over x . Subtracting $D(\omega_{YR} \| (I_Y/2) \otimes \omega_R)$ from the average on the left gives exactly $I(X : Y | R)_{\text{out}}$. Relative entropy is nonnegative. The retained marginal is unchanged by the call, so data processing gives $I(X : R)_{\text{out}} = I(X : R)_{\text{in}} \leq I(X : AR)_{\text{in}}$. Adding these two inequalities proves the claim. For a separable retained register, apply finite-rank compression to R and pass to the limit. The bound is uniform, and mutual information with the finite classical register X is continuous under trace-norm convergence. $\square$

The same calculation applies to any finite classical-output measurement uniformly close to a scalar one. More precisely, if $p_y > 0$, $\sum_y p_y = 1$, and $(1-r)p_y I \preceq E_{x,y} \preceq (1+r)p_y I$, its per-call information increment is at most $\log(1+r^2)$. Indeed, the corresponding collision expression is $1 + \sum_y p_y^{-1} \text{tr}(\Delta_y^2 \rho^{-1}) \leq 1 + r^2$; the linear terms cancel because $\sum_y \Delta_y = 0$. We use only the binary case in what follows.

Theorem S10.2 (A joint dimension and future-loss lower bound). *There is an absolute $c > 0$ such that, for every $d, N \geq 1$, $0 < \delta \leq 2^{-13}$ and $0 < \eta \leq 1/8$, a common learner with*

$$\sup_{I_d/4 \preceq E \preceq 3I_d/4} \mathbb{P}_E\{d_N(\mathcal{M}_E, \mathcal{M}_{\widehat{E}}) > \delta\} \leq \eta$$

requires at least

$$(S10.2) \quad M \geq cd^2 N \delta^{-2}$$

calls in its worst record. The learner may retain arbitrary quantum memory, entangle its training calls, use coherent adaptive controls, and stop publicly. Its estimate may be any legal binary effect; it need not lie in the interior subfamily.

Proof. Put $r = 2048\delta/\sqrt{N} \leq 1/4$. The real vector space of Hermitian matrices has dimension $p = d^2$. A maximal finite $r/4$ -separated set $\{H_x\}_{x=1}^L$ in its operator-norm ball of radius r covers that ball by translates of the radius- $r/4$ ball. Comparing Lebesgue volumes gives $L \geq 4^p$. Such a finite maximal set exists by compactness. Set $E_x = I/2 + H_x$, so every E_x lies between $I/4$ and $3I/4$.

For $x \neq z$, take a unit eigenvector of $H_x - H_z$ achieving its operator norm and repeat this input N times. The two outcome laws are Bernoulli products with parameters $s, t \in [1/4, 3/4]$ and $|s - t| \geq r/4$. Since

$$T_N(s, t) = |s - t| \sqrt{\frac{N}{\max\{s(1-s), t(1-t)\} + N^{-1}}} \geq \frac{1}{2} \sqrt{N} |s - t| \geq 256\delta,$$

Lemma S16.2 and $256\delta \leq 1$ give

$$(S10.3) \quad d_N(\mathcal{M}_{E_x}, \mathcal{M}_{E_z}) \geq 4\delta.$$

The witness may depend on the pair; it is used only to establish separation, not as a common learning rule.

Choose X uniformly from this packing. The learner's initial resources are independent of X . Common intervening channels cannot increase their mutual information with X . Lemma S10.1 applies to each call even when its input and memory already depend on X through previous calls. Pad a stopped experiment by ignored dummy calls to its worst-record budget M . Its final quantum and classical registers therefore have mutual information at most $4Mr^2$ with X .

From the returned effect choose a nearest member of the finite packing, with a fixed tie rule. This is a statistical testing reduction; no computation of an adaptive optimum is required from the learner. The selection is measurable because the hybrid bound makes d_N continuous in each effect. The disjointness in (S10.3) makes the chosen member correct whenever the learning loss is at most δ . Fano's inequality and data processing yield

$$4Mr^2 \geq (1 - \eta) \log L - \log 2 \geq \frac{3}{4} d^2 \log 2.$$

Substituting r proves (S10.2). $\square$

Corollary S10.3 (Joint dimension and accuracy in one-use learning). *At failure probability at most $1/8$, the minimax number of calls needed to learn an arbitrary ordered binary effect to operator-norm error ϵ is*

$$\Theta(d^2\epsilon^{-2}), \quad d \geq 1, \quad 0 < \epsilon \leq 2^{-14},$$

with absolute constants. The lower bound permits arbitrary coherent adaptive training. The upper bound uses independent Choi probes and collective processing of their outputs.

Proof. For binary measurements $d_1(\mathcal{M}_E, \mathcal{M}_F) = 2\|E - F\|_{\text{op}}$, including retained references; see [5, Lemmas III.7–III.8]. Theorem S10.2 with $N = 1$ and $\delta = 2\epsilon$ gives the lower bound. The upper bound is [5, Corollary III.9], at $\eta = 1/8$. Its stated condition $\eta > 4\exp(-4d^2)$ holds for every $d \geq 1$. Thus the lower argument couples dimension and accuracy, while the upper algorithm is the existing one-use estimator. $\square$

Lemma S10.4 (Dimension-free interior comparison). *If $I_d/8 \preceq E, F \preceq 7I_d/8$, then*
(S10.4)

$$\frac{1}{128} \min\{1, \sqrt{N}\|E - F\|_{\text{op}}\} \leq d_N^{\text{na}}(\mathcal{M}_E, \mathcal{M}_F) \leq d_N(\mathcal{M}_E, \mathcal{M}_F) \leq \min\{2, 4\sqrt{N}\|E - F\|_{\text{op}}\}.$$

The constants are independent of d and N .

Proof. The same repeated eigenvector input and Lemma S16.2 give the lower bound. For the upper bound take $G(t) = E + t(F - E)$ and $H = F - E$. Along this path $S(t) = \sqrt{G(t)(I - G(t))} \succeq I/4$. The unregularized Sylvester equation $SK + KS = H$ has solution

$$K = \int_0^\infty e^{-uS} H e^{-uS} du, \quad \|K\|_{\text{op}} \leq 2\|H\|_{\text{op}}.$$

The horizontal calculation (S4.12) bounds the derivative of each fixed common tester's output by $2\sqrt{N}\|K\|_{\text{op}}$. Integrate on $[0, 1]$ and then take the supremum over testers. This proves the upper bound using the same horizontal Kraus and adaptive path methods as Lemma S4.2; see also [22, 23, 24, 26]. $\square$

Theorem S10.5 (Sharp learning on a fixed full-dimensional interior). *For the family $I_d/4 \preceq E \preceq 3I_d/4$, the minimax training cost for future- N loss δ and failure at most $1/8$ is*

$$(S10.5) \quad \Theta(d^2 N \delta^{-2}), \quad d, N \geq 1, \quad 0 < \delta \leq 2^{-13}.$$

The constants are absolute. The upper procedure uses fresh one-call probes and permits collective final processing. The lower bound permits arbitrary coherent adaptive training. Thus the same order holds for every fresh-block cap $1 \leq b \leq N$.

Proof. Theorem S10.2 supplies the lower bound on this very family. For the upper bound apply the binary estimator of [5, Corollary III.9] at operator-norm accuracy $\epsilon = \delta/(8\sqrt{N})$ and failure $1/8$. It uses $O(d^2 N \delta^{-2})$ calls and returns a legal effect F . On its successful event $\|E - F\|_{\text{op}} \leq \epsilon \leq 1/8$, so both endpoints lie in $[I/8, 7I/8]$. Lemma S10.4 then gives $d_N(E, F) \leq \delta/2$. Independent input/reference pairs prepare the Choi copies; all coherent processing acts on completed outputs. This is a fresh one-call experiment in Definition S9.1. $\square$

This establishes a stronger future-loss analysis of an existing tomography estimator on a fixed interior. On the complete effect body, Corollary S9.5 supplies the different quadratic-horizon obstruction for the same acquisition class. The interior comparison is dimension-free, and the interior learning order is joint in d, N, δ ; neither assertion changes the dimension dependence or boundary range of the global comparison and covering theorems. In particular, for $d \geq 2$,

$1 \leq b \leq N$, $0 < \delta \leq \min\{2^{-13}, \delta_*\}$ and $0 < \eta \leq 1/8$, the full-body fresh-block learner has the two simultaneous lower bounds

$$M \geq c\delta^{-2} \left(d^2 N + \frac{N^2}{b} \log(1/\eta) \right),$$

after decreasing an absolute c , while its constructive upper bound remains $Cd^4(N^2/b)\delta^{-2}\log(d/\eta)$. The displayed d^4 upper factor is not matched by this lower bound.

S11. JOINT DIMENSION BOUNDS FOR OPERATIONAL DESCRIPTIONS

The dimension-free comparison of Lemma S10.4 allows a description theorem whose constants are uniform in the matrix dimension. This complements, rather than replaces, the closed-body law of Theorem S5.1. Put

$$\mathfrak{I}_d = \{E = E^* : I_d/4 \preceq E \preceq 3I_d/4\}, \quad p = d^2.$$

Throughout this section, centres are legal ordered binary effects and the operational distance is the unhalved distance d_N of the same consuming, classical-output interface. In particular, the future tester may retain a reference, use coherent adaptive controls, and stop at a public bounded time. All logarithms in description lengths have base two.

Theorem S11.1 (Uniform interior operational entropy). *Let $d, N \geq 1$ and $0 < \delta \leq 2^{-13}$. The least cardinality $\mathcal{C}_N(\mathfrak{I}_d, \delta)$ of a d_N -cover of $\mathfrak{I}_d$, allowing centres anywhere in $\mathfrak{E}_d$, satisfies*

$$(S11.1) \quad \left(\frac{\sqrt{N}}{2048\delta} \right)^p \leq \mathcal{C}_N(\mathfrak{I}_d, \delta) \leq \left(\frac{25\sqrt{N}}{\delta} \right)^p.$$

The upper bound is attained by a public finite exact rational construction when δ is rational. For real δ , a volumetric net gives the stated upper bound. Consequently the optimal deterministic fixed-length description into legal effects is

$$(S11.2) \quad B_{\text{int}}^*(d, N, \delta) = \frac{d^2}{2} \log_2 N + d^2 \log_2(1/\delta) + O(d^2),$$

where the remainder is uniform in all three parameters.

Proof. The real vector space of Hermitian matrices has dimension p , and $\mathfrak{I}_d$ is precisely the translate of its operator-norm ball of radius $1/4$. Set $s = 512\delta/\sqrt{N}$. A maximal subset $\{E_1, \dots, E_L\}$ of $\mathfrak{I}_d$ whose distinct points have operator distance strictly greater than s is finite and its closed s -balls cover $\mathfrak{I}_d$. Translation-invariant Lebesgue measure and homogeneity therefore give

$$Ls^p \geq (1/4)^p, \quad L \geq \left(\frac{\sqrt{N}}{2048\delta} \right)^p.$$

The unit-ball volume cancels; no dimension-dependent volume estimate is used. Lemma S10.4 separates any two selected effects in d_N by at least 4δ , since $512\delta \leq 1/16$. A closed operational ball of radius δ contains at most one selected effect, irrespective of its centre. This uses the triangle inequality for d_N , not a property of a comparison modulus.

For the upper bound choose a maximal operator-norm $\delta/(4\sqrt{N})$ -separated subset of $\mathfrak{I}_d$. It covers the latter at that operator radius and hence at operational radius δ . The disjoint open balls of half that radius lie inside the concentric ball of radius $1/4 + \delta/(8\sqrt{N})$. Their number is at most $(1 + 2\sqrt{N}/\delta)^p$, which is smaller than the asserted bound. The rational construction with a specified finite stopping rule is proved next. A fixed public B -bit decoder has at most 2^B outputs, and indexing a smallest finite cover suffices. Taking logarithms, with one bit for rounding to an integer, proves (S11.2). $\square$

S11.1. An exact public interior dictionary. Only the public parameters (d, N, δ) determine the dictionary. Let $\delta \in \mathbb{Q} \cap (0, 1]$ and set

$$(S11.3) \quad k = \lceil \sqrt{N} \rceil, \quad K = \left\lceil \frac{128dk}{\delta} \right\rceil, \quad t = \frac{\delta}{16k}.$$

The integer k is computed by an integer square-root test. Enumerate Hermitian matrices with diagonal coordinates in $\{0, 1/K, \dots, 1\}$ and real and imaginary off-diagonal coordinates in $\{-1, -1 + 1/K, \dots, 1\}$. Order coordinates as in Theorem S8.1 and retain only the matrices in $[I_d/8, 7I_d/8]$. From this ordered finite stream construct a list $\mathcal{D}$ greedily, accepting G precisely when $\|G - C\|_{\text{op}} > t$ for every previously accepted C . For a Hermitian rational difference H , the complementary test is

$$(S11.4) \quad \|H\|_{\text{op}} \leq t \iff tI_d - H \succeq 0 \text{ and } tI_d + H \succeq 0.$$

Thus exact rational positive-semidefinite tests settle the comparison, including equality, without computing eigenvalues. The Schur test of Theorem S8.1 applies to both inequalities.

Theorem S11.2 (Finite exact interior coding). *For every $d, N \geq 1$ and rational $0 < \delta \leq 1$, the preceding construction terminates and defines a common encoder and decoder. For a supplied Gaussian-rational $E \in \mathfrak{J}_d$, the decoded centre $C \in [I_d/8, 7I_d/8]$ satisfies*

$$(S11.5) \quad d_N(E, C) \leq 5\delta/16.$$

The same conclusion holds for a real $E \in \mathfrak{J}_d$ supplied with a Gaussian-rational Hermitian approximation A certified to satisfy $\|A - E\|_{\text{op}} \leq d/K$. The payload is an index of length $\lceil \log_2 |\mathcal{D}| \rceil$, and

$$(S11.6) \quad |\mathcal{D}| \leq (1 + 12k/\delta)^p \leq (25\sqrt{N}/\delta)^p.$$

All words of that length have legal decodings. No nonpublic matrix coordinates or target-dependent dictionary are free advice.

Proof. Round every real coordinate of A to the nearest multiple of $1/K$, with ties towards positive infinity, and preserve conjugate entries. Call the result G . In every row the diagonal rounding error is at most $1/(2K)$ and each off-diagonal error has modulus at most $1/(\sqrt{2}K)$. The Hermitian row-sum bound gives $\|G - A\|_{\text{op}} \leq d/K$. Therefore

$$\|G - E\|_{\text{op}} \leq 2d/K \leq \delta/(64k) \leq 1/64.$$

For rational E take $A = E$; the same, slightly looser, bound applies. Weyl's inequality puts G in $[I_d/8, 7I_d/8]$. Its diagonal and off-diagonal coordinates also lie in the enumerated ranges, so it occurs in the legal finite stream.

If G is retained, select $C = G$. Otherwise the strict greedy rule supplies an earlier retained centre with $\|G - C\|_{\text{op}} \leq t$; select the first such centre. Hence

$$\|E - C\|_{\text{op}} \leq \frac{\delta}{64k} + \frac{\delta}{16k} = \frac{5\delta}{64k}.$$

Both endpoints lie in the range of Lemma S10.4, which proves (S11.5). Legality and both selection tests use finite exact rational arithmetic.

Distinct dictionary elements have operator distance greater than t , and every element lies within operator radius $3/8$ of $I_d/2$. The open operator balls of radius $t/2$ about them are disjoint and lie in the ball of radius $3/8 + t/2$. Cancelling the common unit-ball volume gives $(1 + 3/(4t))^p = (1 + 12k/\delta)^p$. Since $k \leq 2\sqrt{N}$ and $\sqrt{N}/\delta \geq 1$, the second inequality in (S11.6) follows.

The decoder reconstructs the entire ordered list from the public parameters before interpreting its index. Unused fixed-length words return $I_d/2$. The list is nonempty because the rounded centre $I_d/2$ occurs. This proves termination and total legal decoding. A rational approximation

to a real input is supplied data under the displayed certificate, not an arbitrary exact-real oracle provided by the theorem. $\square$

The raw enumeration has at most $(K+1)^d(2K+1)^{d(d-1)}$ tuples. Exact comparisons against the retained list and the bit lengths of Schur-complement entries are additional reconstruction costs. They are not controlled by (S11.2). The reference implementation rejects a resource-limited incomplete enumeration without emitting a codebook or payload. Thus the theorem is a finite exact construction, not a polynomial-time coding result. Volumetric packing and rational nets are standard mechanisms; the dimension-uniform operational comparison is what transfers them to the present future-use loss.

S11.2. Randomized prefix descriptions. The fixed-length conclusion has a counterpart which permits shared randomness and counts expected transmitted length. This distinguishes an index-cardinality converse from a converse for a randomized communication protocol.

Definition S11.3 (One-way prefix description). The encoder is supplied $E \in \mathfrak{I}_d$. A public random variable R , independent of E , is available to both parties. Conditionally on each public seed, the possible finite binary messages W form a prefix-free set. Private randomization is allowed, and the decoder returns a legal effect C from (R, W) and its independent private randomness. No timing, stopping signal, nonpublic header, or other side channel is uncharged. The cost is $\sup_E \mathbb{E}_E |W|$ and the risk is $\sup_E \mathbb{P}_E \{d_N(E, C) > \delta\}$. The public parameters d, N, δ, η are not part of the message.

Theorem S11.4 (Randomized expected description length). *Let $d, N \geq 1$, $0 < \delta \leq 2^{-13}$ and $0 \leq \eta \leq 1/8$. Write*

$$A_d(N, \delta) = d^2 \log_2(\sqrt{N}/\delta), \quad h_2(\eta) = -\eta \log_2 \eta - (1 - \eta) \log_2(1 - \eta).$$

Here $0 \log_2 0 = 0$. The infimum $\mathcal{B}_\eta$ of the costs in Definition S11.3 with risk at most η satisfies

$$\begin{aligned} \text{(S11.7)} \quad (1 - \eta)(A_d(N, \delta) - 11d^2) - h_2(\eta) &\leq \mathcal{B}_\eta \\ &\leq (1 - \eta)(A_d(N, \delta) + d^2 \log_2 25 + 1). \end{aligned}$$

In particular, $\mathcal{B}_\eta = (1 - \eta)A_d(N, \delta) + O(d^2)$, with absolute uniform constants. For rational δ the achievable nonempty messages are supplied by the finite exact interior dictionary.

Proof. Use the 4δ -separated packing of Theorem S11.1 and let X be uniform on its L labels, independently of R . A decoded effect within δ of the true packing point identifies its label uniquely. Choose a nearest label with a fixed tie rule otherwise. This produces $\hat{X}$ with error probability at most η .

For completeness, if J indicates a classification error, the chain rule gives

$$H(X \mid \hat{X}, R) \leq H(J) + \mathbb{P}(J = 1) \log_2(L - 1) \leq h_2(\eta) + \eta \log_2 L.$$

The final inequality uses $\eta \leq 1/8$ and $L \geq 4$. Since X and R are independent, data processing gives

$$(1 - \eta) \log_2 L - h_2(\eta) \leq I(X; \hat{X} \mid R) \leq I(X; W \mid R) \leq H(W \mid R).$$

The decoder's private randomness is independent conditionally on (R, W) and does not increase this information. For each public seed, Kraft's inequality and nonnegativity of relative entropy give $H(W \mid R = r) \leq \mathbb{E}(|W| \mid R = r)$; integrating proves the same conditional inequality. These statements also apply to countable messages of finite expected length by truncation; infinite cost already satisfies the lower bound. Worst-target expected length is at least its average over the packing. The packing cardinality in (S11.1) now proves the first inequality. This is the usual Fano–Kraft argument, written here to include all the stated randomization and message conventions.

For achievability, use a public coin independent of the target. With probability η transmit the empty word and decode to $I_d/2$. With the complementary probability use the deterministic index code of Theorem S11.2, or an interior net for real δ . For each seed the permitted set is either the singleton empty word or the fixed-length index set, and hence is prefix-free. The decoder knows which case occurs from R , so there is no free transmitted abort flag. Failure can occur only on the public abort event. Equations (S11.5) and (S11.6), with integer length rounding, give the upper bound. The model permits the specified Bernoulli public coin; the zero-failure finite exact dictionary does not require it. $\square$

S11.3. Learning and transmitting at the joint rates.

Theorem S11.5 (Jointly optimal learned interior descriptions). *In the ideal-control query model of Theorem S10.5, let $d, N \geq 1$ and let $0 < \delta \leq 2^{-13}$ be rational. There is one common learner on $\mathfrak{I}_d$ with a finite binary output and a fixed public deterministic decoder such that*

$$(S11.8) \quad \sup_{E \in \mathfrak{I}_d} \mathbb{P}_E\{d_N(E, C) > \delta\} \leq 1/8, \quad M = O(d^2 N \delta^{-2}), \quad B = A_d(N, \delta) + O(d^2).$$

The constants are absolute. The acquisition uses independent one-call input/reference pairs, followed by collective processing of completed outputs. No device call is used for encoding or replay. Any learner with these error and decoder conventions has $M = \Omega(d^2 N \delta^{-2})$ and $B \geq A_d(N, \delta) - 11d^2$.

Proof. Apply the binary estimator of [5, Corollary III.9] at operator accuracy $\epsilon = \delta/(64k)$ and failure $1/8$. It returns a legal effect F with $\|F - E\|_{\text{op}} \leq \epsilon$ on its good event, using $O(d^2 k^2 \delta^{-2})$ calls. Let T be the spectral clipping of F to $[I/4, 3I/4]$. The good event and Weyl's inequality imply $\|T - F\|_{\text{op}} \leq \epsilon$, whence

$$\|T - E\|_{\text{op}} \leq 2\epsilon, \quad d_N(E, T) \leq 8\sqrt{N}\epsilon \leq \delta/8.$$

This argument does not assume that scalar clipping is operator-Lipschitz for a noncommuting pair.

Round the coordinates of T and encode their grid point by the finite dictionary of Theorem S11.2. Its proof gives $d_N(T, C) \leq 5\delta/16$ even though the displayed estimator may initially be real-valued. The resulting map from the original classical outcome to a dictionary index is measurable: spectral clipping is continuous, coordinate rounding is Borel with specified ties, and the final list is finite. Composing the final tomography POVM with this finite partition defines a common finite-output measurement. It needs neither an unknown-effect-dependent measurement nor an exact-real side channel to the decoder. Equivalently, certified finite coordinate approximations can be used with the rounding slack of the coding theorem when supplied by a classical implementation. On the good event the total operational error is at most $7\delta/16 < \delta$. Since $k^2 \leq 4N$, the call bound and the payload upper bound follow.

For the call converse, a decoded word is also an estimator, so Theorem S10.2 applies. For the length converse, positive success probability for every target implies that the finite range of the fixed deterministic decoder is a δ -cover. The first inequality of (S11.1) applies even when those centres lie outside $\mathfrak{I}_d$. $\square$

The last theorem is a query and transmitted-length statement. It inherits the ideal trusted tomography operations of [5]; the measurable finite readout above does not assert an efficient gate synthesis for them. The separate finite-control theorem for our full-body learners remains unchanged. The interior description bounds neither remove the endpoint logarithms on the complete body nor optimize its d^4 learning upper bound. Rather, on the same full-dimensional interior, they identify both training and description costs with constants uniform in dimension, horizon and accuracy.

S12. THE JOINT COST OF DIMENSION, HORIZON, AND CONFIDENCE

The fixed-confidence interior law has a sharp extension to arbitrary confidence. The dependence on dimension and on the confidence logarithm is additive after their respective dimensional factors are included. In this section the unknown measurement is still ordered, binary, consuming, and classical-output. Put $\mathfrak{I}_d = \{E : I_d/4 \preceq E \preceq 3I_d/4\}$ as in Section S11. Logarithms in call bounds are natural unless indicated otherwise.

Theorem S12.1 (Uniform interior learning at every confidence). *There are absolute $c, C > 0$ such that, for every $d, N \geq 1$, $0 < \delta \leq 2^{-13}$, and $0 < \eta \leq 1/8$, the minimax every-record call budget for*

$$(S12.1) \quad \sup_{E \in \mathfrak{I}_d} \mathbb{P}_E \{d_N(\mathcal{M}_E, \mathcal{M}_{\hat{E}}) > \delta\} \leq \eta$$

is between

$$(S12.2) \quad c \frac{N}{\delta^2} \left(d^2 + d \log \frac{1}{\eta} \right) \quad \text{and} \quad C \frac{N}{\delta^2} \left(d^2 + d \log \frac{1}{\eta} \right).$$

The lower bound permits arbitrary coherent adaptive training, references, quantum memory, and public bounded stopping. The estimate may be any legal effect. The upper bound uses independent one-call Choi preparations and collective processing of completed outputs. Consequently the same order, with absolute constants, holds for every fresh-block cap $1 \leq b \leq N$.

The proof combines the dimension lower bound of Theorem S10.2 with a separate confidence obstruction and an amplification of the estimator in [5]. The latter procedure is the tomography primitive throughout; the amplification uses its dimension-dependent tail guarantee.

Lemma S12.2 (Confidence requires observations in every coordinate). *In the parameter range of Theorem S12.1, every learner satisfying (S12.1) has*

$$(S12.3) \quad M \geq \frac{dN}{6 \cdot 512^2 \delta^2} \log \frac{1}{\eta}.$$

This bound already holds when the unknown effects are diagonal in a fixed known basis.

Proof. Set $\Delta = 512\delta/\sqrt{N} \leq 1/16$ and consider

$$E_0 = I/2, \quad E_i = I/2 + \Delta|i\rangle\langle i|, \quad 1 \leq i \leq d.$$

All these effects belong to $\mathfrak{I}_d$. For each pair (E_0, E_i) , repeating the input $|i\rangle$ gives Bernoulli parameters $1/2$ and $1/2 + \Delta$. The comparison quantity in Lemma S16.2 satisfies

$$T_N \geq \frac{1}{2} \sqrt{N} \Delta = 256\delta.$$

That lemma, together with $256\delta \leq 1$, therefore gives

$$(S12.4) \quad d_N(\mathcal{M}_{E_0}, \mathcal{M}_{E_i}) \geq 4\delta.$$

We bound the information used to distinguish all these alternatives from the same null effect. For a diagonal effect with entries $p_1, \dots, p_d$, the reference-assisted output is unchanged if the input is first measured in the known basis, yielding $J = j$, and a Bernoulli bit of parameter p_j is then drawn. We retain the basis label in an extra register for the analysis. Discarding it gives the original device, so any original learner, including its final estimate, is recovered by ignoring these extra registers. This is the diagonal classicalisation used in [5, Lemma IV.19]; the following relative-entropy calculation also treats different quantum inputs under the two hypotheses.

For an input-reference state ρ_{AR} , write $\tau_j = \langle j|\rho_{AR}|j\rangle$. The revealed output is

$$\widetilde{\mathcal{M}}_p(\rho_{AR}) = \sum_{j=1}^d |j\rangle\langle j| \otimes ((1-p_j)|0\rangle\langle 0| + p_j|1\rangle\langle 1|) \otimes \tau_j.$$

Let ρ_0, ρ_i be arbitrary current input-reference states, where the reference includes all retained memory. The common channel $\rho \mapsto \sum_j |j\rangle\langle j| \otimes \tau_j$ cannot increase relative entropy. Appending the coin conditional on j adds its classical relative entropy with weight $\text{tr } \tau_j^0$. Thus the classical-quantum relative-entropy identity gives

$$(S12.5) \quad \begin{aligned} D(\widetilde{\mathcal{M}}_0(\rho_0) \parallel \widetilde{\mathcal{M}}_i(\rho_i)) &\leq D(\rho_0 \parallel \rho_i) + \kappa_\Delta \text{tr } \tau_i^0, \\ \kappa_\Delta &= D(\text{Bern}(1/2) \parallel \text{Bern}(1/2 + \Delta)) \\ &= -\frac{1}{2} \log(1 - 4\Delta^2) \leq 3\Delta^2. \end{aligned}$$

The subscript on $\widetilde{\mathcal{M}}$ now denotes the hypothesis. This calculation uses no equality or independence assumption for ρ_0, ρ_i . Starting from the common initial state, it also establishes finite relative entropy at every stage.

Keep each revealed label without feeding it into the original controls. All intervening operations, including classically controlled or coherent ones, are common channels. Pad a stopped record to its every-record cap M using ignored dummy calls. Let T_i count the revealed labels equal to i among these M calls, and let Ω_0, Ω_i denote the complete final states with the extra labels retained. Iterating (S12.5) yields

$$D(\Omega_0 \parallel \Omega_i) \leq 3\Delta^2 \mathbb{E}_0 T_i, \quad \sum_{i=1}^d \mathbb{E}_0 T_i = M.$$

The same null experiment appears for every i , which is essential for summing these inequalities. General measurable records are included by the relative-entropy chain rule for classical-quantum states; no countability assumption on the learner's observations is required.

By (S12.4), choose hypothesis 0 precisely when the returned effect lies in its closed radius- δ ball. This is a common binary test with both errors α_i, β_i at most η . The test is measurable by the hybrid continuity bound for d_N . Data processing to this test gives

$$\begin{aligned} D(\Omega_0 \parallel \Omega_i) &\geq D(\text{Bern}(1 - \alpha_i) \parallel \text{Bern}(\beta_i)) \\ &\geq (1 - \eta) \log(1/\eta) - \log 2 \geq \frac{1}{2} \log(1/\eta), \end{aligned}$$

where $\eta \leq 1/8$ was used. Summing over i and substituting Δ proves (S12.3). $\square$

Lemma S12.3 (Amplification using a dimension-dependent seed). *For all $d \geq 1$, $0 < \epsilon \leq 1$, and $0 < \eta \leq 1/8$, there is a common estimator on $\mathfrak{E}_d$ with legal output F such that*

$$(S12.6) \quad \sup_{E \in \mathfrak{E}_d} \mathbb{P}_E\{\|F - E\|_{\text{op}} > \epsilon\} \leq \eta, \quad M \leq C\epsilon^{-2} \left(d^2 + d \log \frac{1}{\eta} \right).$$

The acquisition uses independent one-call input/reference pairs. The output is a Borel function of collective measurements on their completed outputs.

Proof. Corollary III.9 of [5] gives a legal binary-effect estimator with operator error u and failure q using $O(u^{-2}(d^2 + d \log(1/q)))$ calls when $q > 4e^{-4d^2}$. Choose the rational seed probability

$$q_d = 2^{-3d}.$$

It satisfies that strict condition for every $d \geq 1$, since $4d^2 - 3d \log 2 - \log 4 > 0$. A seed run with $u = \epsilon/3$ therefore costs at most $Cd^2\epsilon^{-2}$ calls.

Let k be the smallest positive odd integer with $k \geq 2d^{-1} \log_2(1/\eta)$. Make k independent seed runs, obtaining legal effects $F_1, \dots, F_k$. Select the first index j for which more than $k/2$ of the effects have operator distance at most $2\epsilon/3$ from F_j . If there is no such index, return F_1 . Norm continuity and the fixed tie rule make this selection Borel, and it always returns a legal effect.

If more than half the seed estimates are within $\epsilon/3$ of E , a selectable index exists. Every selectable index has a neighbour among these good estimates, and so its distance from E is at most ϵ . The probability of at least $k/2$ failed seeds is at most

$$2^k q_d^{k/2} = 2^{k-3dk/2} \leq 2^{-dk/2} \leq \eta.$$

This proves the risk bound. Since $k < 2d^{-1} \log_2(1/\eta) + 2$, the total call count has the order in (S12.6). Each seed uses fresh Choi preparations; all subsequent joint processing acts on completed outputs. The amplification is a modification of the repetition option in [5, Remark III.4], with seed failure chosen at the dimension scale rather than fixed independently of d . $\square$

Proof of Theorem S12.1. Theorem S10.2 gives $M \geq c_1 d^2 N \delta^{-2}$ throughout the stated range. Lemma S12.2 supplies $M \geq c_2 d N \delta^{-2} \log(1/\eta)$. The maximum of these two bounds is at least half their sum after decreasing the absolute constant, which proves the lower bound in (S12.2).

For the upper bound apply Lemma S12.3 at $\epsilon = \delta/(8\sqrt{N})$. On its good event, $\|F - E\|_{\text{op}} \leq \epsilon \leq 1/8$. Since $E \in \mathfrak{J}_d$, Weyl's inequality puts both endpoints in $[I/8, 7I/8]$. Lemma S10.4 gives $d_N(E, F) \leq 4\sqrt{N}\epsilon = \delta/2$. The call count is the required upper bound. The procedure is in the fresh one-call class, whereas the lower bound imposed no restriction on coherent training. This also proves the assertion for every fresh-block cap. $\square$

For $d = 1$, (S12.2) reduces to $\Theta(N\delta^{-2} \log(1/\eta))$, the scalar order, because $\log(1/\eta) \geq \log 8$. For growing d , the confidence term becomes comparable to the dimension term when $\log(1/\eta)$ is of order d .

Corollary S12.4 (Binary operator-norm learning at every confidence). *For $d \geq 1$, $0 < \epsilon \leq 2^{-14}$, and $0 < \eta \leq 1/8$, the minimax every-record call budget for operator-norm error ϵ on the complete ordered binary effect body is*

$$(S12.7) \quad \Theta\left(\epsilon^{-2} \left(d^2 + d \log \frac{1}{\eta}\right)\right),$$

with absolute constants. The lower bound permits arbitrary coherent adaptive training; the upper bound uses independent Choi preparations and collective completed-output processing.

Proof. The upper bound is Lemma S12.3. For the lower bound restrict to $\mathfrak{J}_d$ and use Theorem S12.1 with $N = 1$ and $\delta = 2\epsilon$, since $d_1(E, F) = 2\|E - F\|_{\text{op}}$. $\square$

On the complete effect body, the confidence obstruction also combines with the independent-block obstruction. For $d \geq 2$, $1 \leq b \leq N$, $0 < \delta \leq \min\{2^{-13}, \delta_*\}$, and $0 < \eta \leq 1/8$, Theorem S9.2 and the interior subfamily give

$$(S12.8) \quad M_b^*(d, N, \delta, \eta) \geq c\delta^{-2} \left[d^2 N + \left(dN + \frac{N^2}{b} \right) \log \frac{1}{\eta} \right].$$

A second upper construction follows from Lemma S12.3 at operator accuracy $\epsilon = \delta/(2N)$. The hybrid inequality $d_N(E, F) \leq 2N\|E - F\|_{\text{op}}$ converts its good event to the required future loss. It uses only fresh one-call preparations, so it is admissible for every block cap b . Combining its public call budget with the construction in Theorem S9.2 gives, in the same parameter range,

$$(S12.9) \quad M_b^*(d, N, \delta, \eta) \leq C \frac{N^2}{\delta^2} \min \left\{ d^2 + d \log \frac{1}{\eta}, \frac{d^4}{b} \log \frac{d}{\eta} \right\}.$$

The learner may choose the construction with the smaller public every-record budget before making any calls. Both constructions are common to all unknown effects. The two-sided bounds (S12.8)–(S12.9) do not in general match when dimension and horizon grow together.

Corollary S12.5 (Jointly optimal learned descriptions at every confidence). *For $d, N \geq 1$, rational $0 < \delta \leq 2^{-13}$, and $0 < \eta \leq 1/8$, one common learner on $\mathfrak{I}_d$ returns a fixed-length binary word whose public deterministic decoder is legal and satisfies (S12.1), with*

$$(S12.10) \quad M = O\left(\frac{N}{\delta^2} \left(d^2 + d \log \frac{1}{\eta}\right)\right), \quad B = d^2 \log_2(\sqrt{N}/\delta) + O(d^2).$$

All constants are absolute. Acquisition uses independent one-call Choi preparations, and encoding uses no additional device calls. Every learner with these risk and decoder conventions requires the lower call order in (S12.2) and $B \geq d^2 \log_2(\sqrt{N}/\delta) - 11d^2$.

Proof. Put $k_N = \lceil \sqrt{N} \rceil$. Apply Lemma S12.3 at $\epsilon = \delta/(64k_N)$ and failure η , and clip its legal output F spectrally to $[I/4, 3I/4]$, obtaining T . On the good event, Weyl's inequality gives $\|T - F\|_{\text{op}} \leq \epsilon$ and hence $\|T - E\|_{\text{op}} \leq 2\epsilon$. This uses only the eigenvalue displacements of F and does not assume operator-Lipschitz spectral clipping. The interior comparison yields $d_N(E, T) \leq \delta/8$.

Use the exact dictionary of Theorem S11.2 at the same (d, N, δ) . As in the proof of Theorem S11.5, spectral clipping and the specified coordinate rounding give a Borel finite-valued index map, which can be incorporated into the final collective measurement. The decoded centre satisfies $d_N(T, C) \leq 5\delta/16$, so its total error is at most $7\delta/16 < \delta$ on the original good event. Thus the finite output does not increase the failure probability or supply an exact-real side channel to the decoder.

The bound $k_N^2 \leq 4N$ and Lemma S12.3 give the call count. Theorem S11.2 gives the payload. Conversely, a decoded word is an estimator, so Theorem S12.1 applies to its calls. For each target, positive success probability forces some decoded effect into its radius- δ ball. The fixed decoder's range is therefore a cover, and Theorem S11.1 gives the stated bit lower bound. $\square$

These upper bounds use the ideal trusted tomography operations of [5]. The Borel finite readout is a statistical construction; it alone supplies no gate synthesis or reconstruction-time bound. The confidence parameter increases the required observations while the same public interior dictionary supplies the transmitted word.

S13. LEARNING WITH FINITE TRUSTED CONTROL SPECIFICATIONS

The learning theorems count unknown-device calls. We now remove exact physical preparation from their hypotheses, while keeping their loss and call orders. A trusted preparation instruction has a finite classical description and a certified trace-norm error; a trusted control instruction has a finite description and a certified diamond-norm error. These are accuracy assumptions on the available controls. They assert neither a universal hardware implementation nor a bound on preparation time, gate count, or classical workspace. All trace norms in this section are unhalved.

Lemma S13.1 (An adaptive control-error budget). *Fix a finite classical decision tree of experiments, with bounded numbers of calls and trusted instructions on every record. Its unknown channel is unchanged. At a scheduled trusted event j , conditional on the preceding classical history h , replace a channel $\mathcal{C}_{j,h}$ by a channel $\tilde{\mathcal{C}}_{j,h}$ such that*

$$(S13.1) \quad \|\tilde{\mathcal{C}}_{j,h} - \mathcal{C}_{j,h}\|_{\diamond} \leq e_j(h), \quad \sum_{j \text{ on } h_*} e_j(h_{<j}) \leq e \quad \text{for every complete record } h_*.$$

For a fresh state preparation, its trace-norm error suffices in place of the diamond norm: conditional on h , the new state is tensor-separated from every carried system in both experiments. The final classical-quantum states satisfy

$$(S13.2) \quad \|\tilde{\rho}_{\text{out}} - \rho_{\text{out}}\|_1 \leq e.$$

In particular, the laws of any unchanged classical output have total variation distance at most $e/2$.

Proof. Include each observed outcome in the classical history and pad stopped branches by identities. Quantum registers of different sizes can be placed in a finite direct sum. Each event is then a channel on the complete classical–quantum state. A quantum instrument means its complete channel $\rho \mapsto \sum_x |x\rangle\langle x| \otimes \mathcal{C}_x(\rho)$: if errors are instead supplied separately for its completely positive branches, their sum is the event’s error charge.

At each event, insert the ideal preceding state between the actual and ideal updates. Contractivity of the actual channel and the diamond-norm bound give

$$D_j \leq D_{j-1} + \sum_h p_{j-1}(h) e_j(h),$$

where D_j is the unhalved trace-norm difference and p_{j-1} is the ideal history law. Unknown-channel calls and unchanged classical calculations have charge zero. Summing the inequality gives the ideal expectation of the pathwise sum in (S13.1), which is at most e . A replacement preparation channel has diamond distance equal to the trace distance of the prepared states, including when another register is retained. Finally, classical readout and deterministic postprocessing are contractions, and classical total variation is one half of the trace norm. No continuity of the decision rules or of the final loss is used. $\square$

We check that the learners to which this lemma is applied have finite, computable control specifications. Take integer dimensions and horizons and rational positive accuracy and confidence targets. All absolute proof constants may be fixed once as sufficiently conservative rational numbers. For sample budgets, replace each $\log x$, $x > 1$ rational, by

$$\ell(x) = \min\{k \in \mathbb{N} : 2^k \geq x\}.$$

Here $\log x \leq \ell(x) \leq 1 + \log x / \log 2$. Thus the concentration bounds and call orders are preserved, while ceilings of the resulting rational sample budgets are decidable by integer arithmetic. This convention avoids requiring an exact comparison of a transcendental expression with an integer.

Calibration frequencies and all complex empirical means are rational. The ordered spectral projections, fixed-order Gram–Schmidt procedure, clipping, normalized frames, and tie rules of Sections S6–S7 use algebraic arithmetic. The notation for phase recovery does not require a transcendental oracle. For a nonzero empirical z passing (S6.20), put $q = z/|z|$ and select the root v of $v^m = q$ with largest real part. It is unique: the passed guard places $\text{Arg } q$ strictly between $-\pi$ and π , so its principal m th root is strictly closer in angle to 1 than every other root. All roots are algebraic, and comparison of their real parts is decidable. Consequently

$$\tan(\text{Arg}(z)/m) = \frac{\Im v}{\Re v}$$

is algebraic; its denominator is positive under the guard. Zero, amplitude, frame-selection, and phase-guard equalities are therefore decidable exactly. The finite rational legalization grid in Theorem S7.1 is in the public input basis; its positivity tests are exact, and its finitely many algebraic objectives can be evaluated with the prescribed slack. The same holds for the exact rational codec. Every record consequently has a finite, terminating classical computation and a legal output.

There are finitely many records at fixed public parameters: the device outputs and preparation signs have finite alphabets and their numbers are bounded on every path. This does not discretize the unknown effect or the retained quantum state. Within a block the state still lies in its full continuous quantum state space. Only the experiment’s classical instructions and observations have finite descriptions.

Finite descriptions can also specify controls with any required positive accuracy. If a computable unit vector $\psi \in \mathbb{C}^D$ is prescribed and $0 < \epsilon \leq 1$, compute a nonzero Gaussian rational vector z

with $\|z - \psi\|_2 \leq \epsilon/8$ and set $\phi = z/\sqrt{\sum_k |z_k|^2}$. Then

$$(S13.3) \quad |||\phi\rangle\langle\phi| - |\psi\rangle\langle\psi|||_1 \leq 2\|\phi - \psi\|_2 \leq 4\|z - \psi\|_2 \leq \epsilon/2.$$

Coordinate accuracy $O(\epsilon/\sqrt{D})$ suffices, so z and its normalization rule have a description of $O(D \log(D/\epsilon))$ bits. The normalization is specified algebraically; rational normalized amplitudes are not assumed. A trusted preparation within $\epsilon/2$ of this finitely specified target has total error at most ϵ from the ideal state.

For completeness, finite specifications can preserve the exact channel constraints as well. Approximate a known algebraic Stinespring isometry V by a Gaussian rational matrix A with $\|A - V\|_{\text{op}} \leq r < 1/2$, and specify $W = A(A^*A)^{-1/2}$. Then $W^*W = I$ and $\|W - V\|_{\text{op}} \leq 2r$: every singular value of A differs from one by at most r . The channels obtained from these isometries differ in diamond norm by at most $4r$, by expansion of $W\rho W^* - V\rho V^*$ and partial-trace contractivity. Keeping a finite outcome flag gives the same construction for instruments. Their actual trusted realizations are charged separately in (S13.1). Independent rounding of Kraus entries without restoring the channel constraints would not suffice.

Theorem S13.2 (Finite-control form of the learning guarantees). *The common matrix learner of Theorem S7.1 and the learner with training-block cap $1 \leq b \leq N$ of Theorem S9.2 admit the preceding finite trusted-control model in their stated ranges. They have the original success target*

$$\sup_{E \in \mathfrak{E}_d} \mathbb{P}_E\{d_N(\mathcal{M}_E, \mathcal{M}_{\hat{E}}) > \delta\} \leq \eta$$

with every-record call orders, respectively,

$$(S13.4) \quad Cd^4N\delta^{-2}\log(d/\eta), \quad Cd^4\frac{N^2}{b}\delta^{-2}\log(d/\eta).$$

Their original block caps and exact legality of the returned classical effect are preserved. The same transfer applies when the classical output is the public codeword. In dimension one the order remains $CN\delta^{-2}\log(1/\eta)$ independently of b .

Proof. Use the corresponding ideal learner at the same accuracy δ and confidence $\eta/2$, with the finite sample-budget convention above. Fix a computable public upper bound M_* on its calls on every record. The explicit constructions use fresh product states or fresh GHZ states embedded into a known two-dimensional subspace. They can be implemented by one preparation event per block, hence at most M_* such events. Given any preceding record, compute that record's prescribed state to trace accuracy η/M_* , using (S13.3) and the trusted realization allowance. Lemma S13.1 gives final output total variation at most $\eta/2$. For each fixed unknown E , apply it to the set of returned effects whose original loss exceeds δ . The ideal probability is at most $\eta/2$, so the actual probability is at most η . Discontinuous decisions, unsuccessful records, and exact equality branches are included in this comparison.

Classical postprocessing is identical in the two experiments; it does not round the returned effect or test a perturbed loss. Thus legality is exact on every actual record and no additional δ -dependent loss allowance is needed. Halving η and upper-rounding sample budgets change only the absolute constants in (S13.4). $\square$

Each approximate preparation above supplies a state on the same $m \leq b$ probe registers as its ideal block; preparation workspace is discarded or reset before the experiment and no quantum system is passed to the next block. Approximation therefore does not enlarge entangled input width or introduce coherent coupling from old memory into fresh probes. In a realization with S additional scheduled control instructions, use a public bound on the total number of trusted events and charge every instruction in (S13.1). Per-event norm precision is then of order $\eta/(M_* + S)$. For an m -probe state the direct coordinate description has $D = d^m$ and size $O(D \log(D(M_* + S)/\eta))$

bits. The precision per coordinate is logarithmic in $(M_* + S)/\eta$, with its dimension dependence explicitly charged; this is not a polynomial preparation bound. Real-valued target tolerances may be replaced by supplied rational targets between one half and the requested values, preserving all orders and the requested guarantees. No representation of an arbitrary real input, and no oracle for the unknown effect, is assumed.

S14. EFFECTIVE SYNTHESIS OF AN OPTIMAL INTERIOR LEARNER

The finite readout in Theorem S11.5 does not by itself give computable entries for its measurement operators. We now construct a finite algebraic readout, and then a finite trusted-control realization, at the joint query and description orders. The construction uses exact quantifier elimination. The tomography estimator of [5] supplies an existence bound; we do not compile its continuous outcome measurement or assume that its coarse-grained operators have computable entries.

S14.1. A decidable collective-readout problem. Fix integers $d, m \geq 1$. Prepare m independent maximally entangled probe-reference pairs and apply the unknown measurement to each probe. Put $E_1 = E$ and $E_0 = I_d - E$. The complete classical-quantum record is

$$(S14.1) \quad \Gamma_m(E) = \bigoplus_{y \in \{0,1\}^m} R_y(E), \quad R_y(E) = d^{-m} \bigotimes_{\ell=1}^m E_{y_\ell}^\top.$$

The transpose is in the public input basis. The blocks are *subnormalized*: their traces are the probabilities of the classical strings y . The real and imaginary parts of every entry of $R_y(E)$ are polynomials with rational coefficients in the d^2 real coordinates of E . No division by an unknown outcome probability occurs.

A final L -outcome readout consists of a POVM $(A_{j,y})_{j=1}^L$ on the d^m retained reference dimensions, selected by the observed classical string y . Its outcome probabilities are

$$(S14.2) \quad p_j(E; A) = \sum_y \text{tr}(A_{j,y} R_y(E)), \quad A_{j,y} \succeq 0, \quad \sum_{j=1}^L A_{j,y} = I_{d^m}.$$

This description uses the outcome bits classically. It entails no coherent access to those bits or to a measurement environment. Conversely, any collective POVM on (S14.1) can be dephased in y without changing its probabilities, giving precisely such a family.

Lemma S14.1 (Effective algebraic readout selection). *Given d, m , finitely many Gaussian-rational Hermitian centres $C_1, \dots, C_L$, and rational $a > 0$, $0 < \alpha < 1$, it is decidable whether a readout (S14.2) satisfies*

$$(S14.3) \quad \sum_{j: \|E - C_j\|_{\text{op}} \leq a} p_j(E; A) \geq 1 - \alpha \quad \text{for every } E \in \mathfrak{I}_d.$$

When it is feasible, one finite deterministic computation returns such a family with algebraic entries, valid for every real $E \in \mathfrak{I}_d$.

Proof. Parameterize each Hermitian matrix by its real coordinates. Positivity is a finite system of polynomial inequalities: use all principal minors, or the equivalent real symmetric representation for a complex Hermitian matrix. Trace and tensor products in (S14.2) yield real polynomials with rational coefficients. The same is true of the input constraint $I_d/4 \preceq E \preceq 3I_d/4$ and of

$$\text{Good}_j(E) : \quad aI_d - (E - C_j) \succeq 0, \quad aI_d + (E - C_j) \succeq 0.$$

For each subset $S \subseteq \{1, \dots, L\}$ impose the implication

$$\left(\bigwedge_{j \in S} \text{Good}_j(E) \right) \wedge \left(\bigwedge_{j \notin S} \neg \text{Good}_j(E) \right) \implies \sum_{j \in S} p_j(E; A) \geq 1 - \alpha.$$

The conjunction of these finitely many implications, universally quantified over $E \in \mathfrak{I}_d$, is exactly (S14.3). Equality belongs to Good_j ; its complement uses the corresponding strict inequalities. Hence boundary cases require no numerical tolerance.

Eliminate the E variables while retaining the readout entries as free variables, and include the positivity and normalization conditions in (S14.2). Effective real quantifier elimination decides whether this rational semialgebraic feasible set is empty. If it is nonempty, the real-closed-field transfer principle gives a point with real algebraic coordinates. One explicit selection rule uses a fixed effective enumeration of algebraic tuples and tests the resulting quantifier-free formula by exact algebraic sign determination. It terminates on this nonempty set. These are standard effective real-algebraic operations; see [38, Theorems 2.77, 2.80 and Chapter 14]. The eliminated formula is equivalent over the reals, so the selected point satisfies the condition for all real effects, not merely algebraic ones. The unknown effect is a quantified variable, never supplied data or an oracle queried by this computation. $\square$

S14.2. Finite controls at the joint optimal rates.

Theorem S14.2 (Effective optimal interior learning and description). *Let $d, N \geq 1$ be integers, and let $0 < \delta \leq 2^{-13}$ and $0 < \eta \leq 1/8$ be rational. There is a common learner on $\mathfrak{I}_d$ whose entire classical specification is computed from (d, N, δ, η) in finitely many steps. In the finite trusted-control model of Section S13, it returns a fixed-length word decoded by a public deterministic decoder into a legal rational effect C , with*

$$(S14.4) \quad \sup_{E \in \mathfrak{I}_d} \mathbb{P}_E\{d_N(E, C) > \delta\} \leq \eta,$$

$$(S14.5) \quad M \leq C_0 \frac{N}{\delta^2} \left(d^2 + d \log \frac{1}{\eta} \right),$$

$$(S14.6) \quad B = \frac{d^2}{2} \log_2 N + d^2 \log_2(1/\delta) + O(d^2).$$

All constants are absolute. The number M and the finite readout are fixed before any unknown-device call. Acquisition uses fresh one-call probe-reference pairs; only completed outputs undergo collective processing. Thus the construction belongs to every fresh-block class $1 \leq b \leq N$.

The call order matches the coherent adaptive lower bound of Theorem S12.1, and the description order matches Theorem S11.1. No bound on classical search time, algebraic-description length, quantum workspace, or elementary known-gate count is asserted.

Proof. Construct the finite rational dictionary $\mathcal{D} = (C_1, \dots, C_L)$ of Theorem S11.2 at parameters (d, N, δ) . Set

$$k = \lceil \sqrt{N} \rceil, \quad a = \frac{\delta}{8k}, \quad \alpha = \eta/2.$$

For $m = 1, 2, \dots$, apply Lemma S14.1 to these centres, a , and α . At the first feasible m , extract its algebraic readout and set $M = m$. Each infeasible candidate is rejected by a terminating decision procedure; there is no unbounded search for a witness at such a candidate.

We prove termination and the call bound without computing any operator of the imported tomography procedure. Apply Lemma S12.3 at operator accuracy $\epsilon = \delta/(64k)$ and failure $\eta/2$. Its independent Choi acquisition uses

$$m_0 \leq C_1 k^2 \delta^{-2} \left(d^2 + d \log \frac{2}{\eta} \right)$$

calls and returns a legal effect F . Clip F spectrally to $[I_d/4, 3I_d/4]$, obtaining T . On the estimation success event, Weyl's inequality gives $\|T - F\|_{\text{op}} \leq \epsilon$ and hence $\|T - E\|_{\text{op}} \leq 2\epsilon$. The rounding and predecessor selection in Theorem S11.2 produce an index j with $\|T - C_j\|_{\text{op}} \leq 5\delta/(64k)$. Therefore

$$(S14.7) \quad \|E - C_j\|_{\text{op}} \leq \frac{7\delta}{64k} < \frac{\delta}{8k} = a$$

on that event.

Clipping, coordinate rounding with its prescribed ties, and selection from a finite list are Borel operations. Composing them with the final tomography measurement defines a finite-outcome POVM on the m_0 Choi records. Its dephased form is a readout (S14.2) satisfying (S14.3). Only its existence is used here; its entries and any defining integrals are not inputs to the algebraic search. Thus the search stops with $M \leq m_0$. Since $k^2 \leq 4N$ and $\log(2/\eta) \leq C_2 \log(1/\eta)$, this proves (S14.5).

The selected ideal readout has probability at least $1 - \eta/2$ of returning a centre at operator distance at most a from E . The target lies in $\mathfrak{I}_d$ and every centre lies in $[I_d/8, 7I_d/8]$. Lemma S10.4 therefore gives

$$d_N(E, C_j) \leq 4\sqrt{N} a \leq \delta/2$$

on this event.

It remains to realize the computed readout with finite trusted controls. For each classical string y , its algebraic POVM has the algebraic isometry

$$V_y \psi = \sum_{j=1}^L |j\rangle_C |j\rangle_Z \sqrt{A_{j,y}} \psi.$$

Tracing out Z and the factor space gives the required classical readout. This is a finite table, including strings of probability zero under some targets. Algebraic positive square roots and the finite descriptions of Section S13 make each entry effectively approximable. In particular, the normalization of a rational approximation to an isometry preserves the exact channel constraints; a fixed final dephasing preserves the classical output type.

There are M fresh-pair preparation events and one final readout event. Require total unhalved error at each event at most

$$(S14.8) \quad e_{\text{event}} = \frac{\eta}{M+1}.$$

For a preparation this is trace-norm error in the fresh joint pair; for the readout it is complete diamond-norm error, uniformly over all y and all retained reference states. Split each allowance equally between approximation by a finite rational specification and its actual trusted realization. Any additional elementary instructions are charged by the same total pathwise budget η . Lemma S13.1 then bounds the total variation between ideal and actual output indices by $\eta/2$. Applying it to the original bad-output event $d_N(E, C_j) > \delta$ proves (S14.4). No continuity of a decision boundary or perturbation of the returned effect is assumed.

Finally transmit the dictionary index using $B = \lceil \log_2 L \rceil$ bits. The exact public decoder, including its legal values on unused words, is unchanged. Theorem S11.2 gives the upper order in (S14.6). Since the dictionary covers $\mathfrak{I}_d$, Theorem S11.1 gives its matching lower order. More generally, positive success probability at every target forces any fixed finite decoder's range to cover the target family. This proves the stated optimality claims. $\square$

The synthesis is an effective construction of a new collective learner at the existing tomography query order. It is not a claim of efficient compilation of the original tomography measurement. At the selected M , there are 2^M possible classical strings, each selecting an L -outcome measurement on dimension d^M . The algebraic search, its coefficients, the corresponding isometry descriptions,

and their preparation work are separate resources from the transmitted index. The per-event norm tolerance in (S14.8) introduces no additional δ/N requirement; dimensions and coordinate precision still depend on δ through M and L . Fresh preparation remains tensor-separated from the stored references, so finite-control approximation preserves the one-call acquisition restriction.

S15. RATIONAL COLLECTIVE READOUTS WITH FINITE RISK CERTIFICATES

The effective selection of Section S14 can be realized using rational arithmetic and finite enumeration. The readout operators themselves may be chosen Gaussian rational. Their uniform guarantee is certified by exact inequalities on an explicit finite net. The net certifies the statistical risk of a fixed readout; it does not discretize the unknown measurement in the learning model.

S15.1. A finite certificate for a collective readout. Keep the public dictionary $\mathcal{D} = (C_1, \dots, C_L)$ of Theorem S11.2, constructed at (d, N, δ) . Put $k = \lceil \sqrt{N} \rceil$ and $a = \delta/(8k)$. For $m \geq 1$, let $D = d^m$ and retain the subnormalized Choi records $\Gamma_m(E)$ of (S14.1). For rational $0 < \eta \leq 1/8$, define

$$(S15.1) \quad r_m = \min \left\{ \frac{\delta}{32k}, \frac{\eta}{16m} \right\}, \quad K_m = \left\lceil \frac{4d}{r_m} \right\rceil, \quad Q_m = \left\lceil \frac{64L^2 D}{\eta} \right\rceil.$$

Let $\mathcal{G}_m$ be the ordered finite set of Gaussian-rational Hermitian matrices with denominator K_m in $\mathfrak{I}_d = [I_d/4, 3I_d/4]$. More explicitly, enumerate diagonal coordinates in $\{0, 1/K_m, \dots, 1\}$ and real and imaginary coordinates above the diagonal in $\{-1, -1 + 1/K_m, \dots, 1\}$, in the public order, and retain exactly those matrices in $\mathfrak{I}_d$. All tests use exact rational positive-semidefinite elimination. For $G \in \mathcal{G}_m$, set

$$J(G) = \{j : \|G - C_j\|_{\text{op}} \leq a\}.$$

These sets are decidable by the two positive-semidefinite tests in (S11.4), including equality.

A rational readout is a family $(A_{j,y} : 1 \leq j \leq L, y \in \{0, 1\}^m)$ of Gaussian-rational Hermitian $D \times D$ matrices satisfying

$$A_{j,y} \succeq 0, \quad \sum_{j=1}^L A_{j,y} = I_D \quad \text{for every } y.$$

Its certificate consists of these exact legality conditions and the finite inequalities

$$(S15.2) \quad \sum_{j \in J(G)} \sum_{y \in \{0, 1\}^m} \text{tr}(A_{j,y} R_y(G)) \geq 1 - \frac{\eta}{4} \quad (G \in \mathcal{G}_m).$$

Every term is rational. A verifier reconstructs the dictionary and net from the public parameters and checks every displayed inequality. A proper prefix of either enumeration does not constitute a certificate.

Theorem S15.1 (Uniform validity of the finite certificate). *Every readout satisfying (S15.2) has, for all real $E \in \mathfrak{I}_d$,*

$$(S15.3) \quad \mathbb{P}_E\{d_N(E, C_J) > 5\delta/8\} \leq 5\eta/16.$$

Here J is the readout index, and C_J is its unchanged public rational decoding. More generally, if a finite operator-norm r -net $\mathcal{G} \subseteq \mathfrak{I}_d$ satisfies the same finite inequalities with right side $1 - \alpha$ and good-label radius a , then

$$(S15.4) \quad \sup_{E \in \mathfrak{I}_d} \mathbb{P}_E\{\|E - C_J\|_{\text{op}} > a + r\} \leq \alpha + mr.$$

The centres may be any fixed finite legal family for this last assertion.

Proof. We first verify that $\mathcal{G}_m$ is an operator-norm r_m -net. Write $K = K_m$, and for any $E \in \mathfrak{J}_d$ form

$$T = \left(1 - \frac{8d}{K}\right)E + \frac{4d}{K}I_d.$$

Since $8d/K \leq 2r_m < 1$, this is a convex displacement towards $I_d/2$. Its spectrum lies in $[1/4 + 2d/K, 3/4 - 2d/K]$ and $\|T - E\|_{\text{op}} \leq 2d/K$. Round each independent real coordinate to the nearest multiple of $1/K$, with the prescribed ties, and preserve Hermitian conjugacy. The row-sum estimate gives $\|G - T\|_{\text{op}} \leq d/K$. Consequently $G \in \mathfrak{J}_d$, its coordinates occur in the specified finite stream, and

$$(S15.5) \quad \|G - E\|_{\text{op}} \leq 3d/K \leq 3r_m/4 \leq r_m.$$

For one Choi record,

$$\|\Gamma_1(E) - \Gamma_1(G)\|_1 = \frac{2}{d}\|E - G\|_1 \leq 2\|E - G\|_{\text{op}}.$$

Tensor-product telescoping between normalized states therefore yields

$$(S15.6) \quad \|\Gamma_m(E) - \Gamma_m(G)\|_1 \leq 2m\|E - G\|_{\text{op}}.$$

Apply the same fixed readout under the two target effects. Their index laws have total variation at most $mr_m \leq \eta/16$. For the particular G in (S15.5), every $j \in J(G)$ satisfies

$$\|E - C_j\|_{\text{op}} \leq a + r_m \leq \frac{5\delta}{32k}.$$

Both effects lie in $[I_d/8, 7I_d/8]$, so Lemma S10.4 gives $d_N(E, C_j) \leq 4\sqrt{N}(a + r_m) \leq 5\delta/8$. The event $\{J \in J(G)\}$ has probability at least $1 - \eta/4 - \eta/16$ under E , by (S15.2). This proves (S15.3). No continuity of the index-selection boundary is used: one fixed subset $J(G)$ is transferred between the two experiments. Replacing the numerical allowances by a, r, α in that same argument proves (S15.4). $\square$

S15.2. A prescribed rational denominator suffices.

Proposition S15.2 (Rational approximation of a finite classical readout). *Let $(B_{j,y})$ be any family of L -outcome POVMs on dimension D , with finitely many classical conditioning values y . For rational $0 < \eta \leq 1/8$, put $s = \eta/16$ and $Q = \lceil 64L^2D/\eta \rceil$. There is a family $(A_{j,y})$ of Gaussian-rational POVMs, with all real and imaginary coordinates in $Q^{-1}\mathbb{Z}$, such that, for every input classical-quantum state, its output-index law differs from that of $(B_{j,y})$ in total variation by at most $3\eta/32$. The rational family belongs to an explicitly finite enumeration determined solely by (D, L, Q) and the conditioning alphabet.*

Proof. When $L = 1$, take $A_{1,y} = I_D$; there is no approximation error. Suppose $L \geq 2$. First put

$$\overline{B}_{j,y} = (1 - s)B_{j,y} + \frac{s}{L}I_D.$$

These effects sum to I_D and are all at least $(s/L)I_D$. For each y , round the independent real coordinates of the first $L - 1$ effects to multiples of $1/Q$, preserving Hermitian conjugacy, and call the rounded matrices $A_{j,y}$. Set

$$A_{L,y} = I_D - \sum_{j < L} A_{j,y}.$$

Thus normalization holds exactly. With $F_{j,y} = A_{j,y} - \overline{B}_{j,y}$, the row-sum estimate gives

$$\|F_{j,y}\|_{\text{op}} \leq D/Q \quad (j < L), \quad \|F_{L,y}\|_{\text{op}} \leq (L - 1)D/Q.$$

Since

$$(L - 1)D/Q \leq \frac{\eta(L - 1)}{64L^2} < \frac{\eta}{16L} = \frac{s}{L},$$

all $A_{j,y}$ are positive. Together with their exact sum, this also implies $A_{j,y} \preceq I_D$. The rounded coordinates therefore lie in the finite ranges $[0, 1]$ on the diagonal and $[-1, 1]$ for each real or imaginary off-diagonal coordinate.

Mixing with the uniform outcome changes every event probability by at most s . Rounding changes it by at most

$$\sum_j \|F_{j,y}\|_{\text{op}} \leq 2(L-1)D/Q \leq 2LD/Q \leq \eta/32.$$

The same bound holds after averaging over any subnormalized classical conditioning blocks. Thus the total variation of the index laws is at most $s + \eta/32 = 3\eta/32$.

For clarity, no exact entries of B are required by the algorithm below. This rounding argument proves that a suitable rational tuple exists in a prescribed finite set. Enumerate the first $L-1$ Hermitian matrices in the indicated coordinate ranges, define the last by the residual formula, and retain only positive tuples. The resulting finite stream contains the family just constructed. $\square$

Theorem S15.3 (Certified effective learning at the joint rates). *Let $d, N \geq 1$, and let $0 < \delta \leq 2^{-13}$ and $0 < \eta \leq 1/8$ be rational. A finite deterministic computation from (d, N, δ, η) returns a query number M , a Gaussian-rational collective readout, and the finite risk certificate (S15.2). In the finite trusted-control model, the resulting common learner on $\mathfrak{I}_d$ satisfies*

$$(S15.7) \quad \sup_{E \in \mathfrak{I}_d} \mathbb{P}_E\{d_N(E, C) > \delta\} \leq \eta,$$

$$(S15.8) \quad M \leq C_0 \frac{N}{\delta^2} \left(d^2 + d \log \frac{1}{\eta} \right),$$

$$(S15.9) \quad B = \frac{d^2}{2} \log_2 N + d^2 \log_2(1/\delta) + O(d^2).$$

All constants are absolute. The acquisition consists of fresh one-call probe-reference pairs, followed by collective processing of completed outputs. The call and description orders are the minimax orders of Theorems S12.1 and S11.1. No classical running-time, workspace, or elementary known-gate bound is asserted.

Proof. Construct the public dictionary once. For each $m = 1, 2, 4, \dots$, compute (S15.1) and exhaust the finite rational readout stream of Proposition S15.2, with 2^m conditioning values and dimension $D = d^m$. For every candidate, check exact POVM legality and every inequality in (S15.2). If one passes, return the first such candidate and set $M = m$. If none passes, the finite enumeration ends and the algorithm moves to $2m$. Thus it does not wait indefinitely at an infeasible query count. All arithmetic, tensor products, comparisons, and positive-semidefinite tests are over the Gaussian rationals.

It remains to show that some stage passes at the claimed call order. Lemma S12.3, used at accuracy $\epsilon = \delta/(128k)$ and failure $\eta/8$, supplies an independent-Choi estimator with legal output F and

$$m_0 \leq C_1 k^2 \delta^{-2} \left(d^2 + d \log \frac{8}{\eta} \right).$$

Clip F spectrally to $[I_d/4, 3I_d/4]$, obtaining T . On its good event, Weyl's inequality gives $\|T - E\|_{\text{op}} \leq 2\epsilon$. The dictionary rounding and predecessor selection give

$$\|T - C_j\|_{\text{op}} \leq \frac{5\delta}{64k}, \quad \|E - C_j\|_{\text{op}} \leq \frac{3\delta}{32k} < a.$$

These operations define a Borel finite partition of the original tomography outcomes. Hence there exists an L -outcome collective POVM, dephased in the classical acquisition strings, with failure at most $\eta/8$ for the operator-radius- a event uniformly on $\mathfrak{I}_d$.

For every $m \geq m_0$, this readout is available on m Choi records by ignoring the last $m - m_0$ records. Equivalently, its effects are tensored with the identity on the extra reference factors and do not depend on the extra outcome bits. Apply Proposition S15.2 at this m . At every $G \in \mathcal{G}_m$, the resulting rational candidate has failure at most

$$\frac{\eta}{8} + \frac{\eta}{16} + \frac{\eta}{32} = \frac{7\eta}{32} < \frac{\eta}{4}.$$

It therefore passes every finite certificate inequality. The first dyadic integer at least m_0 is at most $2m_0$; thus the outer search terminates with $M \leq 2m_0$. This proof uses existence of the tomography readout only. Its entries, continuous outcome integrals, and exact optimization outputs are never inputs to the rational search. Since $k^2 \leq 4N$ and $\log(8/\eta) = O(\log(1/\eta))$, the query bound follows.

By Theorem S15.1, the selected exact readout has original-loss failure at most $5\eta/16$, with successful loss at most $5\delta/8$. Each rational POVM has an effectively computable algebraic Stinespring isometry, obtained from the positive square roots of its effects as in Section S14. Use the normalized rational approximations of Section S13 to specify its trusted implementation. There are M fresh-pair preparation events and one complete classically controlled readout event. Give each event total unhalved error allowance $\eta/(M+1)$, split equally between finite target approximation and actual trusted realization. This final event includes the complete readout, its classical conditioning, and its final dephasing. If it is implemented as several separately charged instructions, subdivide its allowance among those instructions.

The adaptive control-error lemma bounds the total variation of the ideal and actual index laws by $\eta/2$. The exact public decoder is unchanged. Consequently

$$\mathbb{P}_E\{d_N(E, C) > \delta\} \leq \frac{5\eta}{16} + \frac{\eta}{2} = \frac{13\eta}{16} < \eta \quad (E \in \mathfrak{I}_d).$$

Approximate pair preparations remain tensor-separated from every stored reference, so the fresh one-call restriction is preserved. The dictionary index supplies the description order in (S15.9); its lower order follows from the interior covering converse. The unrestricted adaptive learning converse applies to this learner as well. $\square$

The certificate consists entirely of finite exact rational checks. Its continuum implication is Theorem S15.1, whose analytic input is the tensor-product Lipschitz estimate and the dimension-uniform interior metric comparison. Exhaustive rational search replaces real quantifier elimination in this construction. The output arrays have $2^M L d^{2M}$ independent real matrix coordinates before normalization is used; their construction and verification are separate resources from the transmitted dictionary index. At stage m , at most $(2K_m + 1)^{d^2}$ coordinate tuples are inspected for the parameter net, and at most $(2Q_m + 1)^{(L-1)2^m D^2}$ tuples for the readout. At a fixed stage m , the readout array can be stored using $(L-1)2^m D^2 \lceil \log_2(2Q_m + 1) \rceil$ bits in its fixed coordinate order, together with the public parameters; the residual effects are reconstructed exactly. The stage m is included in a finite header or recovered by rerunning the prescribed deterministic search. These explicit finite bounds are not polynomial-time or payload bounds. A software check of a finite instance establishes that instance's certificate, while the stated lemmas establish its uniform interpretation and the termination bound for the complete construction.

S16. THE FULL BINARY EFFECT BODY

The spectral boundary of a biased measurement has two distinct endpoints. A vanishing eigenvalue of one effect need not give a coherent angular scale: the other outcome can still destroy coherence. We therefore retain both endpoint coordinates and determine their joint finite-use geometry.

Let

$$(S16.1) \quad \mathcal{E} = \left\{ (a, x) \in \mathbb{R} \times \mathbb{R}^3 : 0 \leq a - |x| \leq a + |x| \leq 2 \right\}, \quad E_{a,x} = \frac{1}{2}(aI + x \cdot \sigma).$$

We identify a point (a, x) of $\mathcal{E}$ with its effect $E_{a,x}$. Write $\mathcal{M}_{a,x}$ for the measurement with ordered effects $E_{a,x}, I - E_{a,x}$ and no residual quantum output. Its maximum and minimum eigenvalues, spectral gap, and bias are

$$p = \frac{1}{2}(a + |x|), \quad q = \frac{1}{2}(a - |x|), \quad r = p - q = |x|, \quad b = p + q - 1.$$

Thus $0 \leq q \leq p \leq 1$ and $|b| + r \leq 1$. When $r > 0$ put $u = x/r$, so that $E_{a,x} = pP_u + q(I - P_u)$, where $P_u = (I + u \cdot \sigma)/2$. Directions are not identified under outcome exchange.

For an integer $N \geq 1$ set $\tau = N^{-1}$ and define

$$(S16.2) \quad T_N(s, t) = |s - t| \sqrt{\frac{N}{\max\{s(1-s), t(1-t)\} + \tau}}, \quad 0 \leq s, t \leq 1,$$

$$V(p, q) = p(1-p) + q(1-q) = \frac{1}{2}(1 - b^2 - r^2),$$

$$(S16.3) \quad K_N(p, q) = r \sqrt{\frac{N}{V(p, q) + \tau}}.$$

For two effects with data (p, q, u) and (p', q', v) let $h \in [0, \pi]$ be the angle between u, v when both spectral gaps are positive. If either gap is zero, set the following angular term equal to zero, without choosing a direction at the repeated eigenvalue:

$$(S16.4) \quad H_N(E, F) = T_N(p, p') + T_N(q, q') + \min\{K_N(p, q), K_N(p', q')\} h.$$

Theorem S16.1 (Finite-use geometry of binary qubit effects). *For every $E, F \in \mathcal{E}$ and $N \geq 1$,*

$$(S16.5) \quad \frac{1}{8192} \min\{1, H_N(E, F)\} \leq d_N(\mathcal{M}_E, \mathcal{M}_F) \leq \min\{2, 3H_N(E, F)\}.$$

The distance permits all common reference-assisted adaptive testers with at most N calls, including feedback and bounded public stopping, and uses the unhalved trace norm. The constants are independent of both spectra, both directions, and the horizon. In particular,

$$(S16.6) \quad d_1(\mathcal{M}_{a,x}, \mathcal{M}_{a',x'}) = |a - a'| + |x - x'|.$$

The lower comparison is witnessed by pair-dependent nonadaptive tests. It does not require a common estimator of all four target coordinates.

The denominator in (S16.3) involves the two outcome variances. Replacing it by the distance to the nearest spectral face would give an incorrect angular scale on a face such as $q = 0, 0 < p < 1$. The proof below supplies both a scalar-overlap dilation upper bound and explicit finite-distance lower witnesses.

S16.1. The spectral endpoints. Put

$$B_N(s, t) = \|\text{Bern}(s)^{\otimes N} - \text{Bern}(t)^{\otimes N}\|_1.$$

Lemma S16.2 (A Bernoulli estimate at both support endpoints). *For $s, t \in [0, 1]$,*

$$(S16.7) \quad \frac{1}{64} \min\{1, T_N(s, t)\} \leq B_N(s, t) \leq \min\{2, 3T_N(s, t)\}.$$

For a fixed one of the two parameters, B_N increases as the other parameter moves away from it on either side.

Proof. Assume $s < t$, put $\Delta = t - s$, and write $v = \max\{s(1-s), t(1-t)\}$. The identical pair is immediate. A coupling of individual bits gives $B_N \leq 2N\Delta$. The root fidelity $F = \sqrt{st} + \sqrt{(1-s)(1-t)}$ satisfies

$$\begin{aligned} 1 - F &= \frac{\Delta^2}{2} \left(\frac{1}{(\sqrt{s} + \sqrt{t})^2} + \frac{1}{(\sqrt{1-s} + \sqrt{1-t})^2} \right) \\ &\leq \frac{\Delta^2(1+\Delta)}{2t(1-s)} \leq \frac{\Delta^2}{2v}, \end{aligned}$$

whenever $v > 0$. For the last inequality, both endpoint variances are at most $t(1-s)/(1+\Delta)$: after cancellation the required inequalities are $s(1+\Delta) \leq t$ and $(1-t)(1+\Delta) \leq 1-s$. They follow respectively from $\Delta(1-s) \geq 0$ and $t\Delta \geq 0$. The product-fidelity inequality gives $B_N \leq 2\sqrt{N}\Delta/\sqrt{v}$. For $z = Nv$, the inequality $\min\{2, 2/\sqrt{z}\}\sqrt{z+1} \leq 2\sqrt{2} < 3$ combines the two upper bounds. At $v = 0$ use the coupling bound.

For the lower bound, complement both bits if necessary to have $s + t \leq 1$. If $t > 3/4$, then $\Delta \geq 2t - 1 > 1/2$ and $B_N \geq B_1 = 2\Delta > 1$, which suffices. Otherwise $0 < t \leq 3/4$. Set

$$k = \min \left\{ N, \max \left\{ 1, \left\lfloor \frac{1}{2t} \right\rfloor \right\} \right\}, \quad m = \lfloor N/k \rfloor.$$

Then $k \geq \frac{1}{4} \min\{N, t^{-1}\}$, $m \geq N/(2k)$, and $(k-1)t \leq 1/2$. The event of no success in a block of k trials has probabilities $e_s = (1-s)^k$ and $e_t = (1-t)^k$, with

$$e_s - e_t \geq k\Delta(1-t)^{k-1} \geq k\Delta/2.$$

There is a common stochastic processing of this binary event whose success probability is $d + ce$, where

$$c = \frac{1}{\max\{e_s + e_t, 2 - e_s - e_t\}} \geq \frac{1}{2}, \quad d = \frac{1}{2}(1 - c(e_s + e_t)).$$

Indeed $d \geq 0$, $d + c \leq 1$, and the processed pair has parameters $(1 \pm c(e_s - e_t))/2$. Apply Lemma S19.3 to m independent blocks and discard the unused calls. Contraction under this processing gives

$$B_N(s, t) \geq \frac{1}{16\sqrt{2}} \min\{1, \Delta\sqrt{N \min(N, t^{-1})}\}.$$

Since $v \geq t(1-t) \geq t/4$, $T_N(s, t) \leq 2\Delta\sqrt{N \min(N, t^{-1})}$. This proves the stated lower constant.

For monotonicity, the likelihood ratio of the two binomial count laws is monotone in the count. For a fixed larger parameter their total variation is the maximum, over upper count intervals, of the larger-parameter probability minus the smaller-parameter probability. The latter probabilities decrease when the smaller parameter decreases. Fixing the smaller parameter gives the analogous statement with lower count intervals. Endpoint cases follow by continuity of the finite sums. $\square$

Lemma S16.3 (Spectral projection without cancellation). *For arbitrary directions and spectra,*

$$(S16.8) \quad B_N(p, p') \leq d_N(\mathcal{M}_E, \mathcal{M}_F), \quad B_N(q, q') \leq d_N(\mathcal{M}_E, \mathcal{M}_F).$$

If the two directions agree, then

$$(S16.9) \quad d_N(\mathcal{M}_{p,q,u}, \mathcal{M}_{p',q',u}) \leq B_N(p, p') + B_N(q, q').$$

Consequently, with $d = d_N(\mathcal{M}_E, \mathcal{M}_F)$,

$$(S16.10) \quad \begin{aligned} d_N(\mathcal{M}_{p,q,u}, \mathcal{M}_{p,q,v}) &\leq 3d, \\ d_N(\mathcal{M}_{p',q',u}, \mathcal{M}_{p',q',v}) &\leq 3d. \end{aligned}$$

Proof. Suppose first $p \geq p'$. Repeatedly input a maximum-eigenvalue eigenstate of E . Its success probabilities under the two hypotheses are p and some $t \leq p'$. The monotonicity in Lemma S16.2 gives $B_N(p, p') \leq B_N(p, t) \leq d$. If $p < p'$, interchange the hypotheses. For $q \leq q'$, use a minimum-eigenvalue eigenstate of E ; the other success probability is at least q' . Reverse the hypotheses if $q > q'$. These are separate admissible witnesses; no single input is asserted to witness both endpoint changes.

For aligned directions, first make the common projective measurement $\{P_u, I - P_u\}$ and, according to its outcome, generate a bit with success probability p or q . This implements exactly the measurement channel, including on inputs entangled with a reference: the reference blocks are the corresponding partial matrix elements weighted by the displayed success probabilities. Supply in advance N independent p -bits and N independent q -bits; at a call consume the next unused bit from the selected array. Every fixed adaptive tester is a common channel of these two arrays. The triangle inequality for tensor products therefore bounds its output distance by $B_N(p, p') + B_N(q, q')$. This proves (S16.9) with arbitrary references, feedback, and stopping; all unqueried row bits are discarded. Combining it with (S16.8) bounds every aligned spectral change by $2d$. The triangle inequality, aligning at u or at v , then proves (S16.10). $\square$

S16.2. Angular geometry with both eigenvalues retained.

Theorem S16.4 (Angular transition for a biased measurement). *For fixed $0 \leq q \leq p \leq 1$, $N \geq 1$, and directions at angle $h \in [0, \pi]$,*

$$(S16.11) \quad \frac{1}{512} \min\{1, K_N(p, q)h\} \leq d_N(\mathcal{M}_{p,q,u}, \mathcal{M}_{p,q,v}) \leq \min\{2, \sqrt{2}K_N(p, q)h\}.$$

More precisely, put

$$A = \sqrt{p(1-p)} + \sqrt{q(1-q)}, \quad z = (p-q) \sin(h/2).$$

When $A > 0$,

$$(S16.12) \quad d_N \leq \min \left\{ 2, 2Nz, 2\sqrt{1 - \left(\frac{A^2}{A^2 + z^2} \right)^N} \right\}.$$

When $A = 0$, the bound $\min\{2, 2Nz\}$ remains valid.

Proof. The one-use distance is $2z$: the effect difference is traceless with operator norm z , and the same reference trace-norm duality as in Lemma S19.2 applies. Replacing the calls one at a time gives the adaptive bound $2Nz$.

We next construct a dilation with scalar cross overlap. Common unitary conjugation puts the two directions in one real meridian. Let

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad R_t = \exp(tJ), \quad t = h/2, \quad E = \text{diag}(p, q), \quad F = R_t E R_t^\top.$$

A measurement dilation is

$$W_E \psi = |+\rangle_C |+\rangle_L \sqrt{E} \psi + |-\rangle_C |-\rangle_L \sqrt{I - E} \psi.$$

The square-root vectors belong to a further two-dimensional environment. Tracing both environments leaves the ordered classical output and implements the measurement even with a reference. Put

$$f = \sqrt{pq} + \sqrt{(1-p)(1-q)}.$$

For any ϕ , apply the common rotation $R_{\phi-t}$ to the two-dimensional environment of the dilation W_F . The cross overlap is

$$\begin{aligned} W_E^\dagger(I \otimes R_{\phi-t})W_F &= (\sqrt{E} R_\phi \sqrt{E} + \sqrt{I-E} R_\phi \sqrt{I-E})R_{-t} \\ &= (\cos \phi I + f \sin \phi J)R_{-t}. \end{aligned}$$

Here the rotation acts as the identity on the outcome registers. When $f > 0$, choose

$$D = \sqrt{f^2 \cos^2 t + \sin^2 t}, \quad \cos \phi = \frac{f \cos t}{D}, \quad \sin \phi = \frac{\sin t}{D}.$$

Then the displayed overlap is cI , where $c = f/D \in [0, 1]$. This construction also covers $t = 0$ and $t = \pi/2$. To express it in the spectral coordinates, put $T = \sqrt{p(1-q)} + \sqrt{q(1-p)}$. Direct multiplication gives

$$fT = A, \quad (1 - f^2)T^2 = (p - q)^2.$$

If $A > 0$ then $f, T > 0$, and therefore

$$(S16.13) \quad c^2 = \frac{f^2}{f^2 \cos^2 t + \sin^2 t} = \frac{A^2}{A^2 + (p - q)^2 \sin^2 t}.$$

Fix any common adaptive tester. Purify its initial state and all intermediate operations, and retain the measurement environments in this proof. The environments are never accessed by the tester. At each call the scalar identity for the cross overlap multiplies the inner product of the two purifications by c , independently of the current input and reference. The common intervening isometries preserve the inner product. A tester that stops early is padded to N calls on dummy inputs, with their outcomes ignored in its final output. Thus two purifications of that final output have inner product c^N . Trace-norm contraction and the pure-state formula give $d_N \leq 2\sqrt{1 - c^{2N}}$, proving (S16.12). This is a pairwise dilation comparison; it does not require a processor for the entire effect body or an unknown-input environment-seizing procedure.

Since $V \leq A^2 \leq 2V$ and $1 - (1 + y)^{-N} \leq Ny$ for $y \geq 0$, the two nontrivial upper bounds imply

$$d_N \leq \min\{Nrh, \sqrt{N}rh/\sqrt{V}\} \leq \sqrt{2}K_N(p, q)h$$

when $V > 0$. The final inequality follows by putting $z = NV$ and using $\min\{1, z^{-1/2}\}\sqrt{z+1} \leq \sqrt{2}$. If $V = 0$, the only possibilities are scalar deterministic effects or a projective measurement, and the hybrid bound suffices.

For the lower bound, two explicit witnesses suffice. First suppose $r > 0$ and $h > 0$. Input a pure state in the direction $u - v$. Its success probabilities are

$$\frac{1 + b + r \sin(h/2)}{2}, \quad \frac{1 + b - r \sin(h/2)}{2}.$$

Their difference is $r \sin(h/2)$. The larger of their Bernoulli variances is at most $1 - b^2$: after complementing both output laws we may assume $a = 1 + b \leq 1$; both success probabilities lie in $[0, a]$, so their variances are at most $a \leq a(2 - a) = 1 - b^2$. Lemma S16.2, together with $\sin(h/2) \geq h/\pi$, now gives

$$(S16.14) \quad d_N \geq \frac{1}{64\pi} \min \left\{ 1, rh \sqrt{\frac{N}{1 - b^2 + \tau}} \right\}.$$

The repeated pure input is a nonadaptive test.

For the second witness, choose a half-turn of the input qubit about an axis normal to the plane of u, v . It sends both Bloch directions to their negatives. With probability 1/2 use the original measurement, and with probability 1/2 apply this half-turn before the call and swap the two recorded outcomes afterwards. Discard the random choice. The resulting measurement has effects $(I \pm ru \cdot \sigma)/2$ or $(I \pm rv \cdot \sigma)/2$ under the respective hypotheses: the bias has been removed

and the visibility is r . The same random preprocessing and postprocessing are used under both hypotheses and on every call. Any test for this unbiased pair is consequently an admissible test for the original pair. Theorem S19.1 gives

$$(S16.15) \quad d_N \geq \frac{1}{64} \min \left\{ 1, rh \sqrt{\frac{N}{1-r^2+\tau}} \right\}.$$

The witness is the finite entangled-block test of Lemma S19.4, with this common processing. If the directions coincide or are antipodal, the required half-turn is chosen about any perpendicular axis.

Finally, feasibility $|b| + r \leq 1$ implies

$$\min\{1 - b^2, 1 - r^2\} \leq 2(1 - b^2 - r^2) = 4V.$$

For example, write $x = \max\{|b|, r\}$ and $y = \min\{|b|, r\}$; then $x \leq 1 - y$, $0 \leq y \leq 1/2$, and $x^2 + 2y^2 \leq (1 - y)^2 + 2y^2 \leq 1$, which is the asserted inequality. Hence the larger of the two coefficients in (S16.14) and (S16.15) is at least $K_N(p, q)/2$. Their stronger lower bound is at least $(128\pi)^{-1} \min\{1, K_N(p, q)h\}$, which implies (S16.11). If $r = 0$ or $h = 0$, the two channels coincide and the assertion is immediate. $\square$

Proof of Theorem S16.1. For positive gaps use a path whose angular leg has the spectrum with the smaller angular coefficient. Lemma S16.3, followed by Lemma S16.2 and Theorem S16.4, gives

$$d_N(\mathcal{M}_E, \mathcal{M}_F) \leq 3T_N(p, p') + 3T_N(q, q') + \sqrt{2} \min\{K_N(p, q), K_N(p', q')\}h \leq 3H_N(E, F).$$

If one gap vanishes, that scalar effect may be aligned with the other direction at no cost, so the same estimate holds with angular term zero.

For the lower bound, put $A = T_N(p, p')$, $B = T_N(q, q')$, and $C = \min\{K_N(p, q), K_N(p', q')\}h$. The spectral projection lemma gives $d_N \geq 64^{-1} \min\{1, A\}$ and $d_N \geq 64^{-1} \min\{1, B\}$. Its angular projections and Theorem S16.4 give $d_N \geq 1536^{-1} \min\{1, C\}$. The largest of these three bounds is at least $4608^{-1} \min\{1, A + B + C\}$, which proves the weaker displayed constant $1/8192$. At a scalar effect only the two spectral bounds are needed. To check the assertion about lower witnesses, let d_N^{na} be the supremum over nonadaptive tests. The two spectral lower bounds hold with d_N^{na} , and the aligned spectral upper bound is consequently at most $2d_N^{\text{na}}$. Apply the triangle inequality to the output of a nonadaptive angular witness on the three-channel path. Its actual-pair leg is at most d_N^{na} , and its aligned spectral leg is at most $2d_N^{\text{na}}$. The same lower comparison thus holds with d_N^{na} in place of d_N .

It remains to verify the one-use identity. For a reference-assisted input state ρ , the difference has output blocks B_ρ and $-B_\rho$, where $B_\rho = \text{tr}_A[(E - F) \otimes I]\rho$. Duality against Hermitian contractions gives $\|B_\rho\|_1 \leq \|E - F\|_{\text{op}}$. An extremal eigenstate of $E - F$ attains equality. Thus $d_1 = 2\|E - F\|_{\text{op}}$; the two eigenvalues of $E - F$ are $((a - a') \pm |x - x'|)/2$. This proves (S16.6). $\square$

Remark S16.5 (Boundary faces and repeated eigenvalues). On $p = q = t$, the measurement is the input-independent coin $\text{Bern}(t)$ and all directions represent the same effect. Its exact finite-use distance along this scalar interval is $B_N(t, t')$. On the face $q = 0$, $0 < p < 1$, the angular coefficient is

$$K_N(p, 0) = p \sqrt{\frac{N}{p(1-p) + N^{-1}}}.$$

For a fixed p in this interval it has square-root-horizon order, despite the zero eigenvalue. At the projective edge $(p, q) = (1, 0)$ it equals N . At the two deterministic apices $p = q = 0$ and $p = q = 1$, the angular coefficient vanishes while the spectral moduli retain their finite-use support cutoff. These cases are part of the same theorem, without a spectral margin imposed on the target.

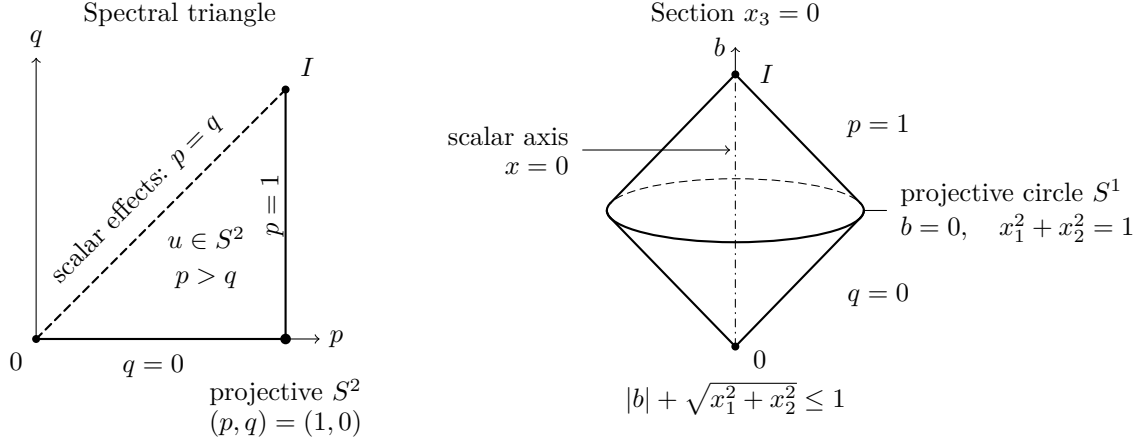


FIGURE 1. The four-dimensional effect body $|b| + |x| \leq 1$. Each point of the spectral triangle with $p > q$ carries a direction $u \in S^2$; on the scalar diagonal all directions represent the same effect. The boundary pieces $q = 0$ and $p = 1$ meet at the projective stratum $(p, q) = (1, 0)$, which carries the full sphere S^2 . The right panel is a projection of the three-dimensional section $x_3 = 0$; its equatorial circle is the intersection of that sphere with the section.

S17. THE DESCRIPTION ENTROPY OF ALL BINARY QUBIT EFFECTS

Let

$$\mathfrak{E} = \{E = E^* \in \mathbb{C}^{2 \times 2} : 0 \leq E \leq I\}.$$

An effect E specifies the ordered measurement with effects $E, I - E$. Write $\mathcal{C}_N(\mathfrak{E}, \delta)$ for its covering number under d_N , allowing arbitrary legal memoryless centres of this interface, and write $\mathcal{C}_N^{\text{rat}}(\mathfrak{E}, \delta)$ when the centres have rational Cartesian coordinates. Both eigenvalues and the eigenspaces are charged. The spectral representation $E = pP_u + q(I - P_u)$, $p \geq q$, represents a scalar effect only once, independently of u .

Theorem S17.1 (The full effect body). *There are absolute constants $c, C, \delta_0 > 0$ such that, for every integer $N \geq 1$ and $0 < \delta \leq \delta_0$,*

$$(S17.1) \quad cN^2 \log(N+2) \delta^{-4} \leq \mathcal{C}_N(\mathfrak{E}, \delta) \leq \mathcal{C}_N^{\text{rat}}(\mathfrak{E}, \delta) \leq CN^2 \log(N+2) \delta^{-4}.$$

Consequently the optimum fixed-length reusable payload is

$$(S17.2) \quad 2 \log_2 N + \log_2 \log(N+2) + 4 \log_2(1/\delta) + O(1).$$

The constants are uniform over the whole closed effect body, including scalar, deterministic and projective effects. For rational accuracy input, rational target data admit an exact rational encoder attaining this order.

The logarithm records accumulation near the projective stratum, where both outcome probabilities can approach their support endpoints. We prove the upper bound by a finite rational construction and the lower bound by separated families of projective-corner boxes.

S17.1. A rational spectral and angular code. For $N \geq 1$ and rational $0 < \delta \leq 1/4$, put

$$(S17.3) \quad B = \left\lceil \sqrt{64N/\delta^2} \right\rceil, \quad s_i = \begin{cases} \frac{i^2}{B^2 + i^2}, & 0 \leq i \leq B, \\ 1 - \frac{(2B-i)^2}{B^2 + (2B-i)^2}, & B < i \leq 2B. \end{cases}$$

Thus $0 = s_0 < s_1 < \dots < s_{2B} = 1$. For $0 \leq j < i \leq 2B$ define

$$(S17.4) \quad K_{ij}^2 = \frac{N(s_i - s_j)^2}{s_i(1 - s_i) + s_j(1 - s_j) + N^{-1}}, \quad A_{ij} = \max \left\{ 1, \left\lceil \sqrt{256K_{ij}^2/\delta^2} \right\rceil \right\}.$$

Use the six signed stereographic charts

$$(S17.5) \quad u_a = \epsilon \frac{1 - |z|^2}{1 + |z|^2}, \quad u_b = \frac{2z_b}{1 + |z|^2} \quad (b \neq a), \quad a \in \{1, 2, 3\}, \quad \epsilon \in \{-1, 1\},$$

with $z \in \{-A_{ij}, \dots, A_{ij}\}^2/A_{ij}$ for the spectral pair (s_i, s_j) . Every chart word decodes to $s_i P_u + s_j(I - P_u)$. A diagonal pair $i = j$ has the single word $s_i I$. The exact number of words, including redundant chart words, is

$$(S17.6) \quad L_{N,\delta} = (2B + 1) + \sum_{0 \leq j < i \leq 2B} 6(2A_{ij} + 1)^2.$$

Theorem S17.2 (Exact rational realization). *The words in (S17.6) form a radius- $\delta/2$ cover of $\mathfrak{E}$ by legal rational effects, and*

$$(S17.7) \quad L_{N,\delta} \leq CN^2 \log(N + 2) \delta^{-4}$$

for an absolute C . If

$$E = \frac{1}{2}((1 + b)I + x \cdot \sigma), \quad b \in \mathbb{Q}, \quad x \in \mathbb{Q}^3, \quad |b| + |x| \leq 1,$$

then a covering word is computable by rational arithmetic and integer comparisons. Its payload is one fixed-length index in the set of (S17.6).

Proof. Write $f(t) = t^2/(1 + t^2)$, $0 \leq t \leq 1$. For a probability in $[0, 1/2]$, round its inverse coordinate t to the nearest multiple of $1/B$; for a probability in $[1/2, 1]$, use the reflected chart $1 - f(t)$. Choose the multiple of smaller magnitude at ties. This defines a nondecreasing quantizer on $[0, 1]$, so separately quantizing $p \geq q$ preserves their order.

The square-root vector associated with $f(t)$ is $(t, 1)/\sqrt{1 + t^2}$; its derivative has norm $1/(1 + t^2) \leq 1$. The reflected chart has the same bound. Hence the Euclidean square-root distance between each probability and its quantization is at most $1/(2B)$. If H is this distance, the product-fidelity inequality gives unhalved distance at most $2\sqrt{N}H$ between the corresponding N -fold Bernoulli laws. The common two-coin programme in Lemma S16.3 therefore gives a spectral error at most

$$2\sqrt{N}/B \leq \delta/4.$$

This argument includes adaptive testers and entangled references.

If the quantized eigenvalues coincide, this completes the encoding. Otherwise choose a signed chart whose distinguished coordinate has maximal absolute value in the target direction. Its inverse coordinate belongs to $[-1, 1]^2$. Rounding its two coordinates to the grid of denominator A_{ij} gives angular error at most $2\sqrt{2}/A_{ij}$, since geodesic distance is bounded by the length of the mapped straight segment and the differential of stereographic projection has norm at most two. The angular upper bound in Theorem S16.4 gives error at most

$$\sqrt{2}K_{ij} \frac{2\sqrt{2}}{A_{ij}} \leq \delta/4.$$

The triangle inequality proves the covering assertion. Formula (S17.5) preserves the unit sphere exactly, so every decoded effect is legal and rational.

We verify exact computability even when the input eigenvalues are irrational. Put $Q = |x|^2 \in \mathbb{Q}$. The eigenvalues are $(1 + b \pm \sqrt{Q})/2$. Each decision in the spectral rounding compares one of these numbers with a rational threshold $f((k + 1/2)/B)$ or its reflection. Such a comparison is the sign of a rational number plus or minus $\sqrt{Q}$; checking signs before squaring makes it an exact rational

comparison. If $Q = 0$, the two quantized eigenvalues coincide. Otherwise choose the smallest index a maximizing $|x_a|$ and put $\epsilon = \text{sgn}(x_a)$. The inverse chart is

$$z_b = \frac{x_b}{\sqrt{Q} + |x_a|} \quad (b \neq a).$$

Its comparison with any rational rounding threshold again reduces to the sign of a rational number plus a rational multiple of $\sqrt{Q}$. The integers B, A_{ij} are computed by integer squaring and rational comparison. Thus no irrational normalization is required. The decoder uses only the displayed rational formulae.

It remains to count the words. Set $m_i = \min(i, 2B - i)$ and $U = B^2/N$. Directly from (S17.3),

$$\frac{m_i^2}{4B^2} \leq s_i(1 - s_i) \leq \frac{m_i^2}{B^2}, \quad K_{ij}^2 \leq \frac{4NB^2}{m_i^2 + m_j^2 + U}.$$

Each value of m_i occurs at most twice. Since $(2A_{ij} + 1)^2 \leq C(1 + K_{ij}^2/\delta^2)$, we obtain

$$(S17.8) \quad L_{N,\delta} \leq CB^2 + \frac{CNB^2}{\delta^2} \sum_{a,b=0}^B \frac{1}{a^2 + b^2 + U}.$$

Here $U \geq 1$. The part with $\max(a, b) \leq \sqrt{U}$ is bounded by an absolute constant. Each successive dyadic square ring contributes at most another absolute constant. There are at most $C \log(N + 2)$ such rings, because $B/\sqrt{U} = \sqrt{N}$. Consequently

$$\sum_{a,b=0}^B \frac{1}{a^2 + b^2 + U} \leq C \log(N + 2).$$

Finally $B^2 \leq CN/\delta^2$, so (S17.8) proves (S17.7). The support cutoff in the denominator is retained throughout the count, including the endpoint grid words. $\square$

S17.2. Packing the projective corner.

Proof of Theorem S17.1. The upper bound follows from Theorem S17.2. For a real δ , choose a rational δ' with $\delta \leq \delta' \leq 2\delta$ and use its radius- $\delta'/2$ code; decrease δ_0 to at most $1/8$. For the lower bound first let $N \geq 16$. For every

$$t = 8^j/N \leq 1/16, \quad j \geq 0,$$

consider the effects with

$$(S17.9) \quad p = 1 - \alpha, \quad q = \beta, \quad t \leq \alpha, \beta \leq 2t, \quad u = u(z), \quad z \in [-1/4, 1/4]^2,$$

where $u(z)$ is the chart in (S17.5) with $a = 3, \epsilon = 1$. On this square, spherical distance is bounded above and below by absolute positive multiples of Euclidean distance in z . Throughout (S17.9), the eigenvalue gap is at least $3/4$ and

$$p(1 - p) \asymp q(1 - q) \asymp t, \quad p(1 - p) + q(1 - q) + N^{-1} \asymp t.$$

The finite-distance comparison of Theorem S16.1 therefore supplies an absolute $c_1 > 0$ such that any two points in this box satisfy

$$(S17.10) \quad d_N \geq c_1 \min \left\{ 1, \sqrt{N/t} |(\alpha, \beta, z) - (\alpha', \beta', z')|_\infty \right\}.$$

Choose an absolute sufficiently large A and then an absolute sufficiently small δ_0 . In each of the four coordinates take a grid of spacing $A\delta\sqrt{t/N}$. Since $Nt \geq 1$, this spacing is at most half the

width of either spectral interval when $\delta \leq \delta_0$. The resulting points have pairwise distance at least 4δ by (S17.10), and their number is at least

$$c \left(\frac{t}{\delta \sqrt{t/N}} \right)^2 \left(\frac{1}{\delta \sqrt{t/N}} \right)^2 = cN^2 \delta^{-4}.$$

Points from different boxes are also uniformly separated. Indeed, if $t' \geq 8t$, then $\alpha' \geq t'$ and $\alpha \leq 2t \leq t'/4$. The spectral modulus of the corresponding upper eigenvalues is bounded below by an absolute positive multiple of $\sqrt{Nt'} \geq 1$. The spectral noncancellation bound in Theorem S16.1 gives $d_N \geq c_2 > 0$. Shrinking δ_0 ensures that this is at least 4δ . There are at least $c \log(N+2)$ permitted boxes. Their union is therefore a 4δ -packing of cardinality $cN^2 \log(N+2) \delta^{-4}$.

For $1 \leq N < 16$, use instead a fixed spectral rectangle $5/8 \leq p \leq 3/4$, $1/4 \leq q \leq 3/8$ and the same angular chart. Theorem S16.1 gives a uniform local Euclidean lower bound, and a grid of spacing $C\delta$ has at least $c\delta^{-4}$ points. The factor $N^2 \log(N+2)$ is bounded on this finite range. This proves the same lower estimate for all $N \geq 1$.

A radius- δ ball about any legal centre contains at most one point of a 4δ -packing, by the triangle inequality. Hence the packing proves the lower bound with the stated centre convention. Taking binary logarithms and rounding the index length proves (S17.2). $\square$

S17.3. The source of the logarithm. The spectral and angular weights also give the following integral form of the same accumulation. Put $v_p = p(1-p)$, $v_q = q(1-q)$ and $\tau = N^{-1}$. Then

$$(S17.11) \quad \int_{0 \leq q \leq p \leq 1} \frac{N^2 (p-q)^2 dp dq}{(v_p + v_q + \tau) \sqrt{(v_p + \tau)(v_q + \tau)}} \asymp N^2 \log(N+2).$$

To verify this, near $p=1, q=0$ put $\alpha = 1-p$, $\beta = q$. After removing the factor N^2 , the integrand is comparable to

$$\frac{1}{(\alpha + \beta + \tau) \sqrt{(\alpha + \tau)(\beta + \tau)}}.$$

For $s = \alpha + \beta$, integration over $0 \leq \alpha \leq s$ is bounded above by an absolute constant times $1/(s + \tau)$. For $s \geq 2\tau$, restriction to $s/4 \leq \alpha \leq 3s/4$ gives the matching lower bound. Integration in s proves logarithmic growth. Near the scalar endpoint $p = q = 0$, the factor $(p-q)^2$ is at most $(p+q)^2$, while $v_p + v_q \asymp p+q$; the remaining integral is uniformly bounded. Complementation gives the same conclusion near $p = q = 1$. Away from these endpoint neighbourhoods, $v_p + v_q$ is bounded below and the factors $v_p^{-1/2}, v_q^{-1/2}$ are integrable. Bounded N is absorbed into the constants. Thus the extra logarithm is contributed by the projective corner. The finite covering construction and packing above also include the collapsed scalar coordinates, where the spectral representation ceases to be a chart.

S18. MEASUREMENT DISCRIMINATION, BLOCK RESOURCES, AND LEARNING

S18.1. Finite-use discrimination and horizontal geometry. The measurement interface determines which discrimination results apply. Here the ordered classical outcome and an arbitrary external reference are retained, while the device returns no postmeasurement quantum system. For projective measurements, Puchała and collaborators [16, Corollary 1 and Theorem 2] give an exact multiple-use formula with ancillary access, used in the observable-readout theorem. Fiurášek–Mičuda [19] study a restricted ancillary interface, with a different comparison of adaptive and parallel probes.

For general POVMs, Krawiec–Pawela–Puchała [20, Section 4] construct a pair of nine-outcome qutrit rank-one measurements admitting perfect adaptive discrimination with two calls and no perfect discrimination by any finite parallel experiment. Datta–Biswas–Saha–Augusiak [21, Theorems 3–5] study entanglement-assisted single-use discrimination, including perfect constructions

and a qubit entanglement advantage; their qubit construction has three outcomes. These results concern retained-reference discrimination for broader outcome alphabets. For ordered binary effects, (S4.23) bounds the adaptive advantage uniformly in the horizon for each fixed input dimension.

The differential argument in Lemma S4.2 uses the horizontal-gauge and channel-extension method [22, 7, 23, 24]. Yuan–Fung give the parallel path estimate [25, Section III, equation (26), PDF version 3], and Sieniawski–Demkowicz–Dobrzański integrate adaptive bounds [26, Section IV.A]. The binary Sylvester choice produces the regularized variance $M(I - M)$, whose path bound can be integrated explicitly and matched by compressed nonadaptive tests. Inverse-Sylvester expressions in Bures geometry [27] and spectral matrix integration [28] provide further antecedents. Here the form acts on the effect displacement H with variance $V = M(I - M)$; the Bures pullback under $M \mapsto V$ would instead act on the differential $H - MH - HM$. Theorem S5.1 combines the resulting geometry with uniform operational-ball estimates and the nested endpoint integral to obtain the binary-effect covering law.

S18.2. One-use tomography and acquisition resources. Mele–Bittel [5, Theorem I.1] give channel tomography with $O(d_{\text{in}}d_{\text{out}}k\epsilon^{-2})$ calls at fixed confidence. Their binary specialization [5, Lemmas III.7–III.8 and Corollary III.9] returns a legal effect with operator error at most ϵ , using

$$M = 1024d^2\epsilon^{-2} + O(d\epsilon^{-2}\log(1/\eta)), \quad 4e^{-4d^2} < \eta < 1, \quad 0 < \epsilon < 1.$$

It prepares independent maximally entangled input–reference pairs and processes the Choi copies collectively [5, Algorithm I.2, Figure 1 and Section III.2]. It therefore belongs to the fresh-block class with $b = 1$, which allows coherent purification and collective output tomography. The source calls all invocations one parallel block; its inputs from distinct invocations nevertheless remain independent. Its amplification [5, Remark III.4, equation (165)] gives $O(d^2\epsilon^{-2}\log(1/\eta))$ calls for all $0 < \eta \leq 1/8$ with the same acquisitions. The sharper confidence extension derived below uses the dimension-dependent tail in the printed Corollary III.9.

Zambrano–Ramos–Calderer–Kueng [29, Definition 1 and Theorem 2] use worst-input one-use outcome total variation. With L outcomes, their global-2-design protocol has sufficient budget

$$M \geq \frac{8(d^2 + \epsilon(d^2 + 1)/6)}{\epsilon^2} \log \frac{2^{L+1}9^{2d}}{\eta}.$$

It probes known inputs separately and returns a legal POVM by least squares and projection. Thus it also has $b = 1$, without a retained reference. When $L = 2$, its loss is $\|E - F\|_{\text{op}}$, whereas $d_1(E, F) = 2\|E - F\|_{\text{op}}$ in our normalization. Their Theorem 9 gives a constant-confidence lower bound $\Omega((d^3 + d^2L)\epsilon^{-2})$ for nonadaptive single-copy measurements on known input states.

S18.3. Interior learning under the future-use loss. Theorem S10.5 applies the Mele–Bittel binary estimator to $I/4 \preceq E \preceq 3I/4$ with $\epsilon = \delta/(8\sqrt{N})$ and $\eta = 1/8$. The horizontal path estimate gives

$$d_N(E, F) \leq 4\sqrt{N}\|E - F\|_{\text{op}} \quad \text{when } \|E - F\|_{\text{op}} \leq 1/8.$$

Consequently $O(d^2N\delta^{-2})$ independent one-call probes suffice at this confidence. Theorem S10.2 gives the matching lower bound under arbitrary coherent adaptive training. This identifies the joint dimension, horizon, and accuracy order on the stated interior.

For all integers $d, N \geq 1$, $0 < \delta \leq 2^{-13}$ and $0 < \eta \leq 1/8$, Theorem S12.1 gives

$$M_{\text{int}}^*(d, N, \delta, \eta) = \Theta\left(N\delta^{-2}[d^2 + d\log(1/\eta)]\right)$$

with absolute constants. The upper bound has $b = 1$ and the lower bound permits unrestricted coherent adaptive training. Corollary S12.4 gives the corresponding binary operator-learning order $\Theta(\epsilon^{-2}[d^2 + d\log(1/\eta)])$ for $0 < \epsilon \leq 2^{-14}$.

The upper bound uses the Mele–Bittel estimator with base failure $q_d = 2^{-3d}$, which satisfies $4e^{-4d^2} < q_d$ in their Corollary III.9. Run it at a fixed fraction of the desired accuracy, and repeat independently a number of times equal to the smallest positive odd integer at least $2 \log_2(1/\eta)/d$. Metric-majority selection and the binomial tail estimate then give $O(\epsilon^{-2}[d^2 + d \log(1/\eta)])$ calls for every $0 < \eta \leq 1/8$. Lemma S12.3 records this confidence extension of the existing estimator.

For the confidence converse, Mele–Bittel’s classicalisation of diagonal multiple-coin channels [5, Lemma IV.19 and Corollary IV.20] is a direct antecedent. Their corollary gives $\Omega(d\epsilon^{-2} \log(d/\eta))$ for small operator error and $0 < \eta < 1/4$. Lemma S12.2 gives a null-versus-coordinate argument under the present coherent adaptive interface and future-use loss. Revealing the sampled diagonal index isolates the hypothesis-dependent Bernoulli coin after a common basis-dephasing map. The classical–quantum relative-entropy calculation permits different current quantum inputs under the two hypotheses. Summation of the testing inequalities gives $\Omega(dN\delta^{-2} \log(1/\eta))$; its maximum with Theorem S10.2 is the asserted sum up to an absolute factor.

Channel-learning converses have different domains. Oufkir–Girardi [6] and Chen–Zhang–Yu [35, Theorems 1.1 and 4.2] treat fully coherent channel learning in one-use diamond loss. The latter gives $\Omega(rd_{\text{in}}d_{\text{out}}\epsilon^{-2})$ when $rd_{\text{out}} \geq 2d_{\text{in}}$ at constant confidence. Quantum output dimension two still allows nonclassical outputs, so that general-channel theorem does not specialize directly to a binary-qc converse. For binary measurements, Mele–Bittel give their projector packing [5, Proposition IV.17] and the diagonal-coin bound above. The near-fair binary information estimate in Lemma S10.1 supplies the self-contained growing-dimension converse used here.

S18.4. The closed body and independent acquisition blocks. On the complete effect body, the hybrid inequality

$$d_N(E, F) \leq 2N\|E - F\|_{\text{op}}$$

converts both one-use tomography guarantees to sufficient budgets $O_d(N^2\delta^{-2} \log(1/\eta))$. Theorem S9.2 and Corollary S9.5 give the corresponding acquisition obstruction: for fixed $d \geq 2$, every fresh $b = 1$ procedure learning the entire ordered binary effect body requires

$$M \geq cN^2\delta^{-2} \log(1/\eta).$$

The converse allows classical adaptive choices, stored references, their collective processing, and public bounded stopping. It therefore applies to every reconstruction from the acquisitions of both cited tomography protocols. The projective boundary forces this full-body horizon order, while the interior calculation above permits the linear order using the same acquisition resource.

More generally, for $d \geq 2$, $1 \leq b \leq N$, $0 < \delta \leq \delta_*$ and $0 < \eta \leq 1/8$, Theorem S9.2 proves

$$c \frac{N^2}{b} \delta^{-2} \log(1/\eta) \leq M_b^*(d, N, \delta, \eta) \leq Cd^4 \frac{N^2}{b} \delta^{-2} \log(d/\eta).$$

Here M_b^* is the worst-record budget for a common classical estimate in the fresh-block model of Section S9. Each block may use a reference and adaptive quantum controls. Stored outputs can be processed jointly, but only the classical record determines the next fresh probe; earlier quantum registers cannot feed its unfinished queries. At $b = N$ this gives the fixed-dimensional linear-horizon order, and at $b = 1$ the quadratic-horizon order. For $d = 1$, the cost is $\Theta(N\delta^{-2} \log(1/\eta))$ independently of b .

The all-confidence operator estimator gives a further full-body upper bound. Use Lemma S12.3 at $\epsilon = \delta/(2N)$, or use the block construction, selecting the smaller public budget before acquisition. This yields (S12.9):

$$M_b^*(d, N, \delta, \eta) \leq C \frac{N^2}{\delta^2} \min \left\{ d^2 + d \log \frac{1}{\eta}, \frac{d^4}{b} \log \frac{d}{\eta} \right\},$$

for $d \geq 2$, $1 \leq b \leq N$, $0 < \delta \leq \min\{2^{-13}, \delta_*\}$ and $0 < \eta \leq 1/8$. The lower and upper dimension dependences for the complete body remain unmatched when the parameters grow jointly.

The block scale has established metrological antecedents. Hyllus et al. [33, Observation 1, equation (9)] and Tóth [34, Observation 3, equation (6)] prove, for b -producible qubit probes under linear phase encoding,

$$F_Q \leq \lfloor M/b \rfloor b^2 + (M - b \lfloor M/b \rfloor)^2 \leq Mb.$$

Our block parameter also permits sequential controls within a fresh block and concerns query acquisition rather than multipartite entanglement of the final state. The finite-pair fidelity argument gives the confidence factor without an unbiasedness assumption. Salek–Hayashi–Winter [9, Section II.2 and Section V.2] formulate classical feed-forward with arbitrary output quantum memory and no input quantum memory, and study asymptotic discrimination exponents. Wilde et al. [14, Proposition 21 and Appendix B] give adaptive divergence converses and fidelity-to-error inequalities. Section S9 establishes the finite-budget conditional fidelity estimate with variable block lengths and stopping. Its upper bound blocks the future tester and applies the common learner at the shorter horizon.

Restrictions on jointly measuring already supplied state copies are a separate resource [36, 37]. The fresh-query restriction used here permits an unrestricted final joint measurement. Mele–Bittel’s blockwise Choi processing [5, Section I.5] concerns the former resource.

The full-body common learner uses multiscale phase recovery [31, 32], with a proved safe-angle induction, bias cancellation, unknown-noise block selection, and conditional confidence allocation. Its matrix extension calibrates both support endpoints and controls variance from data-selected compressions before assembling a legal estimate. These operations provide one procedure at rank changes and repeated eigenvalues; the pair-dependent witnesses of the metric proof serve only to establish operational separation.

S18.5. Exact descriptions and finite collective synthesis. Theorem S8.1 gives an exact rational selection rule with optimal fixed-dimensional payload on the closed body in its small-error range. Theorem S11.1 has explicit dimension-uniform constants on $[I/4, 3I/4]$. Its norm-ball volume argument and the Fano–Kraft proof of Theorem S11.4 use standard geometric and information-theoretic mechanisms to obtain the future-loss description laws. Theorem S11.5 coarsens the Mele–Bittel estimator to a finite output and the exact public dictionary without further device calls; its confidence extension is Corollary S12.5.

Computing the finite readout is a further question: measurable finite coarsening alone does not give computable measurement entries. In Section S14, Lemma S14.1 expresses the risk on independent Choi records by a first-order formula over the reals with rational coefficients. It includes positivity and completeness of the conditional POVMs, the fixed dictionary, and quantification over every real target in the interior. Real quantifier elimination decides feasibility; real-closed-field transfer supplies an algebraic point, found by exact enumeration and sign tests [38, Theorems 2.77 and 2.80 and Chapter 14]. An increasing search of the sample count terminates at the order supplied by the statistical existence bound. The resulting Theorem S14.2 gives an algebraic readout and finite trusted controls with

$$M = O(N\delta^{-2}[d^2 + d\log(1/\eta)]), \quad B = \frac{1}{2}d^2 \log_2 N + d^2 \log_2(1/\delta) + O(d^2).$$

Section S15 gives finite rational certificates for the same uniform risk. If a fixed readout has failure at most α on an operator-norm r -net for good-label radius a , Theorem S15.1 gives

$$\sup_{E \in \mathcal{J}_d} \mathbb{P}_E\{\|E - C_J\|_{\text{op}} > a + r\} \leq \alpha + mr.$$

The analytic input is $\|\Gamma_m(E) - \Gamma_m(G)\|_1 \leq 2m\|E - G\|_{\text{op}}$. For each nearby grid point G , one transfers its fixed set of good labels between the two experiments; no continuity of label selection

is assumed. With $a = \delta/(8\lceil\sqrt{N}\rceil)$ and $r \leq \min\{\delta/(32\lceil\sqrt{N}\rceil), \eta/(16m)\}$, exact grid inequalities at failure $\eta/4$ certify operational error $5\delta/8$ with failure at most $5\eta/16$ for all real interior effects.

Proposition S15.2 mixes an arbitrary L -outcome readout with the uniform outcome and rounds its first $L - 1$ effects at prescribed denominator $Q = \lceil 64L^2d^m/\eta \rceil$. The last effect is their complement, giving exact completeness; the mixing margin preserves positivity. The change of every output-index law is bounded by $3\eta/32$. A finite enumeration therefore contains a certified rational readout whenever the statistical construction supplies the required slack. Exhausting this enumeration at each dyadic call count gives Theorem S15.3 at the same joint call and payload orders. Only existence of the tomography readout is used: its entries and continuous-outcome integrals are not inputs to either synthesis.

Finite-control approximation charges the complete preparation and classically conditioned readout instruments. The pathwise total unhalved error allowance η changes index probabilities by at most $\eta/2$, so the rational construction has actual failure at most $13\eta/16 < \eta$. Acquisition remains fresh and one-call, and the decoder returns the unchanged legal rational centre. Neither construction requires coherent access to the device's classical outputs or environment. Rational verification replaces real quantifier elimination in the latter construction, while both separate device calls and payload from preprocessing, storage, and known-control work.

Finite and efficient state tomography have substantial existing constructions. Pelecanos–Spilecki–Tang–Wright [39, Theorems 1.1–1.3, Section 3.3 and Figures 8–9] give an efficient purification reduction and efficient pure- and mixed-state tomography; their Theorem 4.2 supplies discrete approximate-design circuits used by the pure-state protocol. Preserving the covariance and binary operator-error analysis of the imported channel estimator under a different tomography implementation requires an additional argument. The constructions here instead select a feasible collective readout with a uniform future-loss guarantee and the exact public dictionary. They prove finite computability and, for the rational construction, finite verifiability of the risk conditions. Their exhaustive search, dictionary reconstruction, workspace, and known-gate costs have no efficiency bound.

Result and acquisition	Loss and parameter range	Training-call order
Mele–Bittel with Lemma S12.3; independent Choi probes, $b = 1$	Entire binary body; operator error ϵ ; $0 < \eta \leq 1/8$	$O(\epsilon^{-2}[d^2 + d \log(1/\eta)])$; source estimator with a confidence extension
Zambrano et al.; individual known probes, $b = 1$	Binary global-design protocol; operator error ϵ ; $0 < \eta < 1$	$O(d^2\epsilon^{-2}[d + \log(1/\eta)])$
Theorem S12.1; independent Choi probes, $b = 1$	$I/4 \preceq E \preceq 3I/4$; future error δ ; $d, N \geq 1$, $0 < \eta \leq 1/8$	$\Theta(N\delta^{-2}[d^2 + d \log(1/\eta)])$; coherent adaptive lower bound
Theorem S14.2; synthesized algebraic readout, $b = 1$	Same interior and future loss; rational δ, η	$O(N\delta^{-2}[d^2 + d \log(1/\eta)])$; finite trusted-control specification
Theorem S15.3; certified rational readout, $b = 1$	Same interior and future loss; rational δ, η	$O(N\delta^{-2}[d^2 + d \log(1/\eta)])$; finite rational risk certificate
Theorem S9.2; fresh blocks of size $\leq b$	Entire binary body; future error δ ; $d \geq 2$, $1 \leq b \leq N$, $0 < \eta \leq 1/8$	$\Theta_d((N^2/b)\delta^{-2} \log(1/\eta))$; explicit upper bound in (S12.9)

The future-loss rows use the absolute small-error ranges stated in their theorems. The sharp joint dimension law applies to the displayed interior, while the closed-body law is optimal in

horizon, accuracy, and confidence for fixed dimension. The binary consuming interface is common to the preceding table. The next subsection gives the separate finite-outcome interior analysis; residual device quantum outputs are not covered by either family.

S18.6. Finite-outcome interiors and existing tomography. Mele–Bittel’s general channel theorem [5, Theorem III.3] already applies to channels with any finite classical output alphabet. The present extension is not a claim that learning a multi-outcome POVM is new. Its constructive upper proof uses their binary Corollary III.9 through the confidence amplification and rational realization in Sections S12 and S15. For each label, the public randomization $E_j \mapsto I_d/4 + E_j/2$ supplies exactly the binary Choi record required by that primitive. The operator accuracy is $\delta/(16k\lceil\sqrt{N}\rceil)$, not δ/N . The dimension-independent horizontal estimate supplies this improved horizon conversion. The lower theorem permits coherent adaptive training and keeps its $d\log(1/\eta)$ term.

The general-outcome procedure of Zambrano–Ramos–Calderer–Kueng [29, Theorem 2] can also be evaluated directly under this future-use loss. Their worst-input outcome total variation dominates each $\|E_j - F_j\|_{\text{op}}$: a single label is one of the events in the total-variation supremum. A finite legalization into $\mathfrak{P}_{d,k}$ increases the maximum component error by at most a constant factor, as in the proof of Theorem S23.1. Consequently their stated sufficient budget, with operator target of order $\delta/(k\sqrt{N})$, gives

$$O\left(k^2 N \delta^{-2} d^2 [d + k + \log(1/\eta)]\right)$$

calls in their independent known-input, no-retained-reference model. This is a direct sufficient future-loss analysis of that estimator, not a lower bound against it. Its extra dimension factor is consistent with its different acquisition model. We neither infer that every prior estimator requires the coarse N^2 substitution nor claim that the displayed growing- k bounds are minimax sharp.

The multi-outcome discrimination examples cited above concern broader boundary families. They are not contradicted by an interior comparison under $E_j \succeq I_d/(2k)$. The new tuple entropy uses its affine dimension $(k-1)d^2$ and a norm-ball volume calculation. It is not an extension of the binary endpoint logarithmic law to an unanalysed multi-outcome boundary. All comparisons here use the June 15, 2026 version of Mele–Bittel and the published July 15, 2026 version of Zambrano–Ramos–Calderer–Kueng. An external human priority judgment is separate from this author-side theorem comparison.

S19. A NOISE-UNIFORM READOUT TRANSITION

The dependence on the readout basis can change scale when noise approaches an extremal measurement. We determine that transition in a fixed one-dimensional family, without a positive spectral margin uniform in the noise. The proof is a finite-distance argument; no conclusion is inferred merely from a Fisher-information calculation.

Let σ_1, σ_2 be the first two Pauli matrices. For $\theta \in \mathbb{R}/(2\pi\mathbb{Z})$ and a public visibility $0 \leq \lambda \leq 1$, put

$$(S19.1) \quad E_{\theta,\lambda}^{\pm} = \frac{1}{2}(I \pm \lambda(\cos \theta \sigma_1 + \sin \theta \sigma_2)), \quad \mathcal{M}_{\theta,\lambda}^{\pm}(X) = \text{tr}(E_{\theta,\lambda}^{\pm} X).$$

Both outcome labels are retained. The output quantum system is one-dimensional. In particular, θ and $\theta + \pi$ are not identified by an outcome relabelling. Let $h(\theta, \phi) \in [0, \pi]$ be circular distance and set

$$(S19.2) \quad \eta = 1 - \lambda^2, \quad K_{N,\lambda} = \lambda \sqrt{N \min\{N, \eta^{-1}\}}, \quad N \geq 1,$$

where $\eta^{-1} = +\infty$ at $\lambda = 1$. Thus $K_{N,1} = N$ and $K_{N,0} = 0$. The visibility may depend arbitrarily on N and the requested error; it is not a target coordinate charged in this theorem.

Write $\mathcal{C}_{N,\lambda}(\delta)$ for the least number of legal memoryless binary instruments of this interface whose d_N -balls of radius δ cover all targets in (S19.1). Centres need not belong to the displayed family. Define $\mathcal{C}_{N,\lambda}^{\text{rat}}(\delta)$ by restricting centres to family members with rational $\cos \theta, \sin \theta$; rational visibility then gives rational effects. A constant in the following statement is independent of all three parameters N, λ, δ .

Theorem S19.1 (Noise-uniform finite-use coding). *For all $N \geq 1$, $0 \leq \lambda \leq 1$, and all pairs of angles,*

$$(S19.3) \quad \frac{1}{64} \min\{1, K_{N,\lambda} h(\theta, \phi)\} \leq d_N(\mathcal{M}_{\theta,\lambda}, \mathcal{M}_{\phi,\lambda}) \leq \min\{2, K_{N,\lambda} h(\theta, \phi)\}.$$

There are absolute $c, C > 0$ such that, for $0 < \delta \leq 2^{-10}$,

$$(S19.4) \quad c \left(1 + \frac{K_{N,\lambda}}{\delta}\right) \leq \mathcal{C}_{N,\lambda}(\delta) \leq \mathcal{C}_{N,\lambda}^{\text{rat}}(\delta) \leq C \left(1 + \frac{K_{N,\lambda}}{\delta}\right).$$

Consequently the optimum fixed-length reusable payload is

$$(S19.5) \quad \log_2 \left(1 + \frac{K_{N,\lambda}}{\delta}\right) + O(1).$$

The upper construction is an exact rational algorithm on rational visibility, tolerance and unit-circle input coordinates. At $\lambda = 0$ the family is a singleton and its optimum payload is zero.

The constants and the additive remainder are uniform up to the erased and projective endpoints. In contrast, they do not assert an optimal leading constant, a large-error coherent covering law, or a coding law in which an unknown visibility is also encoded.

S19.1. A two-point programme with its noise dependence retained.

Lemma S19.2 (An adaptive upper modulus). *Let $h = h(\theta, \phi)$ and $a = \lambda \sin(h/2)$. If $\eta > 0$, then*

$$(S19.6) \quad d_N(\mathcal{M}_{\theta,\lambda}, \mathcal{M}_{\phi,\lambda}) \leq \min \left\{ 2, 2Na, 2\sqrt{1 - \left(\frac{\eta}{\eta + a^2}\right)^N} \right\}.$$

For $\eta = 0$ the bound $\min\{2, 2Na\}$ remains valid.

Proof. The difference of the two positive-outcome effects has operator norm a . On a bipartite input state, the two output blocks of the difference are B and $-B$, where $B = \text{tr}_A[(E_{\theta,\lambda}^+ - E_{\phi,\lambda}^+) \otimes I]\rho$. Duality against Hermitian contractions gives $\|B\|_1 \leq a$. An eigenstate of the effect difference attains equality. Hence $d_1 = 2a$, including reference assistance. Replacing the calls one at a time and using trace-norm contraction proves $d_N \leq 2Na$ for every common adaptive tester.

Choose angle representatives with difference h and let u, v be the unit vectors along their midpoint and its oriented perpendicular in the equatorial plane. The two Bloch vectors are $m \pm av$, where $m = \lambda \cos(h/2)u$. Put $b = (1 - \|m\|^2)^{1/2} = (\eta + a^2)^{1/2}$. The two Bloch vectors $m \pm bv$ have norm one, so they define two legal binary projective instruments, say $\mathcal{A}_+$ and $\mathcal{A}_-$. The target pair is exactly the two mixtures of these same instruments with programme laws

$$p_{\pm} = ((1 \pm t)/2, (1 \mp t)/2), \quad t = a/b.$$

This construction depends on the pair, which is permissible in a metric upper bound. It is not one fixed processor for the whole circle.

For a fixed tester, first supply N independent programme bits and, at each requested call, apply the indicated $\mathcal{A}_+$ or $\mathcal{A}_-$. All subsequent operations, including the reference, feedback and a stopping rule, are a common channel of these bits. Unused bits are discarded. The root fidelity of p_+, p_- is $\sqrt{1 - t^2}$. Purifying the classical laws and using contraction gives

$$\|p_+^{\otimes N} - p_-^{\otimes N}\|_1 \leq 2\sqrt{1 - (1 - t^2)^N}.$$

This also follows directly from the classical fidelity inequality. Substitution proves (S19.6). Finally $1 - (1 - t^2)^N \leq Nt^2$ and $a \leq \lambda h/2$ imply

$$d_N \leq \min\{N\lambda h, \sqrt{N/\eta} \lambda h, 2\}.$$

The limiting projective bound is supplied by the hybrid estimate, not by division by zero. $\square$

Classical simulation by common extremal channels and the effect of noise on the Heisenberg scale are established methods [7]. Environment/programme-state processing under adaptive use is also an established framework [13, 14, 15]. The point needed here is the exact two-point expression (S19.6) and its uniform conversion to a family cover, rather than a new claim about the origin of those methods.

S19.2. Finite entangled blocks and a binary product estimate.

Lemma S19.3 (Symmetric Bernoulli products). *Let $P_{\pm}$ be Bernoulli laws with success probabilities $(1 \pm a)/2$, where $0 \leq a \leq 1$. For every integer $m \geq 1$,*

$$(S19.7) \quad \|P_+^{\otimes m} - P_-^{\otimes m}\|_1 \geq \frac{1}{2} \min\{1, a\sqrt{m}\}.$$

Proof. Retain the largest odd number $r = 2j + 1 \leq m$ of coordinates, so $r \geq m/2$. For their majority event write $g_r(a) = P_+(\text{majority}) - P_-(\text{majority})$. Differentiating the finite binomial sum and cancelling consecutive terms gives

$$g'_r(a) = r \binom{2j}{j} 4^{-j} (1 - a^2)^j.$$

For $j \geq 1$, the bound $\binom{2j}{j} 4^{-j} \geq 1/(2\sqrt{j})$ follows by induction: the ratio of consecutive left sides is $(2j+1)/(2j+2) \geq \sqrt{j/(j+1)}$. For $0 \leq a \leq r^{-1/2}$, Bernoulli's inequality gives $(1 - a^2)^j \geq 1 - j/r \geq 1/2$. Thus $g'_r(a) \geq \sqrt{r}/(2\sqrt{2})$ on that interval. The case $r = 1$ is immediate. Integration from zero and monotonicity yield $g_r(a) \geq (2\sqrt{2})^{-1} \min\{1, a\sqrt{r}\}$. Unhalved distance is at least twice this event difference; using $r \geq m/2$ proves the stated bound. $\square$

Lemma S19.4 (A noise-uniform pairwise witness). *The lower inequality in (S19.3) is witnessed by nonadaptive entangled blocks, each of at most N qubits, followed by classical parity and majority processing.*

Proof. A block of k inputs is prepared in $(|0\rangle^{\otimes k} + e^{i\alpha}|1\rangle^{\otimes k})/\sqrt{2}$. Multiplying the k output signs of (S19.1) has mean $\lambda^k \cos(k\theta - \alpha)$. This identity follows by applying $\bigotimes_{i=1}^k (E^+ - E^-)$ to the two off-diagonal terms of the block density matrix. For the two hypotheses choose the one common phase $\alpha = k(\theta + \phi)/2 + \pi/2$. Their parity laws then have opposite biases of magnitude

$$(S19.8) \quad a_k = \lambda^k |\sin(kh/2)|.$$

Repeating $m = \lfloor N/k \rfloor$ independent blocks and ignoring remaining calls is an admissible tester. Its unhalved output distance is at least $\frac{1}{2} \min\{1, \sqrt{m} a_k\}$ by Lemma S19.3. The phase and block size may depend on the pair, but never on the unknown hypothesis or a private outcome.

Assume $\lambda > 0$ and put $T = \min\{N, \eta^{-1}\}$, $k_0 = \lfloor T \rfloor$. Then $T/2 \leq k_0 \leq T$. For every $1 \leq k \leq k_0$,

$$(S19.9) \quad \lambda^k \geq \lambda/2.$$

Indeed, when $0 < \eta < 1$, the inequality $\log(1 - \eta) \geq -\eta/(1 - \eta)$ and $k - 1 \leq \eta^{-1} - 1$ give $(k - 1) \log \lambda \geq -1/2$. At $\eta = 0$ the assertion is immediate.

If $h \leq 1/k_0$, choose $k = k_0$. Since $\sin(kh/2) \geq kh/\pi$ and $m \geq N/(2k)$, we obtain

$$\sqrt{m} a_k \geq \frac{\lambda \sqrt{N k_0}}{2\pi \sqrt{2}} h \geq \frac{K_{N,\lambda}}{4\pi} h.$$

This gives a lower constant $1/(8\pi)$. If $k_0 = 1$, use $k = 1$ for all $h \leq \pi$ instead. Now $T < 2$ unless $T = 1$, and $\sqrt{N}\lambda \sin(h/2) \geq K_{N,\lambda}h/(\pi\sqrt{2})$; this is stronger than required.

It remains that $k_0 \geq 2$ and $h > 1/k_0$. Here $\lambda \geq 1/\sqrt{2}$. Choose $k = \max\{1, \lfloor 1/h \rfloor\}$. Then $k \leq k_0$ and $1/2 \leq kh \leq \pi$. Equations (S19.8) and (S19.9) give $a_k \geq 1/(4\pi\sqrt{2})$. Even one such block has distance at least $1/(8\pi\sqrt{2}) > 1/64$. The three cases prove the claim. For $\lambda = 0$ both sides of the desired inequality vanish. $\square$

S19.3. Covering and exact rational realization.

Proof of Theorem S19.1. The metric assertion follows from the preceding lemmas. For a lower cover choose points on the circle with circular separation at least $256\delta/K_{N,\lambda}$ whenever that quantity is at most π . An equally spaced packing has at least a fixed constant times $K_{N,\lambda}/\delta$ points. Every two points have d_N -distance at least 4δ , because $256\delta \leq 1/4$. No radius- δ ball, even about an arbitrary legal instrument, can contain two such points. If the required angular separation exceeds π , the trivial lower bound one suffices, with a smaller absolute constant. This proves the first inequality in (S19.4).

For the upper bound use the two rational semicircle charts

$$(S19.10) \quad z_s(t) = s \left(\frac{1-t^2}{1+t^2}, \frac{2t}{1+t^2} \right), \quad s \in \{-1, 1\}, \quad -1 \leq t \leq 1.$$

Each chart has angular speed $2/(1+t^2) \leq 2$. Round t to the nearest j/B , $-B \leq j \leq B$, with a fixed rule at ties. The circular error is at most $1/B$. Thus any integer $B \geq \max\{1, K_{N,\lambda}/\delta\}$ yields a cover with at most $4B + 2$ centres. For rational inputs, an entirely rational choice is

$$(S19.11) \quad B = \max \left\{ 1, \min \left\{ \left\lceil \frac{4N\lambda}{\delta} \right\rceil, \left\lceil \sqrt{\frac{16N\lambda^2}{\eta\delta^2}} \right\rceil \right\} \right\},$$

omitting the second entry when $\eta = 0$. Integer squaring and rational comparison compute the ceiling square root exactly. Both entries are ceilings of the corresponding real bounds, so their minimum is $\lceil 4K_{N,\lambda}/\delta \rceil$. For $z = (x, y)$ on the rational unit circle choose $s = 1$ when $x \geq 0$ and $s = -1$ otherwise, and set $t = sy/(1+sx)$. Its denominator is at least one, so it is rational and belongs to $[-1, 1]$.

The chart digit (s, j) determines a legal effect pair with eigenvalues $(1 \pm \lambda)/2$. No approximate positivity test, irrational normalization or search over real angles is needed. A fixed-length index over $4B + 2$ slots includes the chart sign. Its length gives (S19.5). Redundant chart endpoints cause at most a constant overhead. At $\lambda = 0$ dispense with the chart and emit the single fair, input-erasing measurement. $\square$

Corollary S19.5 (A continuous family of horizon exponents). *Fix $0 < \delta \leq 2^{-10}$. If $\lambda_N \rightarrow 1$ and $1 - \lambda_N^2 = N^{-\alpha+o(1)}$ for $\alpha \geq 0$, then*

$$\frac{\log_2 \mathcal{C}_{N,\lambda_N}(\delta)}{\log_2 N} \rightarrow \frac{1 + \min\{\alpha, 1\}}{2}.$$

If $\lambda_N = N^{-\beta+o(1)} \rightarrow 0$, with $\beta \geq 0$, the limit is $\max\{1/2 - \beta, 0\}$. The projective and erased endpoints have coefficients one and zero, respectively.

Proof. Substitute the two regimes in (S19.2) and use (S19.4). The term $1 + K/\delta$ handles a vanishing or subpolynomial scale without taking the logarithm of zero. $\square$

S20. RETAINED MULTI-OUTCOME GEOMETRY AND CONSEQUENCES

S21. REGULARIZATION TRANSPORTED BY THE OUTPUT CHANNEL

The fixed Euclidean ridge in Theorem 2.1 is tied to the displayed outcome coordinates. Its contraction under arbitrary stochastic relabeling does not follow from the unregularized data-processing theorem. A ridge obtained from a public probability vector has a different, covariant property.

For a probability vector $w = (w_j)$, put $W = (w_j I_d)_j$ and $\mathcal{B}_w = \mathcal{C}_W$. On the full real Hermitian tuple space this operator is

$$(S21.1) \quad (\mathcal{B}_w K)_j = w_j \left(K_j - \sum_{\ell} w_{\ell} K_{\ell} \right).$$

For a zero-sum direction H and $\tau > 0$ define the extended energy

$$(S21.2) \quad \mathcal{E}_{E,w,\tau}(H) = \sup_{K \in \mathcal{H}_d^k} \{2\langle H, K \rangle - \langle K, (\mathcal{C}_E + \tau \mathcal{B}_w) K \rangle\}.$$

It equals the pseudoinverse quadratic form on the range and is $+\infty$ off the range. Constant tuples have zero energy and zero pairing with H . If every $w_j > 0$, the supremum is finite and may be evaluated by restricting to $\mathcal{T}$.

Theorem S21.1 (Processing-covariant regularized energy). *For every column-stochastic output map T , every $\tau > 0$, and every zero-sum Hermitian tuple H ,*

$$(S21.3) \quad \mathcal{E}_{TE,Tw,\tau}(TH) \leq \mathcal{E}_{E,w,\tau}(H).$$

Zero entries of Tw are allowed, with the extended-energy convention in (S21.2). If a stochastic map S satisfies $ST = \text{Id}$, equality holds in (S21.3). In particular, reversible outcome splitting leaves this energy unchanged when the public weights are split by the same rule.

Proof. The matrix Jensen identity in Proposition 2.4 is an inequality on the full tuple space, not only on its zero-sum subspace. It gives

$$\langle L, \mathcal{C}_{TE} L \rangle \geq \langle T^* L, \mathcal{C}_E T^* L \rangle.$$

Apply the same inequality to the scalar measurement W to get $\langle L, \mathcal{B}_{Tw} L \rangle \geq \langle T^* L, \mathcal{B}_w T^* L \rangle$. Also $\langle TH, L \rangle = \langle H, T^* L \rangle$. Thus each expression in the supremum defining the left side of (S21.3) is bounded by a permissible expression on the right. Taking suprema proves the claim, including infinite values. No assumption that $T^* L$ has zero sum is needed. If $ST = \text{Id}$, application of the same inequality to S gives the reverse bound. $\square$

Proposition S21.2 (Relation to the operational upper certificate). *If $w_j > 0$ and $w_{\max} = \max_j w_j$, then for every pair of legal measurements, with midpoint M and difference H ,*

$$(S21.4) \quad D_N(E, F) \leq \min \left\{ 2, 2\sqrt{k+1} \sqrt{N \mathcal{E}_{M,w,1/(Nw_{\max})}(H)} \right\}.$$

For $w_j = 1/k$, one has $\mathcal{B}_w|_{\mathcal{T}} = \text{Id}/k$, and $\mathcal{E}_{M,w,k/N}(H) = \mathcal{Q}_N(E, F)^2/N$ exactly.

Proof. The variance expression gives $0 \preceq \mathcal{B}_w \preceq w_{\max} \text{Id}$. On $\mathcal{T}$, all coordinates of w positive make $\mathcal{B}_w$ positive definite. Inverse order therefore gives

$$\langle H, (\mathcal{C}_M + N^{-1} \text{Id})^{-1} H \rangle \leq \mathcal{E}_{M,w,1/(Nw_{\max})}(H).$$

Use Theorem 2.1. The uniform formula follows by substitution into (S21.1). $\square$

Theorem S21.1 compares the same τ and *transported* weights. Replacing Tw by a new uniform vector, or changing τ to the new alphabet size divided by N , is a different operation. Consequently this result does not retroactively assert arbitrary-processing monotonicity of the original fixed ridge. Exact evaluation with positive rational weights uses the tangent Gram system of Proposition 2.6, with the ridge stiffness $\tau\mathcal{B}_w$ in place of G/N .

S22. FINITE-OUTCOME GEOMETRY AND EXACT DESCRIPTIONS

The interior geometry does not require two outcomes. We now allow any fixed finite alphabet, while retaining the input-consuming, ordered, classical-output interface. The effects below need not commute. The binary results on the complete effect body remain separate: no positive lower spectral bound is imposed in those results.

For integers $d \geq 1$ and $k \geq 2$, put

$$(S22.1) \quad \mathfrak{P}_{d,k} = \left\{ \mathbf{E} = (E_1, \dots, E_k) : E_j = E_j^*, \quad \sum_{j=1}^k E_j = I_d, \quad \frac{I_d}{2k} \preceq E_j \preceq \frac{3I_d}{2k} \right\}.$$

An element specifies the channel $\mathcal{M}_{\mathbf{E}}(\rho) = \sum_j \text{tr}(E_j \rho) |j\rangle\langle j|$. It returns no residual quantum system. Write D_N for its N -use distance, with precisely the reference, feedback, bounded public stopping, and unhalved trace-norm conventions defining d_N . The superscript na denotes nonadaptive experiments, not a restriction to unentangled inputs. Set

$$r(\mathbf{E}, \mathbf{F}) = \max_j \|E_j - F_j\|_{\text{op}}, \quad p = (k-1)d^2.$$

The affine space $\sum_j E_j = I_d$ has real dimension p . In its translated vector space $\sum_j H_j = 0$, the set $\mathfrak{P}_{d,k}$ is exactly the closed r -ball of radius $R = 1/(2k)$ about $\mathbf{U} = (I_d/k, \dots, I_d/k)$.

Theorem S22.1 (A dimension-independent interior comparison). *For $d, N \geq 1$, $k \geq 2$, and $\mathbf{E}, \mathbf{F} \in \mathfrak{P}_{d,k}$,*

$$(S22.2) \quad \frac{1}{128} \min\{1, \sqrt{N} r(\mathbf{E}, \mathbf{F})\} \leq D_N^{\text{na}}(\mathbf{E}, \mathbf{F}) \leq D_N(\mathbf{E}, \mathbf{F}) \leq \min\{2, 2k\sqrt{N} r(\mathbf{E}, \mathbf{F})\}.$$

The lower bound holds for arbitrary legal ordered k -outcome measurements, without the spectral bounds in (S22.1). The upper constants are independent of d and N ; dependence on k is displayed and is not asserted sharp.

Proof. Let $H_j = F_j - E_j$ and $A_j(t) = E_j + tH_j$ for $0 \leq t \leq 1$. Each $A_j(t)$ is positive definite. Solve the matrix differential equation

$$(S22.3) \quad \dot{X}_j(t) = \frac{1}{2} X_j(t) A_j(t)^{-1} H_j, \quad X_j(0) = \sqrt{E_j}.$$

Its coefficients are continuous. Differentiating $X_j^* X_j$ shows that it solves the linear equation

$$\dot{B}_j = \frac{1}{2} H_j A_j^{-1} B_j + \frac{1}{2} B_j A_j^{-1} H_j, \quad B_j(0) = E_j.$$

The function A_j solves this same equation, so uniqueness gives $X_j^* X_j = A_j$. Consequently

$$(S22.4) \quad X_j^* \dot{X}_j = \frac{1}{2} H_j, \quad \dot{X}_j^* \dot{X}_j = \frac{1}{4} H_j A_j^{-1} H_j.$$

This argument neither diagonalizes the tuple nor assumes that A_j and H_j commute.

Use the Stinespring isometry

$$W_t \psi = \sum_{j=1}^k |j\rangle_Y |j\rangle_Z X_j(t) \psi.$$

Tracing out Z and the last factor gives $\mathcal{M}_{A(t)}$. The environment in this dilation is used only in the proof and is not supplied by the device. Since $\sum_j H_j = 0$, (S22.4) yields

$$(S22.5) \quad W_t^* W_t = I_d, \quad W_t^* \dot{W}_t = 0, \quad \dot{W}_t^* \dot{W}_t = \frac{1}{4} \sum_j H_j A_j(t)^{-1} H_j.$$

For completeness, purify a common adaptive tester, retaining a fresh inaccessible environment for each call. Its controls are isometries independent of t . Differentiate the final purified state. It is a sum of at most N terms, with $\dot{W}_t$ replacing one occurrence of W_t in each term. In the inner product of two distinct terms, cancel the later common isometries. At the later differentiated slot the remaining contraction is $W_t^* \dot{W}_t$ or its adjoint, hence zero. Each diagonal term has norm at most $\|\dot{W}_t\|_{\text{op}}$. The derivative of the purified state therefore has norm at most $\sqrt{N}\|\dot{W}_t\|_{\text{op}}$. The derivative of its rank-one density operator has trace norm at most twice this number. Partial trace and all final processing contract trace norm. Bounded stopping is covered by a coherent stopping flag and dummy calls on a fixed discarded input after stopping; the resulting final state of the original tester is a common postprocessing of this padded experiment.

Integrating the derivative bound gives the more informative inequality

$$(S22.6) \quad \begin{aligned} D_N(\mathbf{E}, \mathbf{F}) &\leq \sqrt{N} \int_0^1 \left\| \sum_j H_j A_j(t)^{-1} H_j \right\|_{\text{op}}^{1/2} dt \\ &\leq \sqrt{2kN} \left\| \sum_j H_j^2 \right\|_{\text{op}}^{1/2} \leq \sqrt{2k} \sqrt{N} \max_j \|H_j\|_{\text{op}}. \end{aligned}$$

We used $A_j(t)^{-1} \preceq 2kI_d$ and the order-preserving map $B \mapsto H_j B H_j$; commutation is unnecessary. The weaker constant $2k$ in (S22.2) is convenient below. The trace distance is always at most two.

For the lower bound, choose a label j and a unit eigenvector of $F_j - E_j$ attaining its operator norm. Send that same vector in every call and replace the observed label by the indicator of j . The two records are products of Bernoulli laws with parameters $s, t \in [0, 1]$ and $|s - t| = \|F_j - E_j\|_{\text{op}}$. Since $\max\{s(1-s), t(1-t)\} + N^{-1} \leq 5/4$, the quantity in (S16.2) is at least $\frac{1}{2}\sqrt{N}|s - t|$. Lemma S16.2 gives the lower constant $1/128$. Maximize over j . No interior assumption was used in this part. $\square$

S22.1. The affine norm ball and its entropy. Let $\mathcal{C}_{N,k}(\delta)$ be the least number of radius- δ D_N -balls covering $\mathfrak{P}_{d,k}$. Their centres may be arbitrary legal memoryless ordered k -outcome measurements.

Theorem S22.2 (Dimension-uniform finite-outcome entropy). *For $d, N \geq 1$, $k \geq 2$, and $0 < \delta < 1/128$,*

$$(S22.7) \quad \max \left\{ 1, \left(\frac{\sqrt{N}}{256k\delta} \right)^p \right\} \leq \mathcal{C}_{N,k}(\delta) \leq \left(\frac{3\sqrt{N}}{\delta} \right)^p, \quad p = (k-1)d^2.$$

In particular, uniformly in d, N, δ for each fixed k ,

$$(S22.8) \quad \log_2 \mathcal{C}_{N,k}(\delta) = \frac{p}{2} \log_2 N + p \log_2(1/\delta) + O(p \log(k+1)).$$

Proof. If $D_N(\mathbf{E}, \mathbf{F}) \leq \delta < 1/128$, the lower bound of Theorem S22.1 implies $r(\mathbf{E}, \mathbf{F}) \leq 128\delta/\sqrt{N}$. Lebesgue volume in the p -dimensional translated affine space gives the lower covering bound, because the target is its r -ball of radius $1/(2k)$. This argument also applies when the centre lies outside $\mathfrak{P}_{d,k}$.

Take a maximal ϵ -separated subset of that ball, where $\epsilon = \delta/(2k\sqrt{N})$. It is an r -net. The open r -balls of radius $\epsilon/2$ about its points are disjoint and lie in the ball of radius $R + \epsilon/2$. Their number is at most $(1 + 2R/\epsilon)^p = (1 + 2\sqrt{N}/\delta)^p$. By the upper comparison it is a D_N -cover at radius δ . This proves (S22.7); taking logarithms gives (S22.8), including the range where the displayed power in the lower bound is less than one. $\square$

S22.2. A finite grid with exact normalization. An independent rounding of all k effects need not preserve their sum. The following grid preserves both normalization and the two spectral inequalities simultaneously.

Lemma S22.3 (Rational nets of the balanced body). *Let $0 < t \leq 1/(8k)$ be rational and choose*

$$(S22.9) \quad K = \left\lceil \max \left\{ \frac{4(k-1)d}{t}, 8k(k-1)d \right\} \right\rceil.$$

Enumerate the first $k-1$ Hermitian matrices using diagonal coordinates $0, 1/K, \dots, 1$ and real and imaginary coordinates above the diagonal in $K^{-1}\mathbb{Z} \cap [-1, 1]$. Set the last matrix equal to I_d minus their sum and retain exactly the legal tuples in $\mathfrak{P}_{d,k}$. This finite ordered grid, denoted $\mathcal{G}(t)$, is an r -net of radius t . All membership and pairwise norm-threshold tests are finite exact rational positive-semidefinite tests.

Proof. Put $e = (k-1)d/K$ and $\gamma = 4ke \leq 1/2$. For a target tuple form $E'_j = (1-\gamma)E_j + \gamma I_d/k$. Each E'_j has margin $2e$ from both spectral faces in (S22.1), and $\|E'_j - E_j\|_{\text{op}} \leq 2e$. Round the independent real coordinates of the first $k-1$ matrices to nearest multiples of $1/K$, with a fixed tie rule and Hermitian conjugacy. The row-sum bound gives operator error at most d/K for each rounded matrix. Define the last matrix by exact subtraction from I_d ; its error is at most $(k-1)d/K = e$. Thus every rounded effect still has margin at least e , their sum is exactly I_d , and their distance from the target is at most $3e \leq 3t/4 < t$. The rounded coordinates occur in the stated enumeration.

Legality is checked on $E_j - I_d/(2k)$ and $3I_d/(2k) - E_j$, for every j . The relation $r(\mathbf{E}, \mathbf{F}) \leq s$ is decided by checking $sI_d \pm (E_j - F_j) \succeq 0$ for every j . Exact Schur elimination, including the zero-pivot case, decides these tests over the Gaussian rationals. Equality is included. $\square$

Theorem S22.4 (A public exact code). *For integers $d, N \geq 1$, $k \geq 2$, and rational $0 < \delta < 1/128$, there is a finite deterministic dictionary with centres in $\mathfrak{P}_{d,k}$ and covering error at most $\delta/2$. Its fixed-length index uses*

$$(S22.10) \quad B = \frac{p}{2} \log_2 N + p \log_2(1/\delta) + O(p \log(k+1))$$

bits, with absolute implied constants. A fixed public decoder into arbitrary legal k -outcome measurements cannot improve this order for worst-case radius δ . The construction is finite and exact; no polynomial running-time or workspace assertion is made.

Proof. Put $n = \lceil \sqrt{N} \rceil$ and $s = \delta/(8kn)$, and enumerate $\mathcal{G}(s)$ from Lemma S22.3. Retain a point precisely when its r -distance from every earlier retained point is strictly greater than s . The list is finite and nonempty. It is an r -cover of the grid at radius s ; together with the grid approximation it covers the body at radius $2s$. Its D_N -covering error is therefore at most $2k\sqrt{N}(2s) \leq \delta/2$.

Distinct retained points have r -distance greater than s . The volume argument of Theorem S22.2 gives dictionary size at most

$$(1 + 2R/s)^p = (1 + 8n/\delta)^p \leq (17\sqrt{N}/\delta)^p.$$

The lower bound follows from the same theorem because the dictionary covers at radius δ , even allowing arbitrary legal centres. Taking the ceiling of the logarithm changes the length by at most one.

Encoder and decoder reconstruct the same ordered dictionary from (d, k, N, δ) ; only its index is transmitted. Unused binary words decode to the fixed uniform tuple. For a supplied rational target, scan the list for the first centre within $2s$, using the exact tests in Lemma S22.3. For a real target supplied with certified rational approximations, search for a grid point at certified strict distance less than s and use its assigned predecessor. Such a point exists at distance at most $3s/4$ by the grid lemma, so progressively finer certified input approximations make the strict test terminate. An unrestricted oracle for arbitrary real coordinates is not part of the encoding model. The covering assertion itself concerns every real target, irrespective of its computational representation. $\square$

The grid has at most $[(K+1)^d(2K+1)^{d(d-1)}]^{k-1}$ candidates before the legality tests. Reconstructing this list and comparing its retained points can cost far more than the transmitted index. The estimate concerns the description of an unknown measurement, not free communication of a target-dependent dictionary.

S23. JOINT LEARNING AND FINITE CERTIFICATES FOR A FINITE ALPHABET

We next construct one learner common to all unknown elements of $\mathfrak{P}_{d,k}$. Its training experiments do not use the pair-dependent eigenvectors in the geometric lower bound. The outcome count k is public, as are d, N, δ, η . The upper procedure uses independent one-call probe-reference pairs; coherent collective processing is confined to completed outputs. The converse permits arbitrary coherent adaptive training, with an every-record budget and bounded public stopping.

Theorem S23.1 (Joint finite-outcome learning and coding). *Let $d, N \geq 1$, $k \geq 2$, $0 < \delta \leq 2^{-24}$, and $0 < \eta \leq 1/8$. Write $M_{d,k}^*(N, \delta, \eta)$ for the minimum every-record call budget with worst-case failure at most η under D_N -error δ on $\mathfrak{P}_{d,k}$. There are absolute constants $c, C > 0$ such that*

$$(S23.1) \quad c \frac{N}{\delta^2} \left(d^2 + d \log \frac{1}{\eta} \right) \leq M_{d,k}^*(N, \delta, \eta) \leq C \frac{k^3 N}{\delta^2} \left(d^2 + d \log \frac{k}{\eta} \right).$$

For each fixed k , this is a matching order jointly in d, N, δ, η , including $d = 1$:

$$M_{d,k}^*(N, \delta, \eta) = \Theta_k \left(N \delta^{-2} [d^2 + d \log(1/\eta)] \right).$$

The displayed dependence on growing k is not claimed optimal.

For rational accuracy and confidence, the upper procedure has a finite deterministic specification, returns an index with a fixed public decoder into $\mathfrak{P}_{d,k}$, and simultaneously uses

$$(S23.2) \quad B = \frac{(k-1)d^2}{2} \log_2 N + (k-1)d^2 \log_2(1/\delta) + O((k-1)d^2 \log(k+1))$$

bits. On its good event the decoded error is at most $5\delta/8$. The same call order is available with finite trusted controls. Classical search, readout storage, known-state preparation, and known-control realization are separate resources.

Proof. We first give the constructive upper bound for rational parameters. Set $n = \lceil \sqrt{N} \rceil$ and

$$(S23.3) \quad \epsilon = \frac{\delta}{16kn}.$$

For each j consider the binary effect $F_j = I_d/4 + E_j/2$. It belongs to $[I_d/4, 3I_d/4]$. One call to $\mathcal{M}_E$ simulates a call to $\mathcal{M}_{F_j}$: conditional on the original label x , return bit one with probability $3/4$ when $x = j$ and with probability $1/4$ otherwise. This is an equality of channels, including their action on a retained reference, because the bit-one effect is

$$\frac{3}{4}E_j + \frac{1}{4} \sum_{x \neq j} E_x = \frac{1}{4}I_d + \frac{1}{2}E_j.$$

Only public classical randomness is added. Each simulated call uses exactly one call to the unknown device.

Apply Theorem S15.3 to each F_j at future horizon one, accuracy ϵ , and failure η/k , using fresh records for different j . Its output $\hat{F}_j$ is Gaussian rational and legal. The binary identity $d_1(F_j, \hat{F}_j) = 2\|F_j - \hat{F}_j\|_{\text{op}}$ implies that, with probability at least $1 - \eta$, all the Hermitian matrices $B_j = 2\hat{F}_j - I_d/2$ obey

$$(S23.4) \quad \max_j \|B_j - E_j\|_{\text{op}} \leq \epsilon.$$

Only a union bound is needed; no independence assertion concerning selected quantum readout outcomes is used.

The matrices B_j need not sum to I_d . Reconstruct the finite legal grid $\mathcal{G}(\epsilon/2)$ and choose its first tuple $\mathbf{G}$ satisfying

$$\|G_j - B_j\|_{\text{op}} \leq 2\epsilon \quad \text{for every } j.$$

Each comparison is an exact rational positive-semidefinite test. On (S23.4), such a tuple exists by Lemma S22.3; it has $r(\mathbf{G}, \mathbf{E}) \leq 3\epsilon$. On every other record, if the finite search finds no tuple, return $\mathbf{U}$. Thus every record has a legal output and the same predetermined training budget. The geometric upper bound gives

$$D_N(\mathbf{E}, \mathbf{G}) \leq 6k\sqrt{N}\epsilon \leq 3\delta/8.$$

Encode $\mathbf{G}$ with the dictionary of Theorem S22.4 at parameter $\delta/2$. Its additional error is at most $\delta/4$, so the decoded error is at most $5\delta/8$. It uses no further device calls and has length (S23.2).

There are k binary learners, each using at most $C_0\epsilon^{-2}[d^2 + d\log(k/\eta)]$ calls. Since $n^2 \leq 4N$, this proves the upper bound. The inherited rational synthesis and finite-control theorem applies to the simulated binary channels as to any channels of the same form. The public $1/4, 3/4$ relabeling is exact classical control. The error guarantee used in (S23.4) is the inherited actual finite-control guarantee, not an idealized readout subsequently assumed exact. The construction terminates in finite classical time, with no assertion that this time is polynomial. For nonrational error parameters, choose rational $\delta' \in [\delta/2, \delta]$ and $\eta' \in [\eta/2, \eta]$ to obtain the stated existence bound; these choices change only absolute constants.

For the lower bound let $m = \lfloor k/2 \rfloor$ and $q = 2m/k \geq 2/3$. Embed any binary interior effect $F \in [I_d/4, 3I_d/4]$ by the tuple

$$(S23.5) \quad E_j(F) = \begin{cases} 2F/k, & 1 \leq j \leq m, \\ 2(I_d - F)/k, & m < j \leq 2m, \\ I_d/k, & j = k = 2m + 1. \end{cases}$$

This belongs to $\mathfrak{P}_{d,k}$. A query to this tuple is simulated using one binary F query: with probability q , retain the binary outcome and select a uniform label from its corresponding group of m labels; otherwise erase the bit and return the extra label. In the odd case the erased reference state is the sum of its two subnormalized binary states, as required by the effect I_d/k . Thus the simulation is valid for entangled inputs and adaptive controls, not merely for outcome probabilities on pure states. For even k there is no erasure.

Suppose a k -outcome learner returns any legal tuple $\hat{\mathbf{E}}$ within D_N -error δ . Coarse-graining the first m labels gives a binary effect $H = \sum_{j \leq m} \hat{E}_j$ satisfying $d_N(qF, H) \leq \delta$. This follows by executing the same coarse-graining inside any binary tester of the two induced channels. The Bernoulli lower bound in Theorem S22.1, applied to these two binary effects, implies

$$\|H - qF\|_{\text{op}} \leq 128\delta/\sqrt{N},$$

since $\delta < 1/128$. Spectrally clip H/q to $[I_d/4, 3I_d/4]$, obtaining $\hat{F}$. Weyl's inequality bounds the clipping displacement by $\|H/q - F\|_{\text{op}}$; the triangle inequality therefore gives

$$\|\hat{F} - F\|_{\text{op}} \leq \frac{256}{q} \frac{\delta}{\sqrt{N}} \leq \frac{384\delta}{\sqrt{N}}.$$

No operator-Lipschitz claim about spectral clipping is used. The last accuracy is at most 2^{-14} under the stated cap. The proof of Corollary S12.4 gives the binary operator-loss lower bound already on this interior, with arbitrary coherent adaptive training. Applying it to the simulation proves the left inequality of (S23.1), including the contribution $d \log(1/\eta)$ and the scalar case. The same simulation preserves every-record budgets and public stopping. $\square$

The length in (S23.2) has the optimal leading order also among learners with a fixed-length, fixed public decoder. Indeed, if failure is less than one at every target, at least one word must decode inside that target's loss ball. The decoder's image is therefore a covering of the entire target family. The lower bound in Theorem S22.2 applies regardless of the number of training calls. This is separate from the statistical query lower bound. Shared-randomness expected-prefix conventions are different and are not part of this assertion.

S23.1. Finite uniform-risk certificates. The certification mechanism also survives a finite outcome alphabet. For an ideal one-call maximally entangled acquisition, its normalized classical-quantum record is

$$(S23.6) \quad \Omega(\mathbf{E}) = \frac{1}{d} \bigoplus_{j=1}^k E_j^\top.$$

The classical labels remain ordered. A collective readout after M independent acquisitions can be described by one POVM $(A_{c,x})_c$ on dimension d^M for each classical string $x \in \{1, \dots, k\}^M$. Here $A_{c,x} \succeq 0$ and $\sum_c A_{c,x} = I_{d^M}$. Its output label c is decoded by a fixed finite family $\mathbf{C}_c$ of legal k -outcome tuples.

Theorem S23.2 (Finite-alphabet certificate transfer). *Let $\mathcal{G} \subset \mathfrak{P}_{d,k}$ be a complete finite r_0 -net in the tuple norm r . Suppose a fixed readout satisfies*

$$(S23.7) \quad \sum_{c: r(\mathbf{G}, \mathbf{C}_c) \leq a} \sum_{x \in \{1, \dots, k\}^M} \text{tr} \left[A_{c,x} d^{-M} \bigotimes_{i=1}^M G_{x_i}^\top \right] \geq 1 - \alpha \quad (\mathbf{G} \in \mathcal{G}).$$

Then every real $\mathbf{E} \in \mathfrak{P}_{d,k}$ satisfies

$$(S23.8) \quad \mathbb{P}_{\mathbf{E}}\{r(\mathbf{E}, \mathbf{C}_J) > a + r_0\} \leq \alpha + \frac{Mkr_0}{2}.$$

If all decoding centres are in $\mathfrak{P}_{d,k}$, the same failure bound holds for D_N -error greater than $2k\sqrt{N}(a + r_0)$. For Gaussian-rational data, rational a, r_0, α , and the net of Lemma S22.3, the hypotheses admit a finite exact verification. A proper prefix of the net or of the classical string list is not a certificate.

Proof. The block-diagonal formula (S23.6) gives

$$\|\Omega(\mathbf{E}) - \Omega(\mathbf{G})\|_1 = d^{-1} \sum_j \|E_j - G_j\|_1 \leq k r(\mathbf{E}, \mathbf{G}).$$

Tensor telescoping between normalized states gives

$$(S23.9) \quad \|\Omega(\mathbf{E})^{\otimes M} - \Omega(\mathbf{G})^{\otimes M}\|_1 \leq Mkr(\mathbf{E}, \mathbf{G}).$$

Apply the same readout to both states. Its index laws differ in total variation by at most $Mkr_0/2$ when $r(\mathbf{E}, \mathbf{G}) \leq r_0$. Choose such a $\mathbf{G}$. The fixed set of labels $\{c : r(\mathbf{G}, \mathbf{C}_c) \leq a\}$ has probability

at least $1 - \alpha$ at $\mathbf{G}$, by (S23.7); every label in this set has $r(\mathbf{E}, \mathbf{C}_c) \leq a + r_0$. Transferring this one event proves (S23.8). There is no assumption that the success-label sets vary continuously with the unknown target.

The geometric upper comparison proves the assertion about D_N . For exact verification, reconstruct the whole legal net and every classical string. All traces in (S23.7) are Gaussian-rational expressions with rational real values. Check complete POVM positivity and normalization at every string, use $sI_d \pm (G_j - C_{c,j}) \succeq 0$ to determine the good labels including equality, and check each rational inequality. These are finite computations. Their soundness on the real parameter body is supplied by (S23.9), not by a numerical sampling argument. $\square$

The certificate describes the ideal specified rational readout. When implemented controls have a separately certified output-law error ζ , the right side of (S23.8) increases by at most ζ . Such an error must be budgeted, not silently identified with zero. The finite-control guarantee in Theorem S23.1 instead uses the already budgeted binary procedures. The exact certificate transfer and the actual-control learning guarantee thus have explicitly distinct hypotheses.

S24. SEARCH-FREE LEGALIZATION OF BALANCED ESTIMATES

The exact public dictionary and the statistical legalization step have different purposes. A dictionary exploits global packing to reduce transmitted length. Legalizing a collection of component estimates need not reconstruct such a dictionary.

Theorem S24.1 (Affine repair with a uniform error bound). *Let $B = (B_1, \dots, B_k)$ be Hermitian matrices and $a > 0$. Define*

$$(S24.1) \quad A_j = B_j + \frac{I_d - \sum_i B_i}{k},$$

$$(S24.2) \quad L_a(B)_j = \frac{A_j + 4aI_d}{1 + 4ka}.$$

Suppose $E \in \mathfrak{P}_{d,k}$, the balanced interior from (S22.1), and $\max_j \|B_j - E_j\|_{\text{op}} \leq a$. Then $L_a(B) \in \mathfrak{P}_{d,k}$ and

$$(S24.3) \quad \max_j \|L_a(B)_j - E_j\|_{\text{op}} \leq \frac{4a}{1 + 4ka} \leq 4a.$$

For rational input and rational a , this is a rational map. Testing its balanced legality and returning the uniform tuple upon failure defines a total legal procedure on every Hermitian input record. It uses $O(kd^2)$ rational arithmetic operations apart from $2k$ exact positive-semidefinite tests of order d , and polynomial bit complexity. There is no grid enumeration.

Proof. Let $\Delta_j = B_j - E_j$. Then $A_j - E_j = \Delta_j - k^{-1} \sum_i \Delta_i$, whose norm is at most $2a$, and $\sum_j A_j = I_d$. Consequently

$$(1/(2k) - 2a)I_d \preceq A_j \preceq (3/(2k) + 2a)I_d.$$

Adding $4aI_d$ and dividing by $1 + 4ka$ gives exactly the two balanced spectral inequalities. The sum of the repaired effects is I_d . Finally,

$$L_a(B)_j - E_j = \frac{A_j - E_j + 4ka(I_d/k - E_j)}{1 + 4ka}.$$

The balanced hypothesis gives $\|I_d/k - E_j\|_{\text{op}} \leq 1/(2k)$, proving (S24.3). Rationality and the operation count follow from the formula. The exact legality test makes no good-event assumption; if it fails the uniform tuple is an explicitly specified legal output. On the good event it never fails. $\square$

Corollary S24.2 (A common learner with polynomial legalization). *Under the range and access conventions of Theorem S23.1, replace its tuple-grid legalization by the guarded map in Theorem S24.1. Taking*

$$a = \frac{\delta}{32k\lceil\sqrt{N}\rceil}$$

and allocating failure η/k to each fresh binary component learner gives, on every record,

$$M \leq Ck^3N\delta^{-2}[d^2 + d\log(k/\eta)]$$

for an absolute C , with future-use error at most $\delta/4$ on a common event of probability at least $1 - \eta$. Encoding with the public dictionary at parameter $\delta/2$ gives error at most $\delta/2$ on the same event, without further calls, and retains the optimal fixed- k fixed-length description order. Only the legalization cost is made polynomial; neither dictionary construction nor general collective-readout synthesis is asserted to be efficient.

Proof. Use precisely the reference-preserving binary simulation and actual finite-control component learners from Theorem S23.1, at the displayed smaller accuracy. Its conditional good-event argument gives component error at most a with total failure at most η . Affine repair then gives tuple error at most $4a$. The balanced metric upper bounds the future-use error by $8k\sqrt{N}a \leq \delta/4$. The dictionary adds at most $\delta/4$. Fresh calls and all finite-control budgets are charged exactly as in the component theorem; decreasing the component accuracy by a factor two changes its call bound only by an absolute factor. The guarded map is total even off the good event, so the recordwise call bound and public stopping convention are unchanged. $\square$

REFERENCES

- [1] J. Watrous, *The Theory of Quantum Information*, Cambridge University Press, 2018. Theorems 2.22 and 2.26 (Choi criteria), Corollary 3.40 (trace-norm contraction), and Section 3.3, equation (3.306) (hybrid comparison).
- [2] T. Surawy-Stepney, J. Kahn, R. Kueng and M. Guță, Projected least-squares quantum process tomography, *Quantum* **6** (2022), 844; arXiv:2107.01060v2, 18 October 2022.
- [3] G. Gutoski, On a measure of distance for quantum strategies, *J. Math. Phys.* **53** (2012), 032202; arXiv:1008.4636v4, 14 March 2012.
- [4] S. G. Naik, M. Zartab, N. Gisin and M. Banik, No-Go Theorem for Generic Simulation of Qubit Channels with Finite Classical Resources, arXiv:2501.15807v2 (16 July 2025), Theorems 2 and 5.
- [5] A. A. Mele and L. Bittel, Optimal learning of quantum channels in diamond distance, arXiv:2512.10214v3 (2026), Theorem I.1, Lemmas III.7–III.8, Corollary III.9 and Sections III.1.1, IV.1.1.
- [6] A. Oufkir and F. Girardi, Improved Lower Bounds for Learning Quantum Channels in Diamond Distance, arXiv:2601.04180v4 (2026).
- [7] R. Demkowicz-Dobrzański, J. Kołodyński and M. Guță, The elusive Heisenberg limit in quantum-enhanced metrology, *Nature Communications* **3** (2012), 1063. DOI: 10.1038/ncomms2067; arXiv:1201.3940v2, equations (4)–(6) (classical simulation), equations (10)–(13) (channel extension), and the classical-simulation argument in Methods.
- [8] Y. Yang, G. Bai, G. Chiribella and M. Hayashi, Compression for quantum population coding, *IEEE Transactions on Information Theory* **64** (2018), no. 7, doi:10.1109/TIT.2017.2788407; arXiv:1701.03372v8.
- [9] F. Salek, M. Hayashi and A. Winter, Usefulness of adaptive strategies in asymptotic quantum channel discrimination, *Phys. Rev. A* **105** (2022), 022419; arXiv:2011.06569v3 (17 March 2024).
- [10] T. Cooney, M. Mosonyi and M. M. Wilde, Strong converse exponents for a quantum channel discrimination problem and quantum-feedback-assisted communication, arXiv:1408.3373v3 (26 February 2016), p. 5 and Sections 4.1–4.2.
- [11] A. Acín, Statistical distinguishability between unitary operations, *Phys. Rev. Lett.* **87** (2001), 177901. DOI: 10.1103/PhysRevLett.87.177901.
- [12] R. Duan, Y. Feng and M. Ying, Entanglement is not necessary for perfect discrimination between unitary operations, *Phys. Rev. Lett.* **98** (2007), 100503. DOI: 10.1103/PhysRevLett.98.100503.
- [13] S. Das and M. M. Wilde, Quantum reading capacity: General definition and bounds, *IEEE Trans. Inform. Theory* **65** (2019), 7566–7583; arXiv:1703.03706v3; doi:10.1109/TIT.2019.2929925.

- [14] M. M. Wilde, M. Berta, C. Hirche and E. Kaur, Amortized channel divergence for asymptotic quantum channel discrimination, *Lett. Math. Phys.* **110** (2020), 2277–2336; arXiv:1808.01498v2; doi:10.1007/s11005-020-01297-7.
- [15] X. Wang and M. M. Wilde, Resource theory of asymmetric distinguishability for quantum channels, *Phys. Rev. Research* **1** (2019), 033169; arXiv:1907.06306v2; doi:10.1103/PhysRevResearch.1.033169.
- [16] Z. Puchała, Ł. Paweła, A. Krawiec, R. Kukulski and M. Oszmaniec, Multiple-shot and unambiguous discrimination of von Neumann measurements, *Quantum* **5** (2021), 425; arXiv:1810.05122v4; doi:10.22331/q-2021-04-06-425.
- [17] M. Sedláč and M. Ziman, Optimal single-shot strategies for discrimination of quantum measurements, *Phys. Rev. A* **90** (2014), 052312; arXiv:1408.0934; doi:10.1103/PhysRevA.90.052312. Section VI, Example 4, equation (37).
- [18] Z. Puchała, Ł. Paweła, A. Krawiec and R. Kukulski, Strategies for optimal single-shot discrimination of quantum measurements, *Phys. Rev. A* **98** (2018), 042103; arXiv:1804.05856; doi:10.1103/PhysRevA.98.042103. Theorem 1.
- [19] J. Fiurášek and M. Mičuda, Optimal two-copy discrimination of quantum measurements, *Phys. Rev. A* **80** (2009), 042312; arXiv:0909.2940; doi:10.1103/PhysRevA.80.042312.
- [20] A. Krawiec, Ł. Paweła and Z. Puchała, Discrimination of POVMs with rank-one effects, *Quantum Information Processing* **19** (2020), 428; arXiv:2002.05452v1; doi:10.1007/s11128-020-02883-3.
- [21] C. Datta, T. Biswas, D. Saha and R. Augusiak, Perfect discrimination of quantum measurements using entangled systems, *New Journal of Physics* **23** (2021), 043021; arXiv:2012.07069v2; doi:10.1088/1367-2630/abecaf.
- [22] A. Fujiwara and H. Imai, A fibre bundle over manifolds of quantum channels and its application to quantum statistics, *J. Phys. A: Math. Theor.* **41** (2008), 255304; doi:10.1088/1751-8113/41/25/255304. Theorems 4 and 5.
- [23] R. Demkowicz-Dobrzański and L. Maccone, Using entanglement against noise in quantum metrology, *Phys. Rev. Lett.* **113** (2014), 250801; arXiv:1407.2934v3; doi:10.1103/PhysRevLett.113.250801.
- [24] S. Kurdziałek, W. Górecki, F. Albarelli and R. Demkowicz-Dobrzański, Using adaptiveness and causal superpositions against noise in quantum metrology, *Phys. Rev. Lett.* **131** (2023), 090801; arXiv:2212.08106v2; doi:10.1103/PhysRevLett.131.090801.
- [25] H. Yuan and C.-H. F. Fung, Fidelity and Fisher information on quantum channels, arXiv:1506.00819v3 (2017), Section III, equation (26).
- [26] S. Sieniawski and R. Demkowicz-Dobrzański, Adaptive quantum channel discrimination using methods of quantum metrology, *New Journal of Physics* **28** (2026), 024502; arXiv:2510.15506v3; doi:10.1088/1367-2630/ae3edd.
- [27] J. Dittmann, Explicit formulae for the Bures metric, *J. Phys. A: Math. Gen.* **32** (1999), 2663–2670; doi:10.1088/0305-4470/32/14/007; arXiv:quant-ph/9808044, entitled *Note on explicit formulae for the Bures metric*.
- [28] H.-J. Sommers and K. Życzkowski, Bures volume of the set of mixed quantum states, *J. Phys. A: Math. Gen.* **36** (2003), 10083–10100; arXiv:quant-ph/0304041v3; doi:10.1088/0305-4470/36/39/308.
- [29] L. Zambrano, S. Ramos-Calderer and R. Kueng, Fast quantum measurement tomography with optimal error bounds, *Quantum* **10** (2026), 2162; arXiv:2507.04500v3; doi:10.22331/q-2026-07-15-2162.
- [30] A. Bisio, G. M. D’Ariano, P. Perinotti and M. Sedláč, Quantum learning algorithms for quantum measurements, *Physics Letters A* **375** (2011), 3425–3434; arXiv:1103.0480v2; doi:10.1016/j.physleta.2011.08.002.
- [31] B. L. Higgins, D. W. Berry, S. D. Bartlett, M. W. Mitchell, H. M. Wiseman and G. J. Pryde, Demonstrating Heisenberg-limited unambiguous phase estimation without adaptive measurements, *New Journal of Physics* **11** (2009), 073023; arXiv:0809.3308v4; doi:10.1088/1367-2630/11/7/073023.
- [32] S. Kimmel, G. H. Low and T. J. Yoder, Robust calibration of a universal single-qubit gate set via robust phase estimation, *Phys. Rev. A* **92** (2015), 062315; doi:10.1103/PhysRevA.92.062315; erratum, *Phys. Rev. A* **104** (2021), 069901, doi:10.1103/PhysRevA.104.069901; arXiv:1502.02677v3.
- [33] P. Hyllus, W. Laskowski, R. Krischek, C. Schwemmer, W. Wieczorek, H. Weinfurter, L. Pezzé and A. Smerzi, Fisher information and multiparticle entanglement, *Phys. Rev. A* **85** (2012), 022321; arXiv:1006.4366v3; doi:10.1103/PhysRevA.85.022321. Section III.A, Observation 1, equation (9).
- [34] G. Tóth, Multipartite entanglement and high-precision metrology, *Phys. Rev. A* **85** (2012), 022322; arXiv:1006.4368v3; doi:10.1103/PhysRevA.85.022322. Observation 3, equation (6).
- [35] K. Chen, Z. Zhang and N. Yu, Optimal lower bound for quantum channel tomography in away-from-boundary regime, arXiv:2601.10683v1 (15 January 2026), Theorems 1.1, 3.2 and 4.2.
- [36] S. Chen, J. Li and A. Liu, An optimal tradeoff between entanglement and copy complexity for state tomography, arXiv:2402.16353v1 (26 February 2024).
- [37] A. Nayak and X. Zhou, Optimal Low-Rank Quantum State Tomography with Bounded-Sample Joint Measurements, arXiv:2609.10514v1 (9 September 2026), Theorems 1.1–1.2.

- [38] S. Basu, R. Pollack and M.-F. Roy, *Algorithms in Real Algebraic Geometry*, second edition, Algorithms and Computation in Mathematics, vol. 10, Springer, Berlin, Heidelberg, 2006; doi:10.1007/3-540-33099-2. Theorems 2.77 and 2.80 and Chapter 14.
- [39] A. Pelecanos, J. Spilecki, E. Tang and J. Wright, Mixed state tomography reduces to pure state tomography, arXiv:2511.15806v1 (19 November 2025). Theorems 1.1–1.3 and 4.2, Section 3.3, Figures 8–9.
- [40] J. F. Bonnans, R. Cominetti and A. Shapiro, Second order optimality conditions based on parabolic second order tangent sets, *SIAM J. Optim.* **9** (1999), 466–492.
- [41] A. W. Harrow, A. Hassidim, D. W. Leung and J. Watrous, Adaptive versus non-adaptive strategies for quantum channel discrimination, *Phys. Rev. A* **81** (2010), 032339.
- [42] F. Salek, M. Hayashi and A. Winter, Usefulness of adaptive strategies in asymptotic quantum channel discrimination, *Phys. Rev. A* **105** (2022), 022419.
- [43] Y. Li, C. Hirche and M. Tomamichel, Sequential quantum channel discrimination, arXiv:2210.11079.
- [44] Z. Huang, J. J. Meyer, T. Nuradha and M. M. Wilde, Query complexities of quantum channel discrimination and estimation: A unified approach, arXiv:2511.10832v2 (2026).
- [45] P. Ghosal, P. Halder, A. Patra and A. Sen, Quantum hypothesis testing of non-mixed-unitarity: A multifaceted hierarchy of quantum channel discrimination, arXiv:2609.40355v1 (2026).
- [46] T.-A. Ohst, S. Zhang, H. C. Nguyen, M. Plávala and M. T. Quintino, Characterising memory in quantum channel discrimination via constrained separability problems, *Quantum* **10** (2026), 1988; doi:10.22331/q-2026-01-28-1988; arXiv:2411.08110v2.